Document

May 14, 2026

1 Benchmark of search times in various data structures

The benchmark looks at the search speeds of the following data structures:

- A simple binary tree
- A binary tree which stores its data in a contiguous array
- std::map (typically implemented as a red-black tree)
- std::unordered_map with the following bucket counts:
  - The dynamically chosen bucket count
  - 100 buckets
  - 1000 buckets
  - 5000 buckets
  - 10000 buckets
  - 20000 buckets
  - 40000 buckets
  - 70000 buckets

Each data structure was examined under the following loads:

- 1000 Nodes
- 10000 Nodes
- 100000 Nodes
- 1000000 Nodes
- 10000000 Nodes

One Node contains a single key of type uint64_t, and two pointers t its child nodes. These nodes were also used in data structures which do not require them for the sake of consistency.

2 Benchmark enviroment

The benchmark was performed in the following enviroment:

- Processor: 11th Gen Intel Core i3
  - Clock speed: 2.00 GHz
  - cores: 4
  - logical units: 2
  - Cache: 320KB/5MB/8MB
- Operating system: Ubuntu under WSL
- Language: C++ compiled with GCC

3 Software Libraries

- nanobench: Used for performing the benchmark. Its Random number generation capabilities were also used for load generation
- C++ standard library: The std::map and std::unordered_map data structures being tested are from this library.

4 The Two Trees

Two of the data strutures are my own implementation. They are both binary search trees with a very simple interface:

- The constructor takes an std::vector of keys and generates a tree from it.
- The Search method takes a key and returns a pointer to the desired Node

The only difference between the two trees is their methid if storage. The BinTree class simply allocates Nodes on the heap, whereas the BinTreeCont class stores them using an std::vector.

5 Performing the Benchmark

The structure of the benchmark is as follows:

The program takes 6 arguments:

- The number of Nodes to generate
- The seed used for generating Nodes
- The seed used for generatin search keys
- The number of iterations the benchmark performs
- The number of buckets the std::unordered_map with the custom bucket count will have
- The file to output the results to

If no parameters are given, the preogram will continue with default values.

The program then generates the test cases, and performs the bencmark, outputting the result in JSON format.

A warmup period of 1000 iterations take place before each test to ensure correct measurements.

Using a bash script, the program can be run several times with different configurations.

The std::unordered_map with the custom bucket count was only tested once per bucket count and node count combination. This was done due to the lower bucket counts causing very long run times. All other data structures, including the dynamically sizzed std::unordered_map were tested several times, and their results are an average of the median of each test.

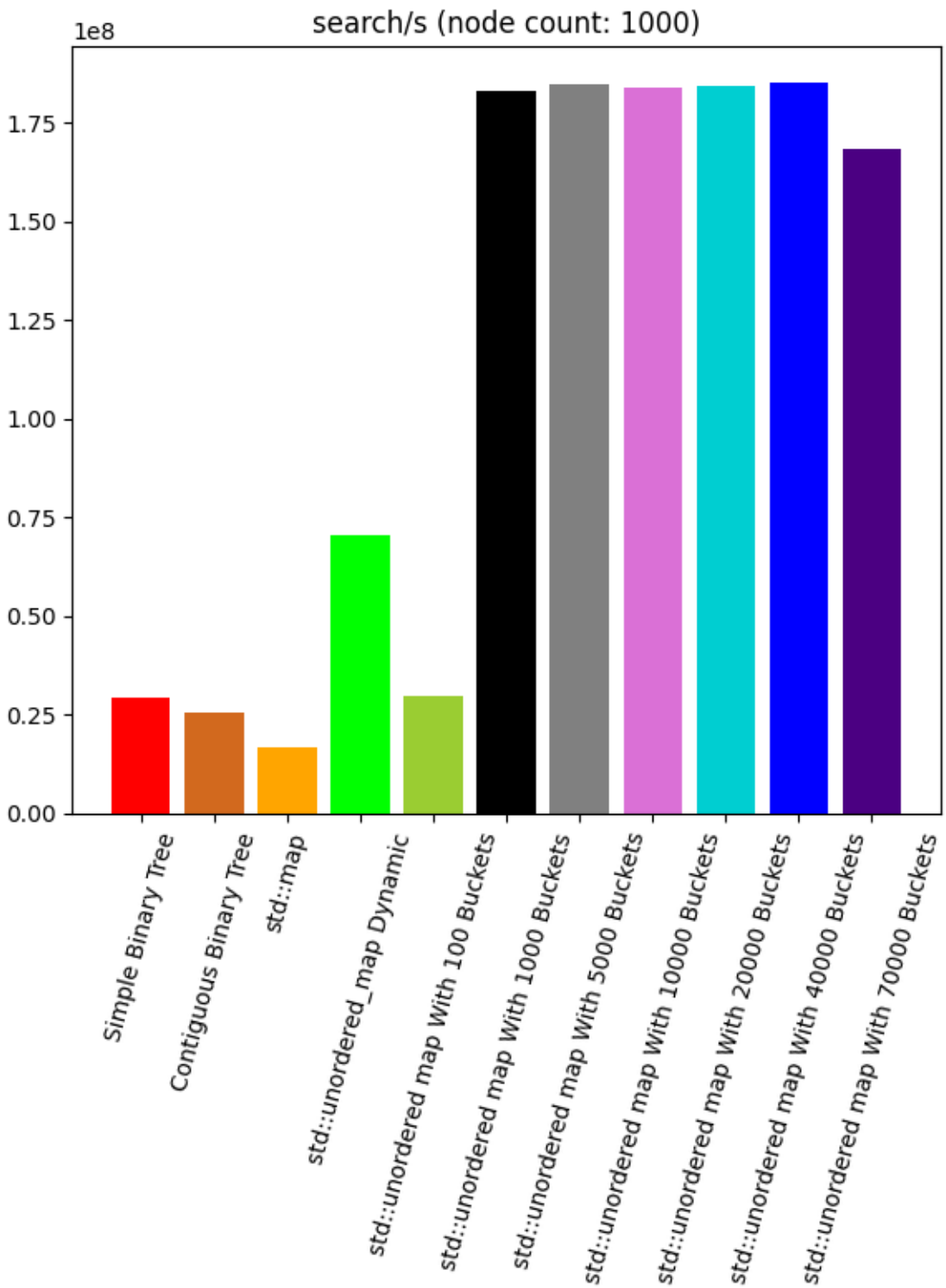
6 Units of measurement

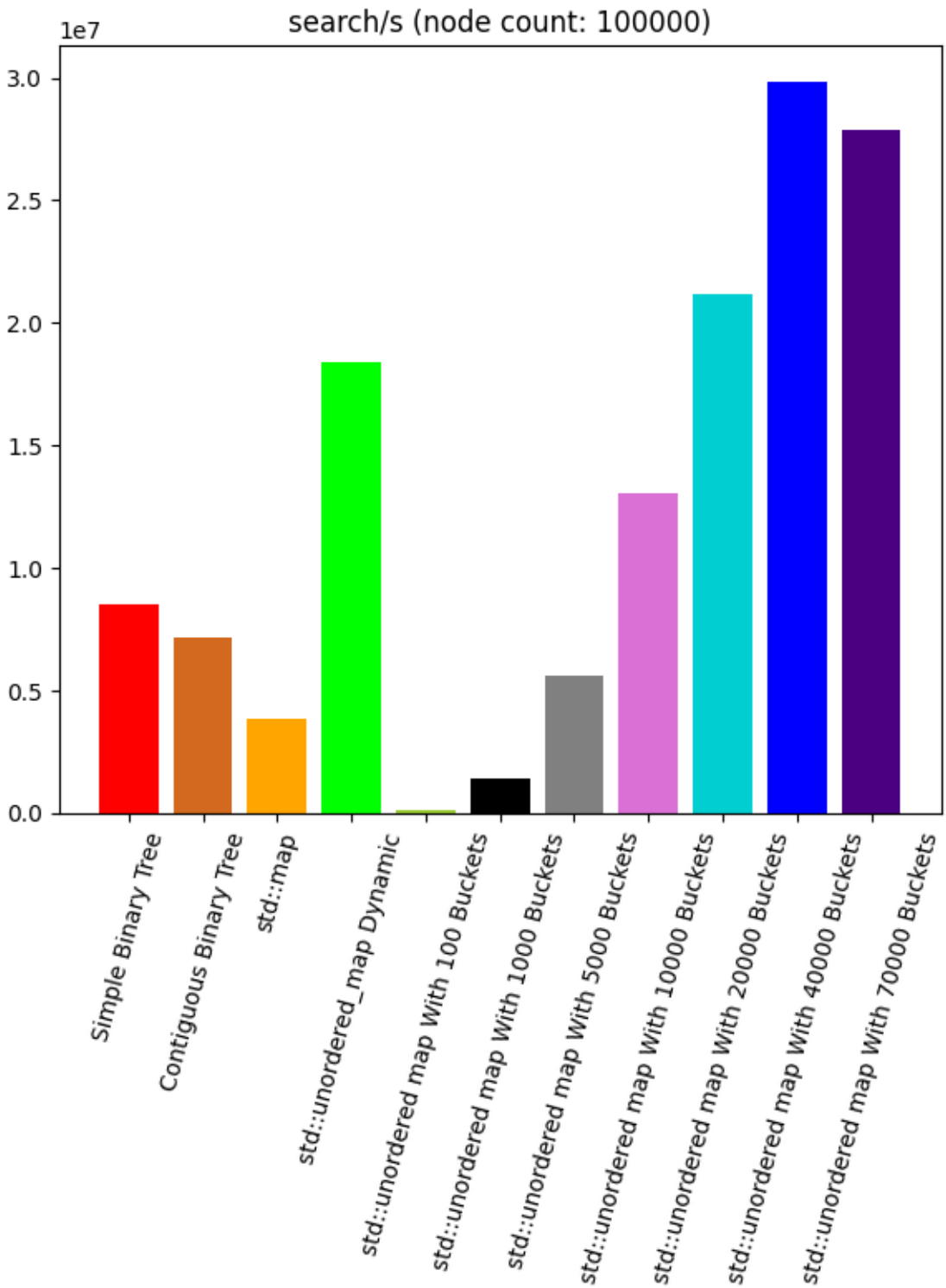
- searches per second (search/s)
- search time in seconds (elapsed)
- CPU instructions per search (instructions)
- CPU cycles per search (cycles)

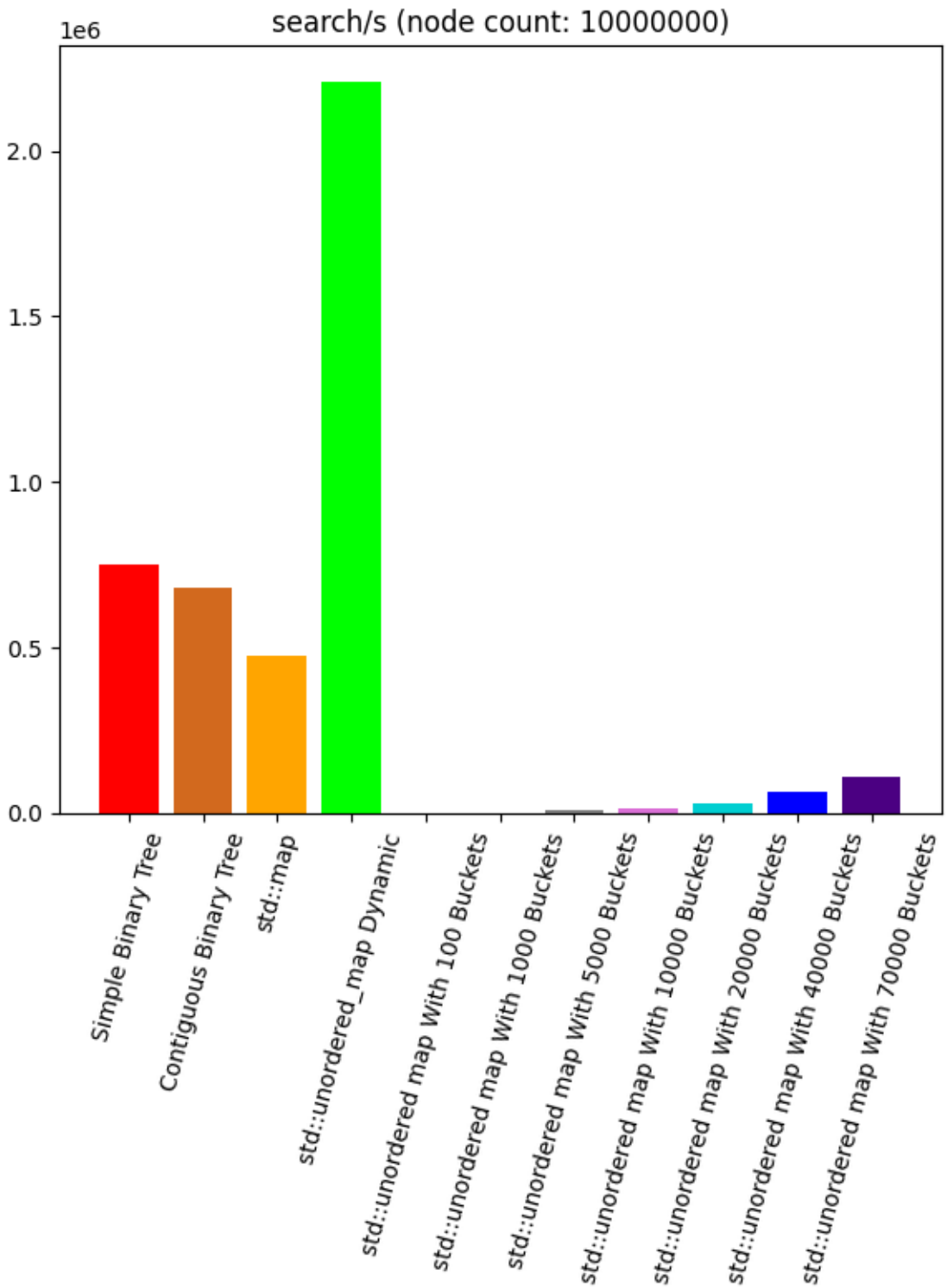
- CPU cycles per instruction (cycles/ins)
- branch instructions (branches)
- branch misses (ranch misses)
- branch misses per branch instruction (misses/branch)

7 Results

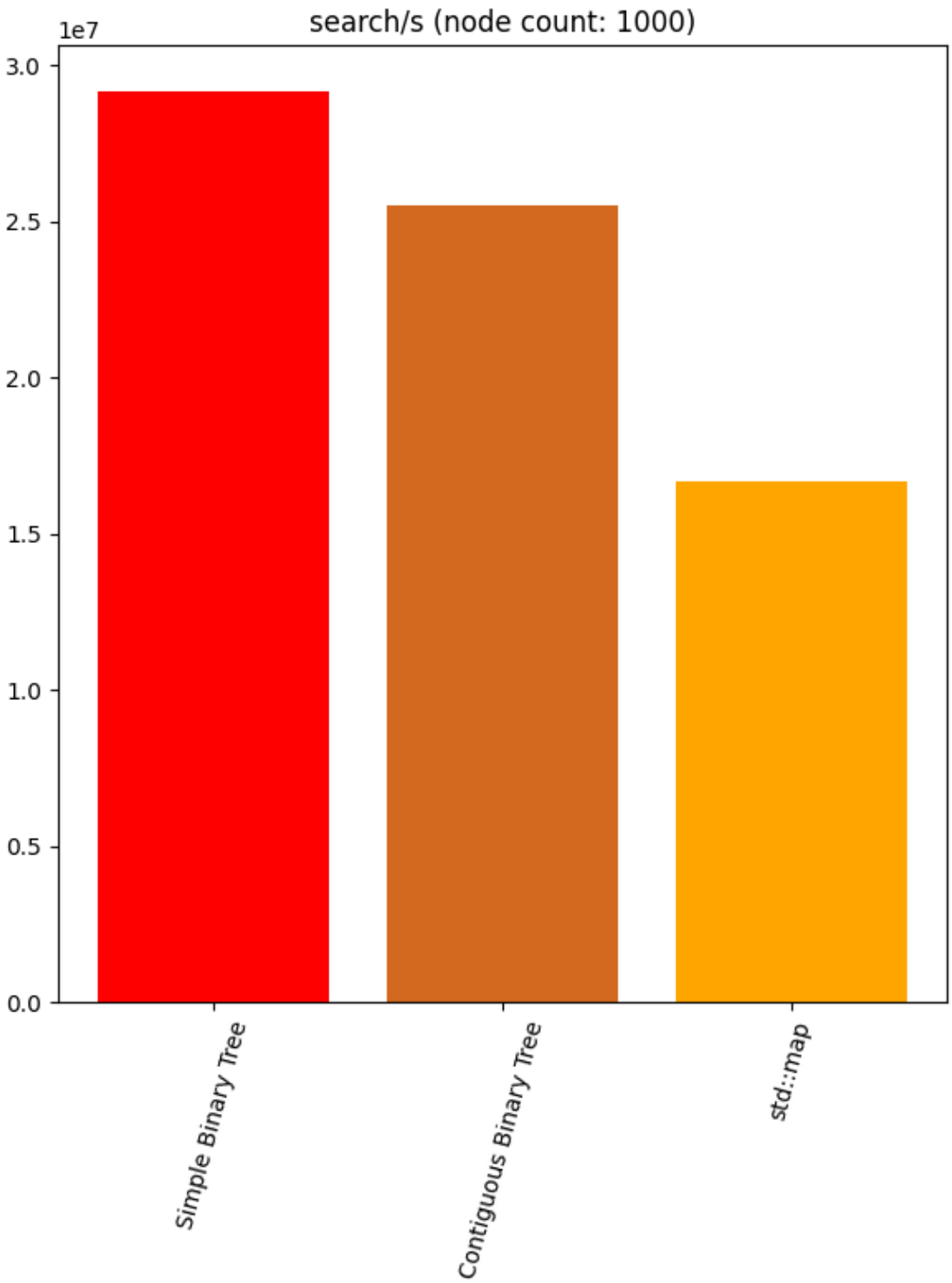
The results show that an std::unordered_map with more buckets than stored elements can find an element in near-constant time, vastly outperforming any other data structure. On the other hand, if the bucket count becomes too low, the search speed plummets. The dynamic std::unordered_map matches the bucket count to its size really well, consistently outdoing the other data structures.

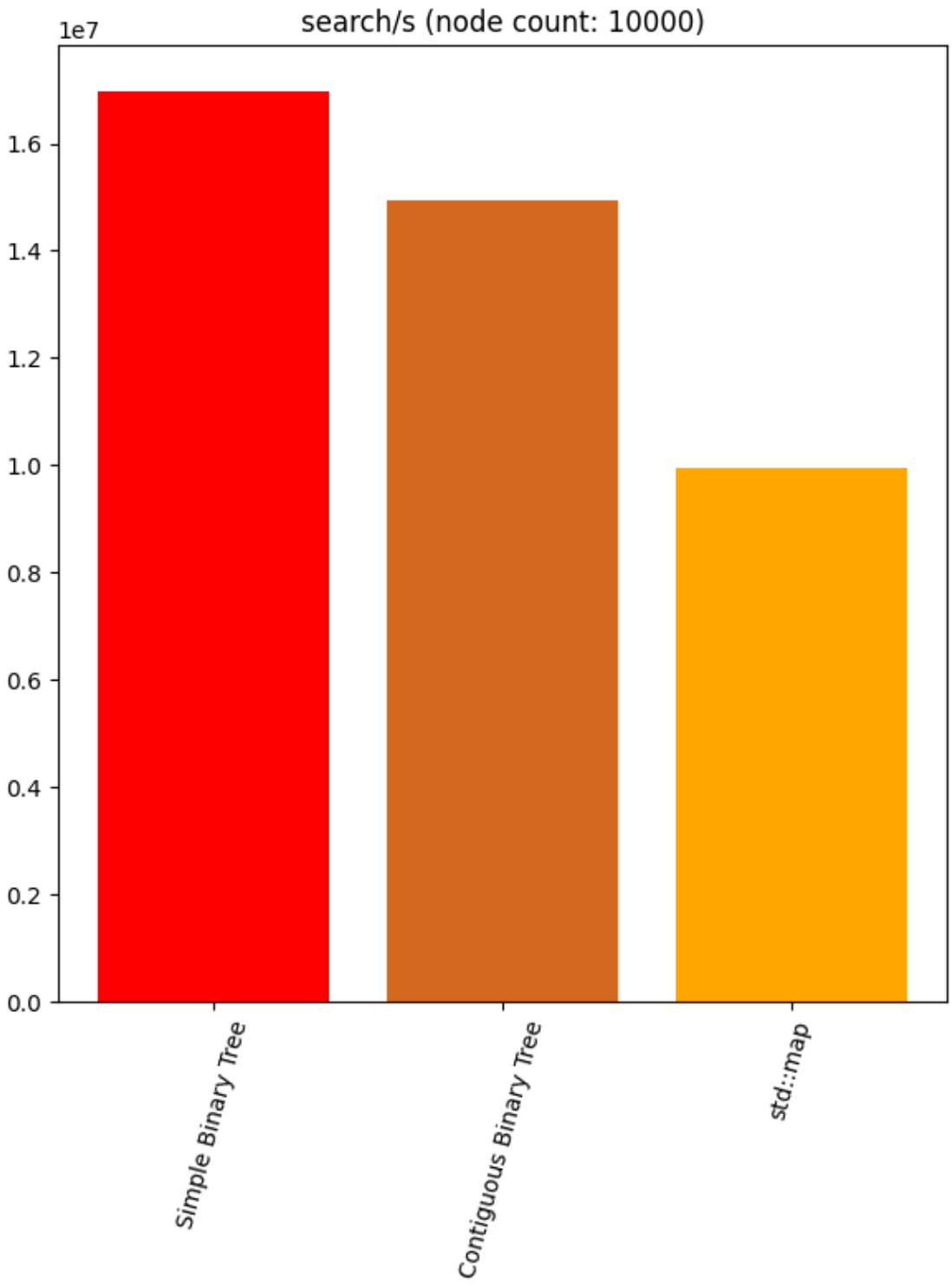

search/s (node count: 1000)
1e8
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
Simple Binary Tree
Contiguous Binary Tree
std::map
std::unordered_map Dynamic
std::unordered map With 100 Buckets
std::unordered map With 1000 Buckets
std::unordered map With 5000 Buckets
std::unordered map With 10000 Buckets
std::unordered map With 20000 Buckets
std::unordered map With 40000 Buckets
std::unordered map With 70000 Buckets

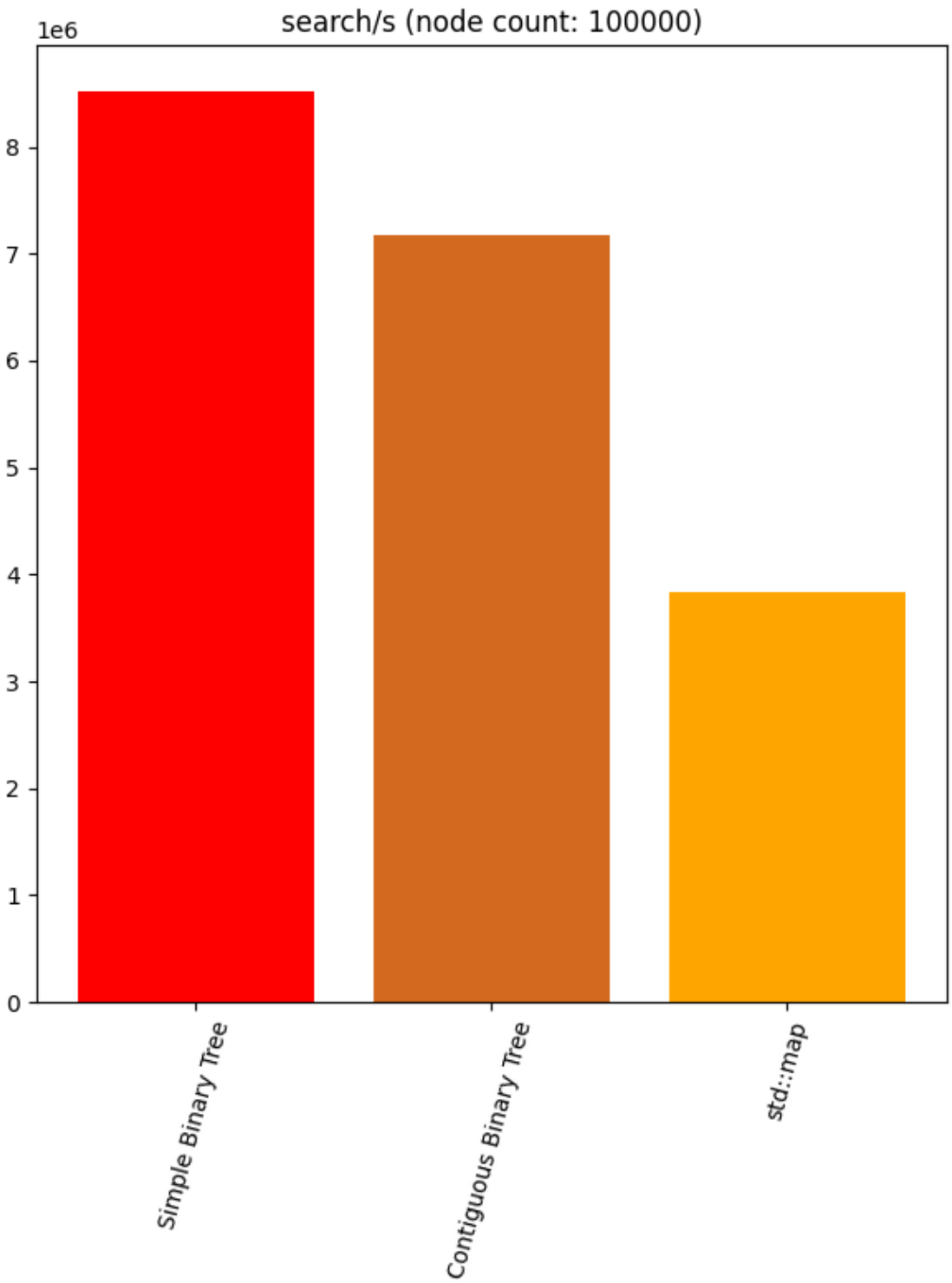

search/s (node count: 100000)
1e7
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Simple Binary Tree
Contiguous Binary Tree
std::map
std::unordered_map Dynamic
std::unordered map With 100 Buckets
std::unordered map With 1000 Buckets
std::unordered map With 5000 Buckets
std::unordered map With 10000 Buckets
std::unordered map With 20000 Buckets
std::unordered map With 40000 Buckets
std::unordered map With 70000 Buckets



My simple binary tree seems to be consistently about twice as fast as std::map. The exact reasons for this are unclear.

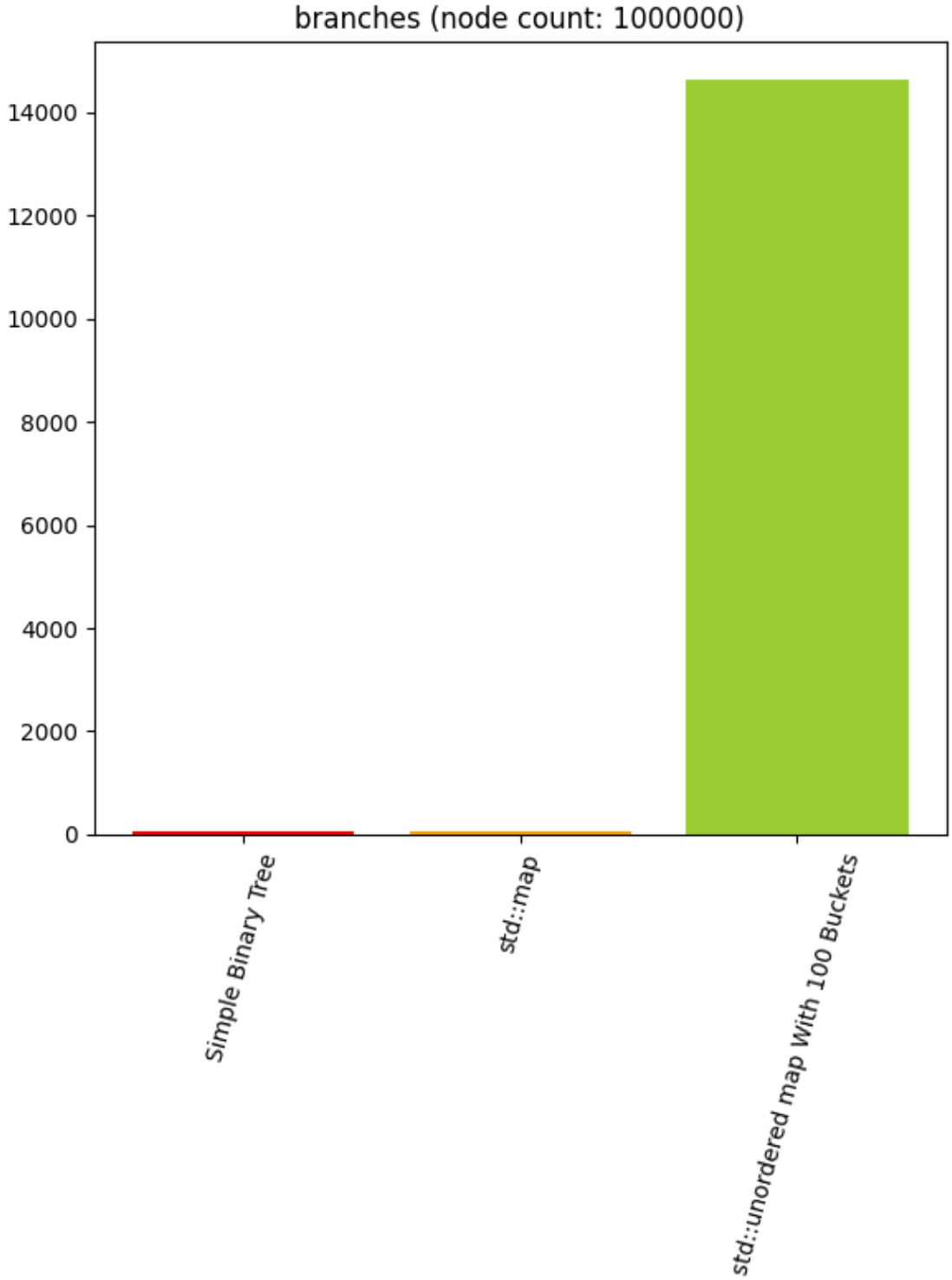

search/s (node count: 1000)
1e7
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Simple Binary Tree
Contiguous Binary Tree
std::map

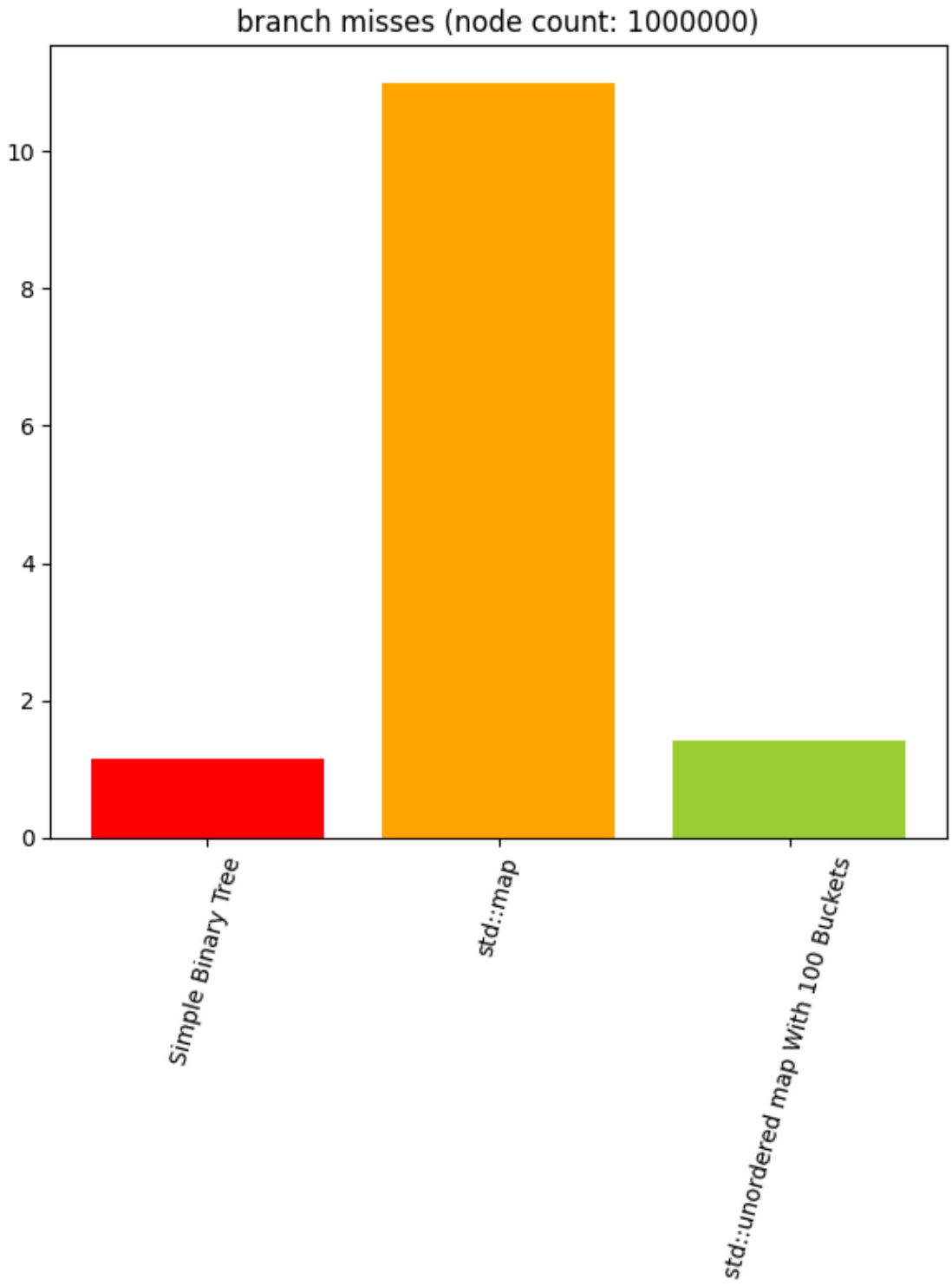

search/s (node count: 10000)
1e7
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
Simple Binary Tree
Contiguous Binary Tree
std::map

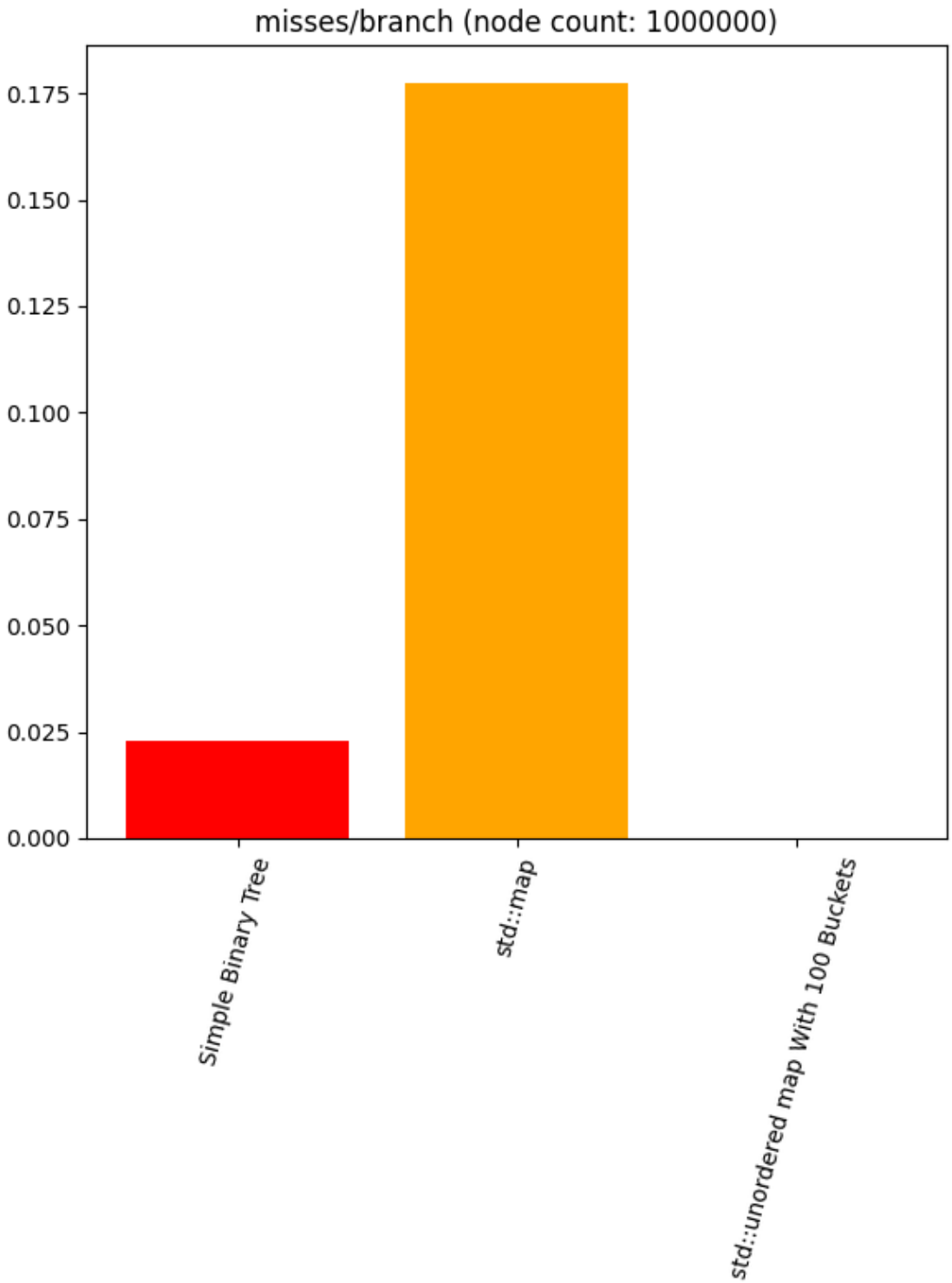


At low bucket counts compared to the number of stored elements, the number of branches during an std::unordered_map's search skyrockets. These branches however, are extremely consistent, the branch predictor almost never makes a mistake. This makes sense, as with a bucket count his low,

most of the search is spent traversing a linked list. Compared to this, std::map misses about 10% of its branches, indicating that something more complex is going on behind the scenes.

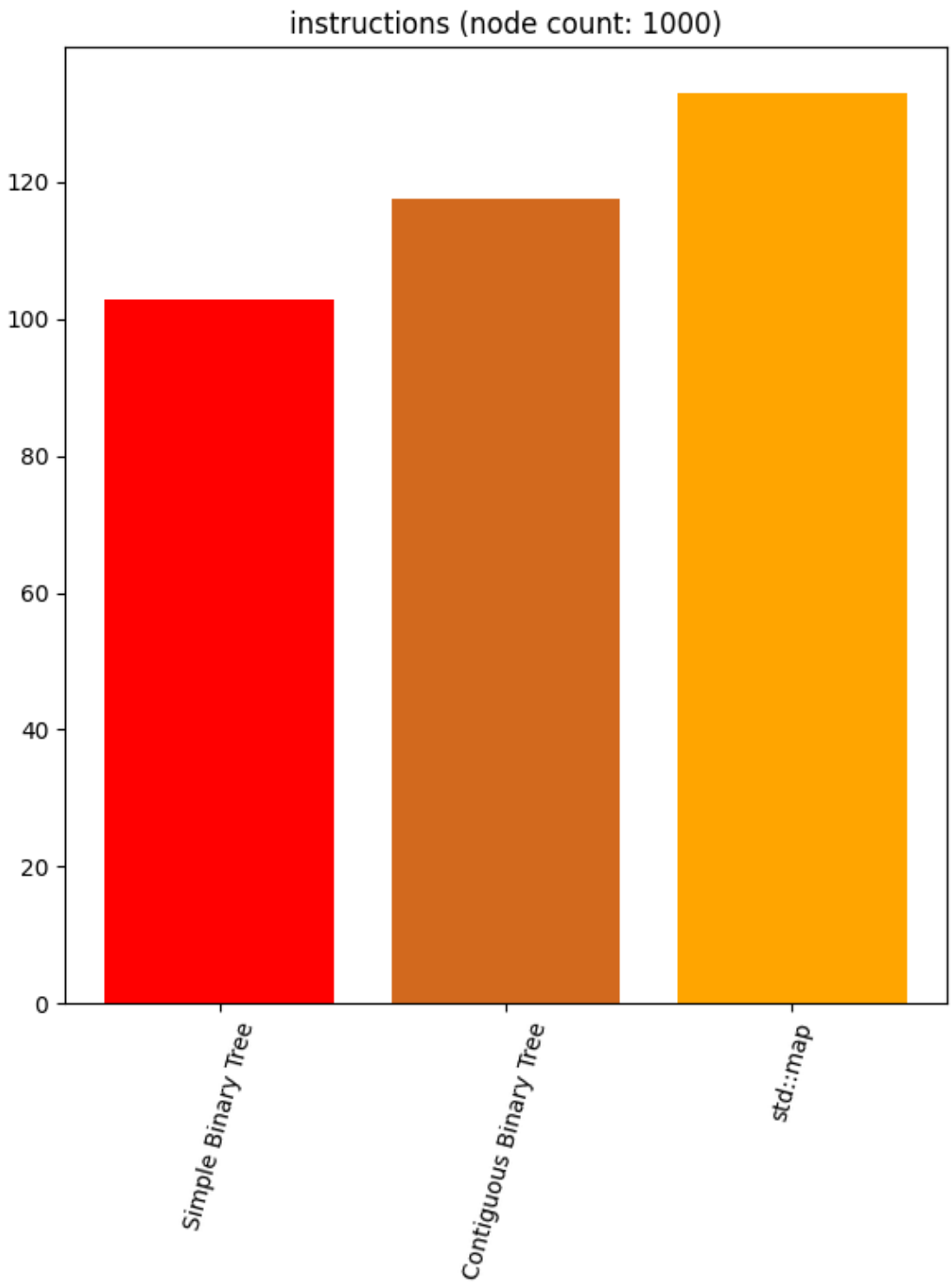


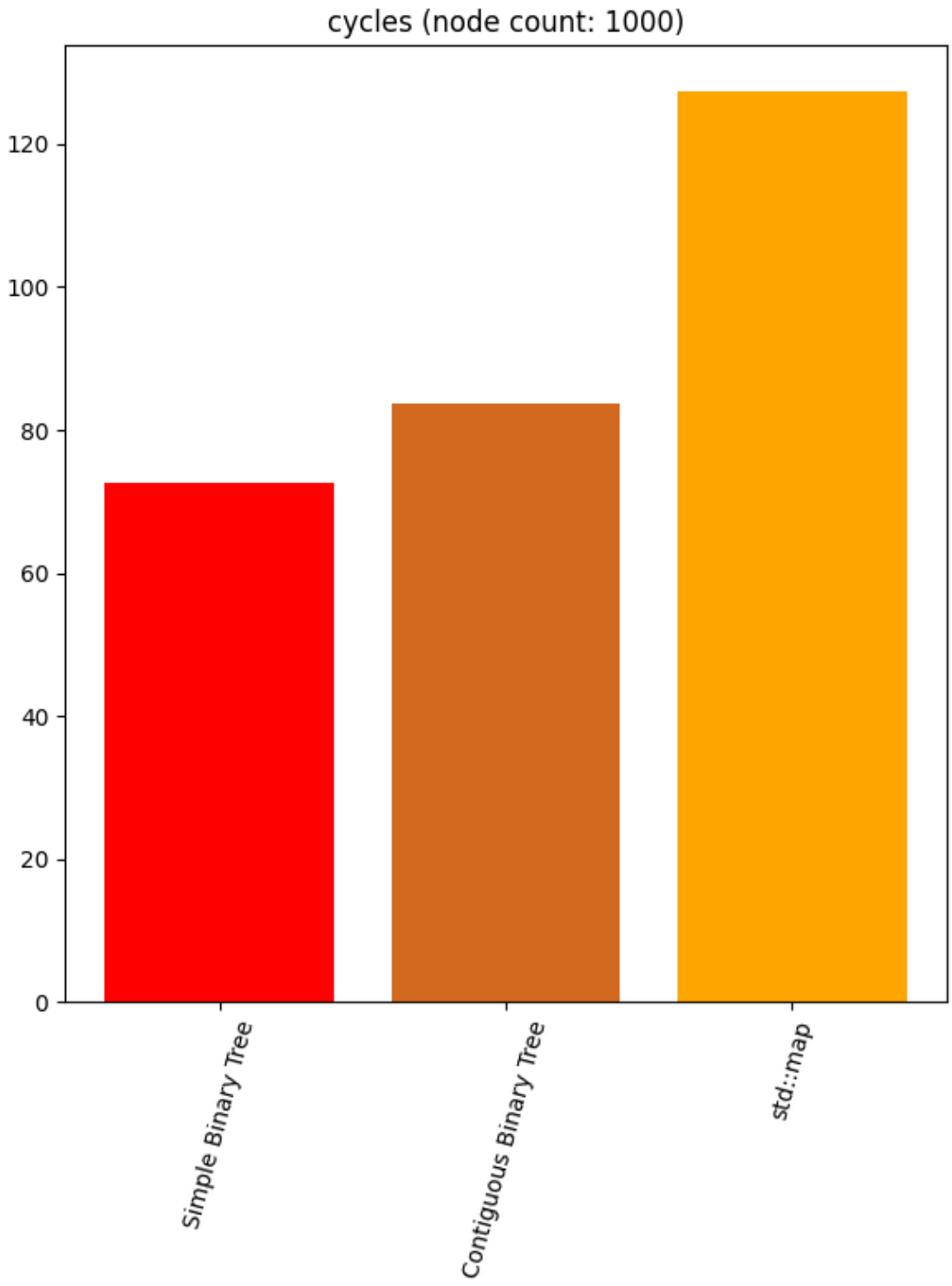

branch misses (node count: 1000000)
0
2
4
6
8
10
Simple Binary Tree
std::map
std::unordered map With 100 Buckets

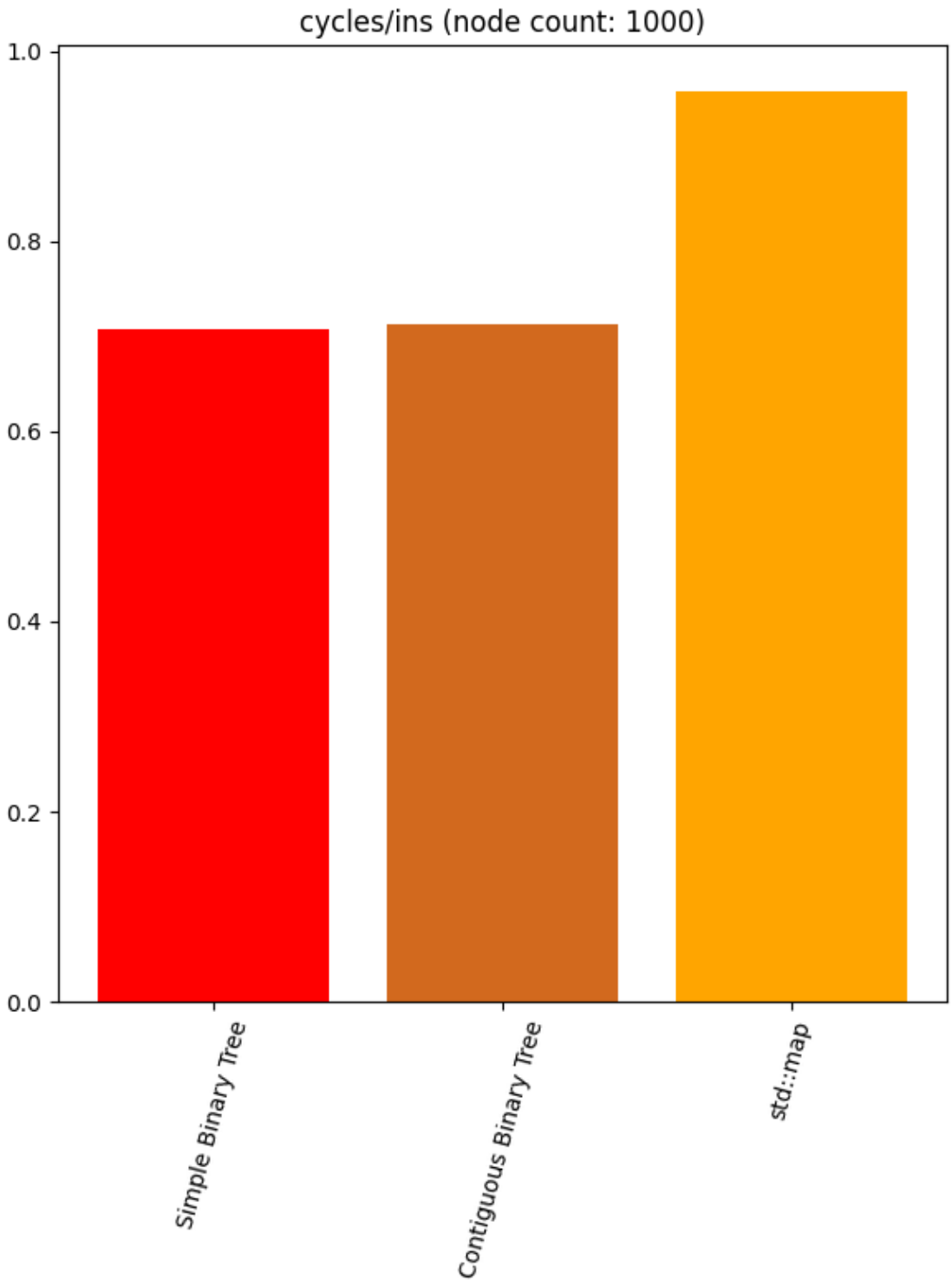


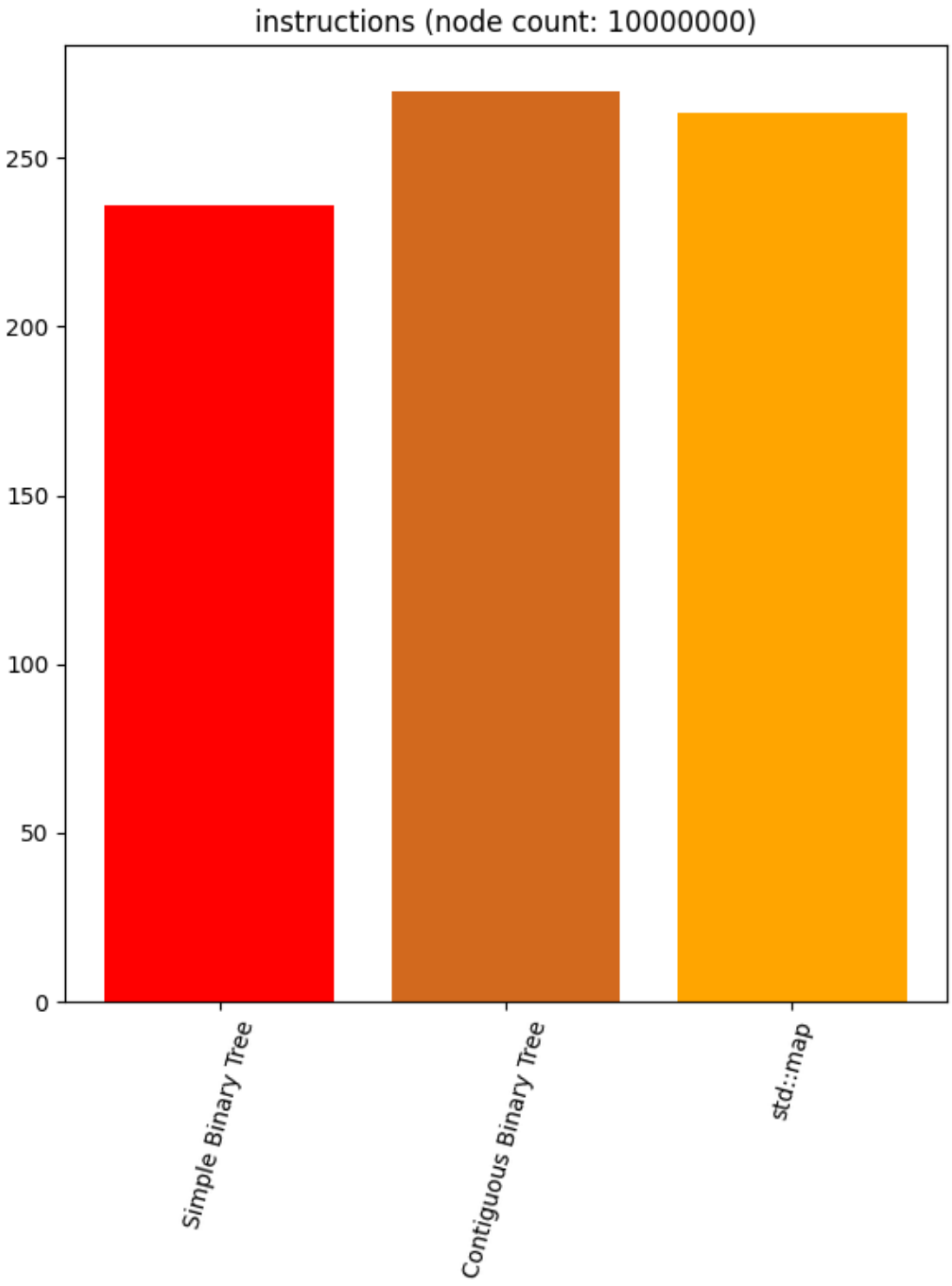
Similarly, we can see that std::map not only performs more work during a search than my own tree, but the work it performs also seems more complex, as each instruction takes more cycles. This seems to stay consistent regardless of tree size.

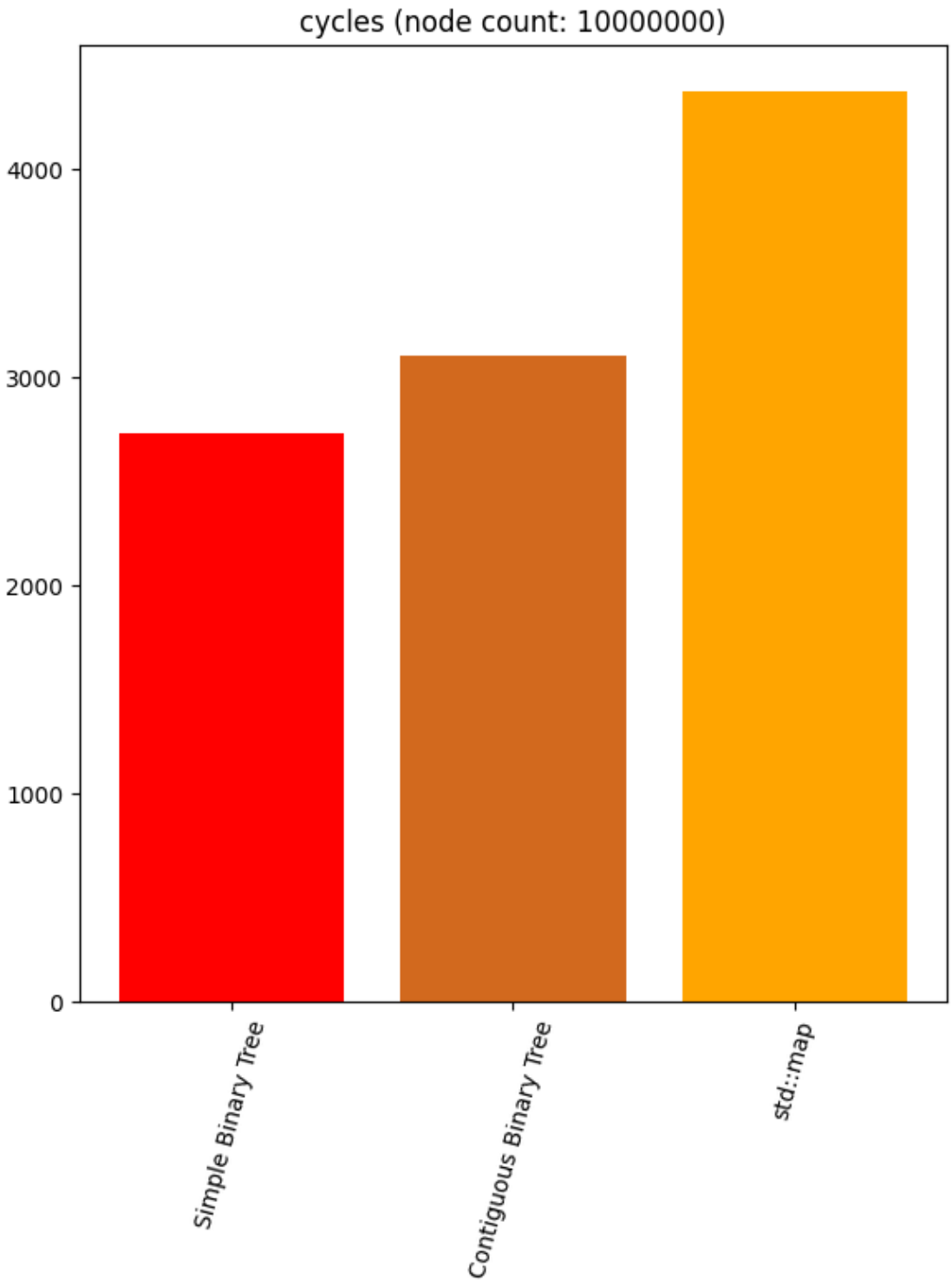
In this chart we can also see that BinTreeCont storing nodes contiguously seems to come with a slight overhead, despite the search operation being identical to what BinTree performs.

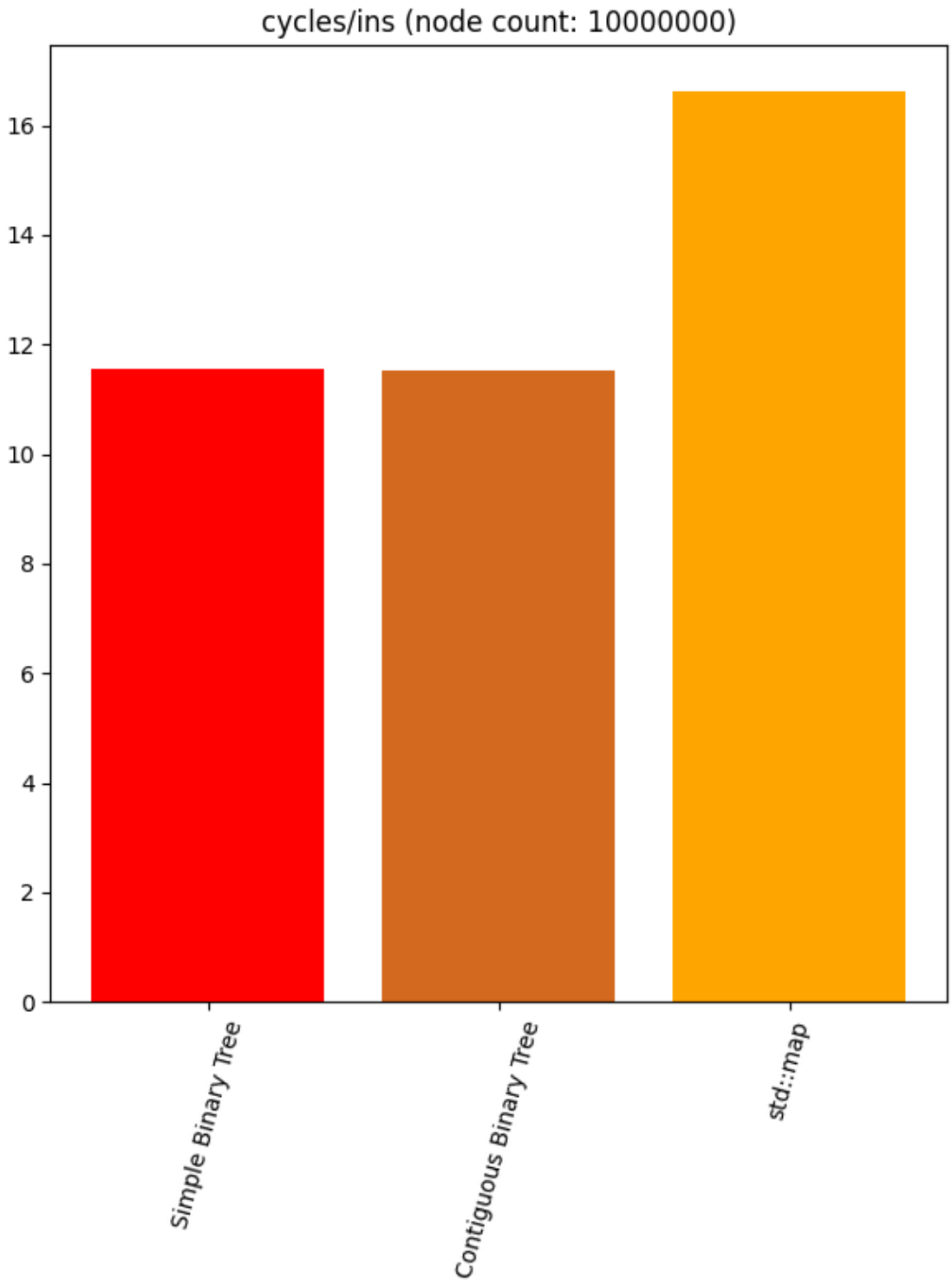



cycles (node count: 1000)
0
20
40
60
80
100
120
Simple Binary Tree
Contiguous Binary Tree
std::map


cycles/ins (node count: 1000)
1.0
0.8
0.6
0.4
0.2
0.0
Simple Binary Tree
Contiguous Binary Tree
std::map


instructions (node count: 10000000)
250
200
150
100
50
0
Simple Binary Tree
Contiguous Binary Tree
std::map


cycles (node count: 10000000)
4000
3000
2000
1000
0
Simple Binary Tree
Contiguous Binary Tree
std::map


cycles/ins (node count: 10000000)
0
2
4
6
8
10
12
14
16
Simple Binary Tree
Contiguous Binary Tree
std::map

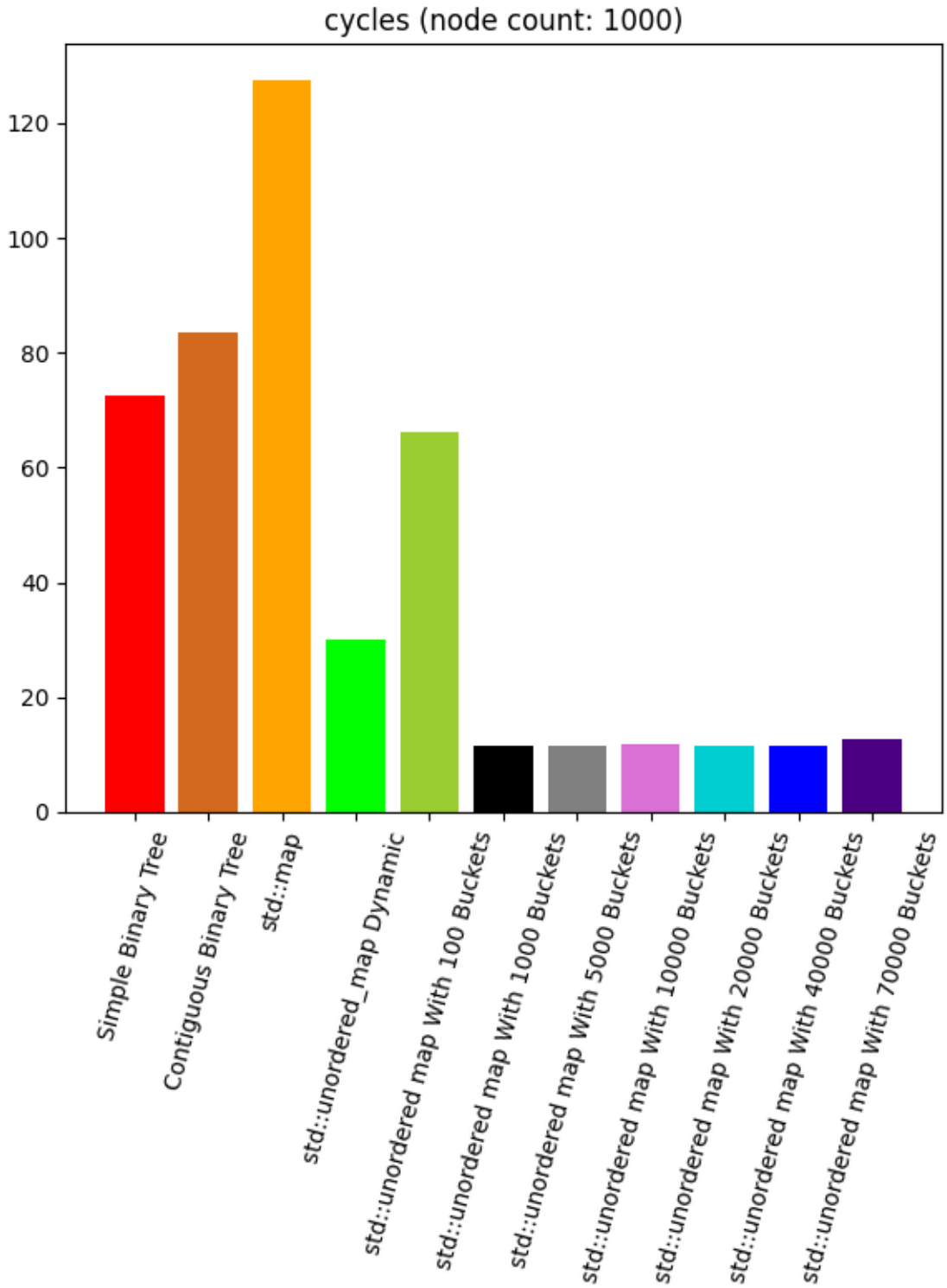
8 Conclusion

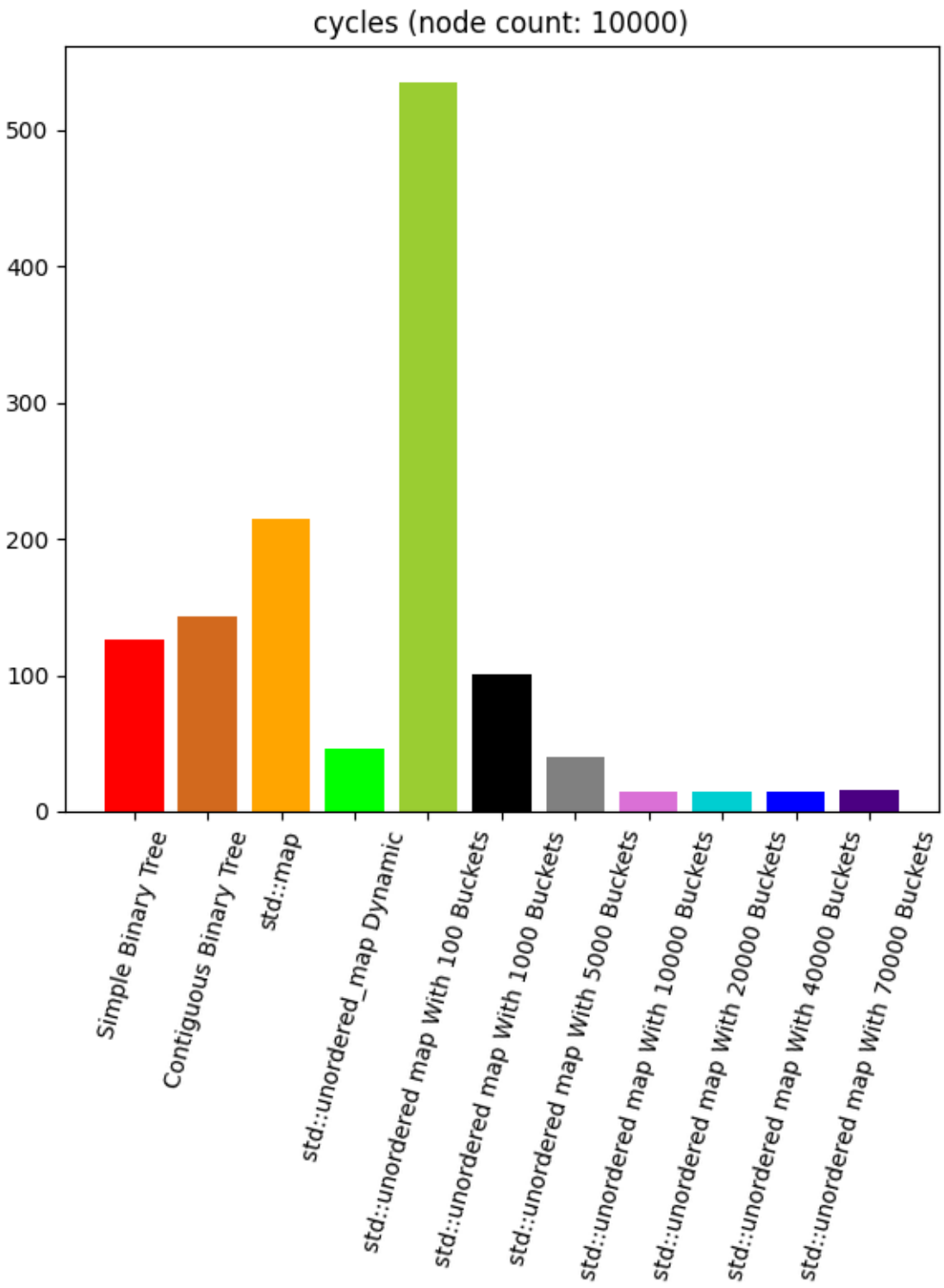
Overall, std::unordered_map seems to scale really well regardless of how much data it needs to store, and consistently yields great results. Finding out that a simple binary tree is faster than std::map definitely came as a surprise, the exact reason behind that needs further investigation.

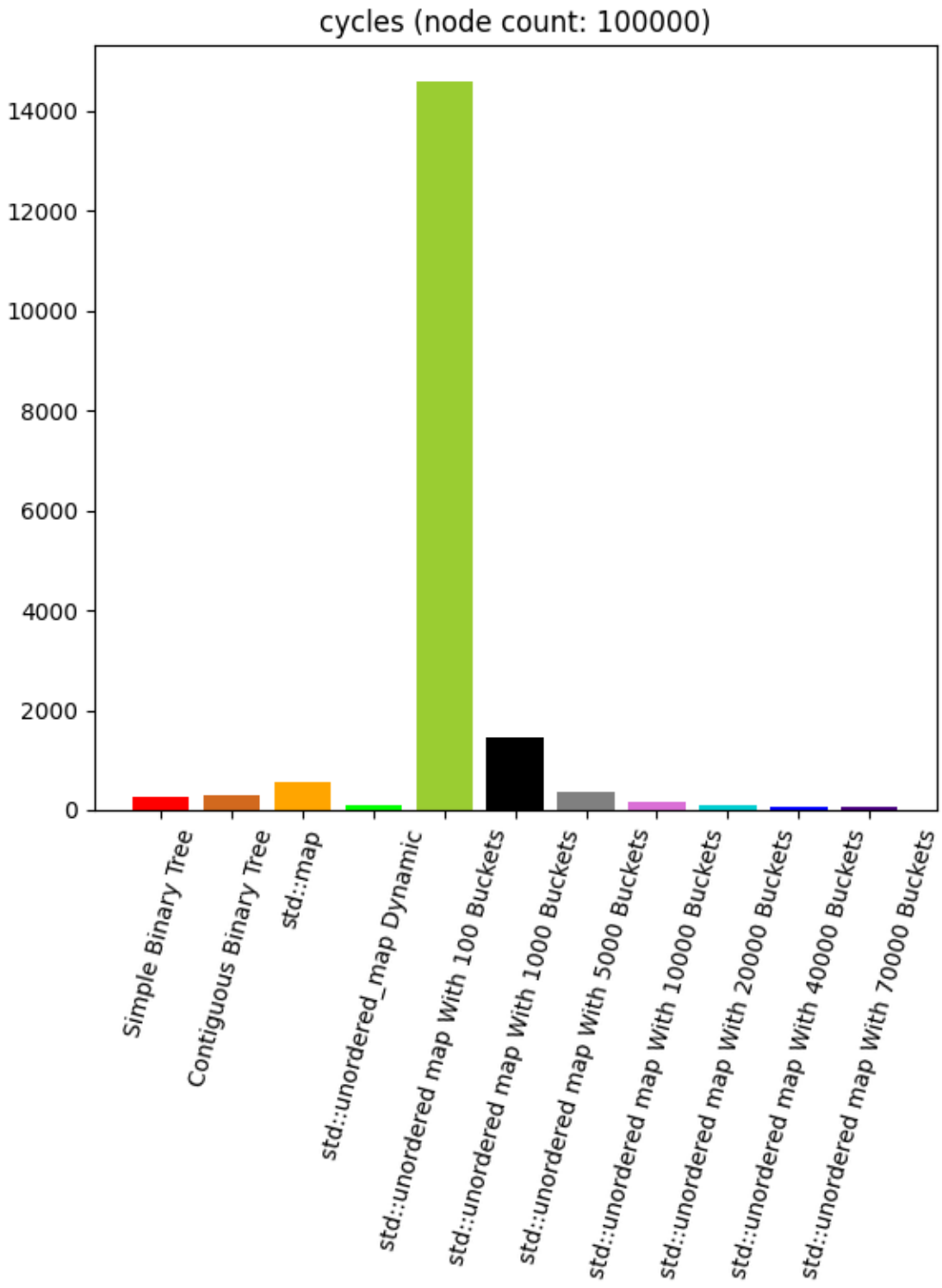
The benchmark suite could definitely use some improvements, currently there is a lot of repeated code and hard-coded values both in the software itself and the script used to run it. Ironing out these details would allow for more precise control over the ways the benchmark could be performed.

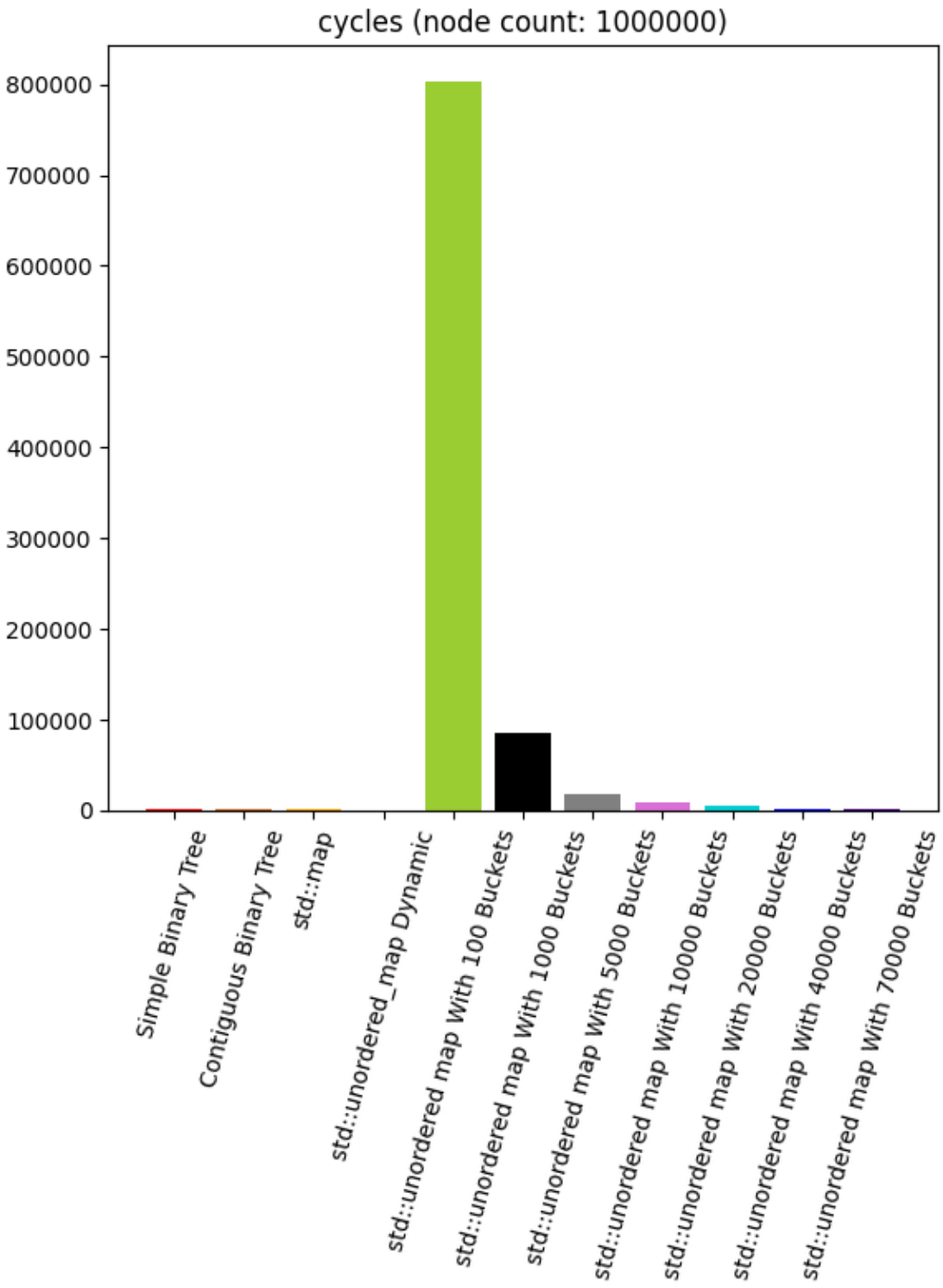
	Search time in seconds	1000 Nodes	10000000 Nodes
0	Simple Binary Tree	3.427458e-08	1.333370e-06
1	Contiguous Binary Tree	3.916635e-08	1.463927e-06
2	std::map	6.003871e-08	2.100107e-06
3	std::unordered_map Dynamic	1.422204e-08	4.532744e-07
4	std::unordered map With 100 Buckets	3.348581e-08	5.199030e-03
5	std::unordered map With 1000 Buckets	5.470532e-09	5.319181e-04
6	std::unordered map With 5000 Buckets	5.410372e-09	1.171406e-04
7	std::unordered map With 10000 Buckets	5.436593e-09	5.970849e-05
8	std::unordered map With 20000 Buckets	5.426845e-09	3.101795e-05
9	std::unordered map With 40000 Buckets	5.407252e-09	1.577713e-05
10	std::unordered map With 70000 Buckets	5.939181e-09	9.285400e-06

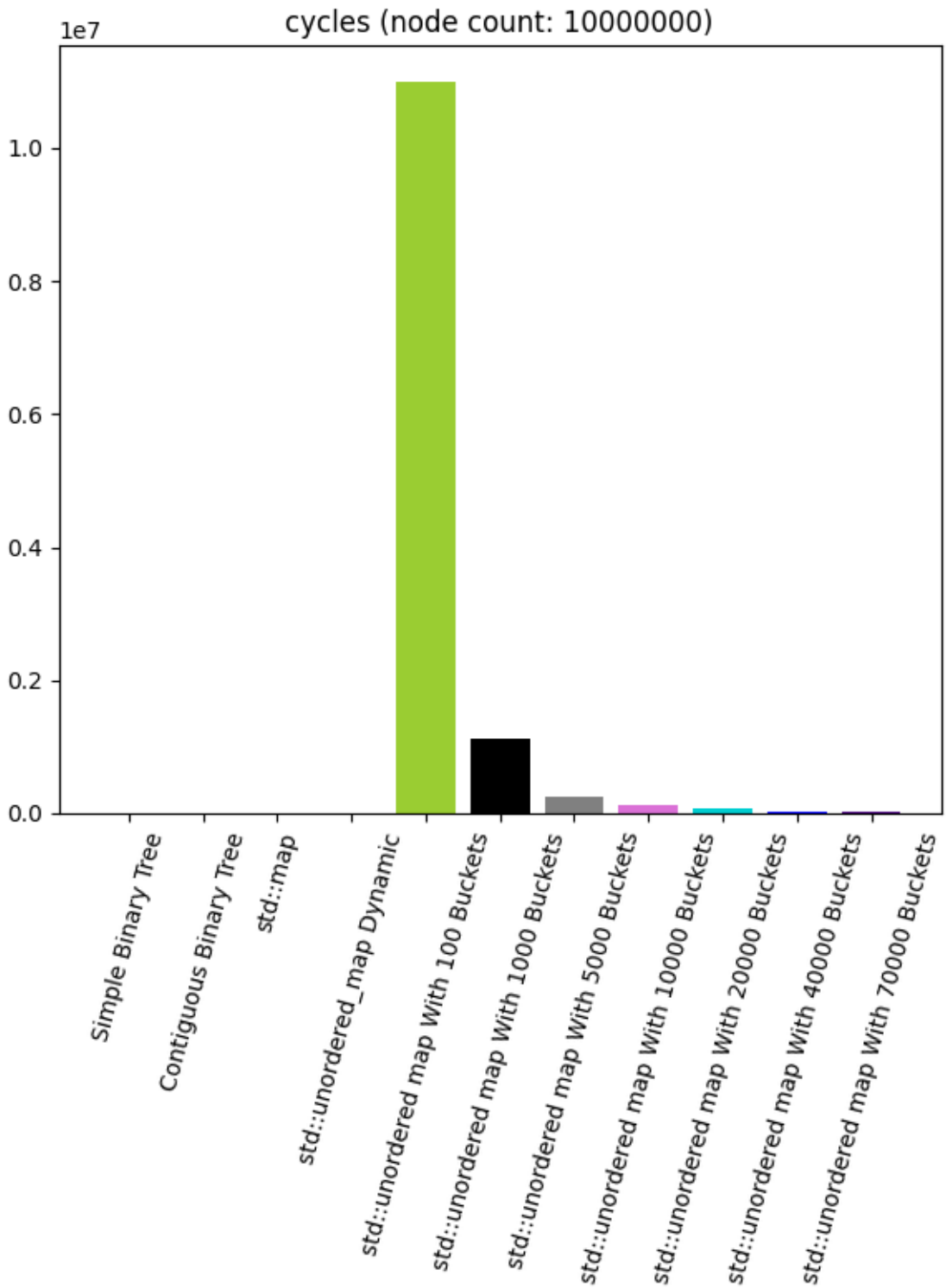
9 Charts and Tables

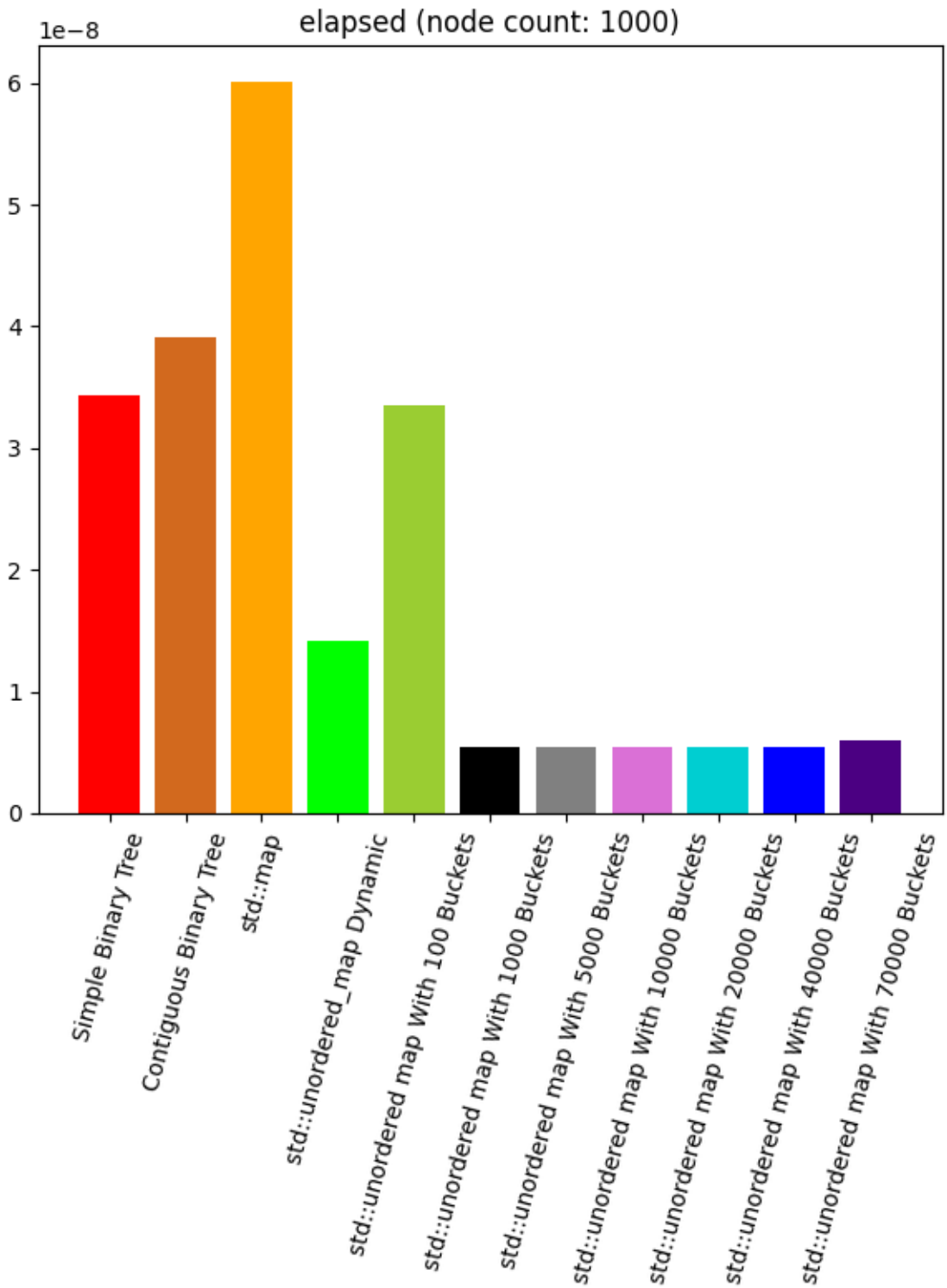


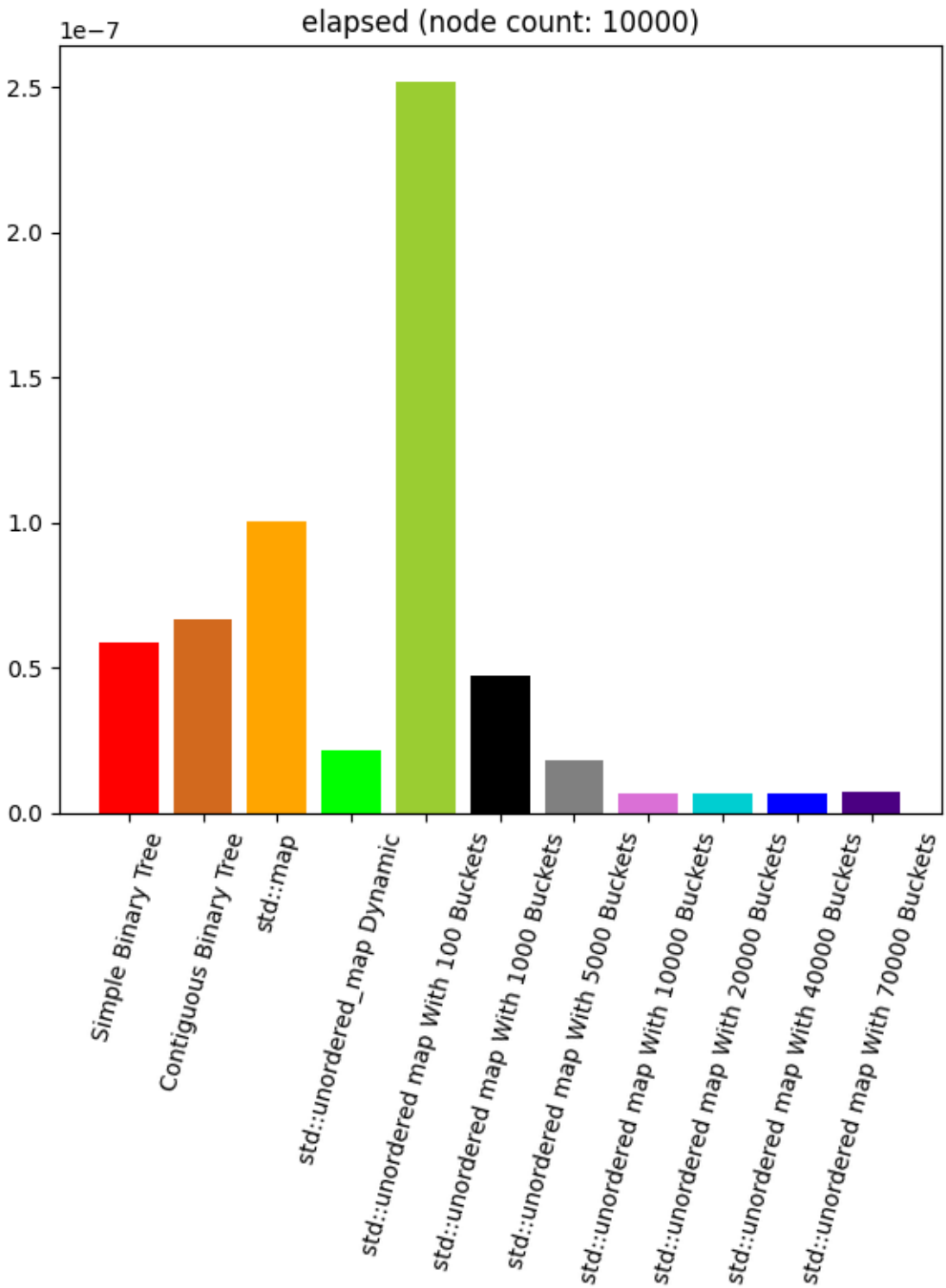

cycles (node count: 10000)
0
100
200
300
400
500
Simple Binary Tree
Contiguous Binary Tree
std::map
std::unordered_map Dynamic
std::unordered map With 100 Buckets
std::unordered map With 1000 Buckets
std::unordered map With 5000 Buckets
std::unordered map With 10000 Buckets
std::unordered map With 20000 Buckets
std::unordered map With 40000 Buckets
std::unordered map With 70000 Buckets

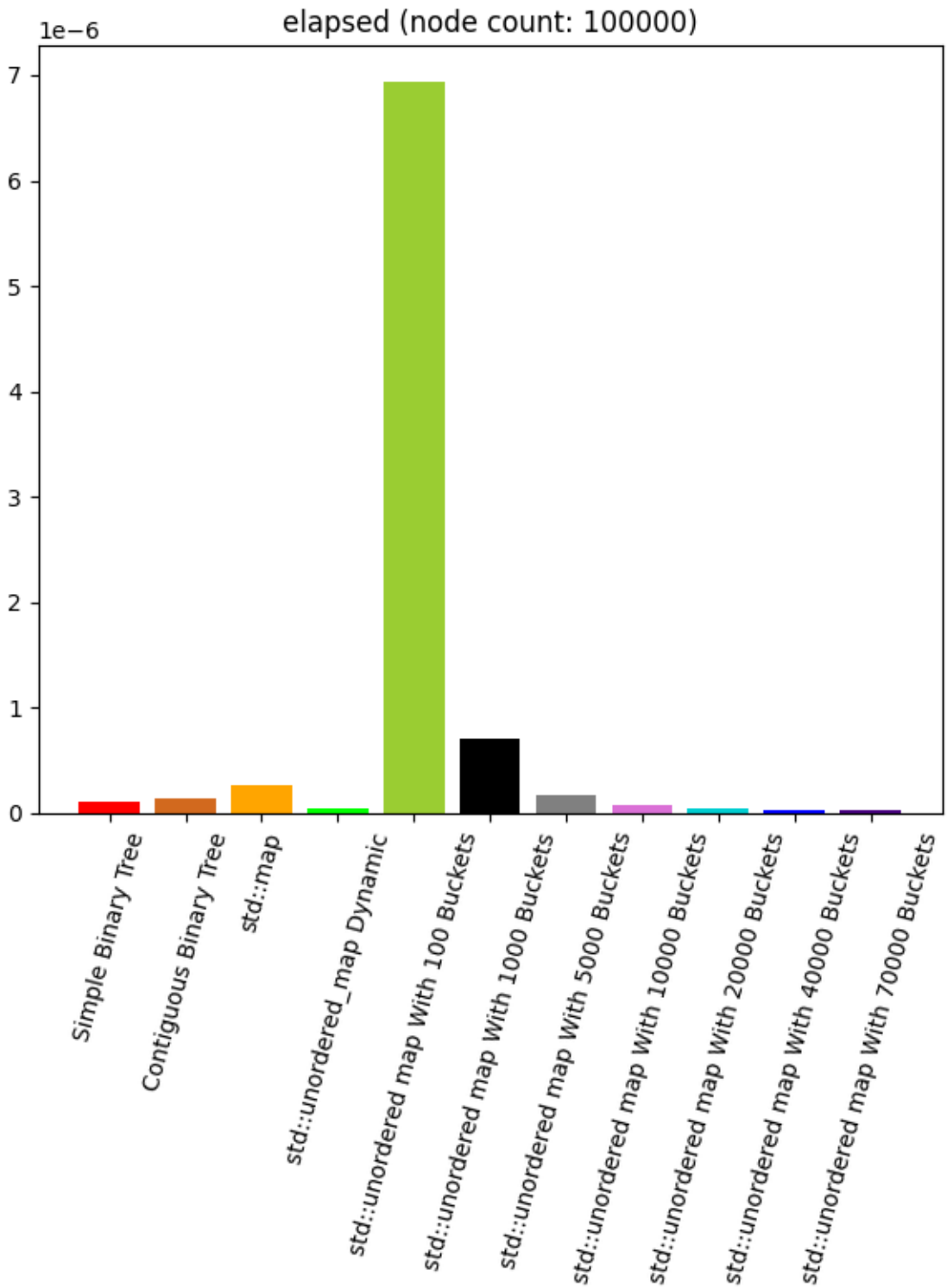

cycles (node count: 100000)
14000
12000
10000
8000
6000
4000
2000
0
Simple Binary Tree
Contiguous Binary Tree
std::map
std::unordered_map Dynamic
std::unordered map With 100 Buckets
std::unordered map With 1000 Buckets
std::unordered map With 5000 Buckets
std::unordered map With 10000 Buckets
std::unordered map With 20000 Buckets
std::unordered map With 40000 Buckets
std::unordered map With 70000 Buckets

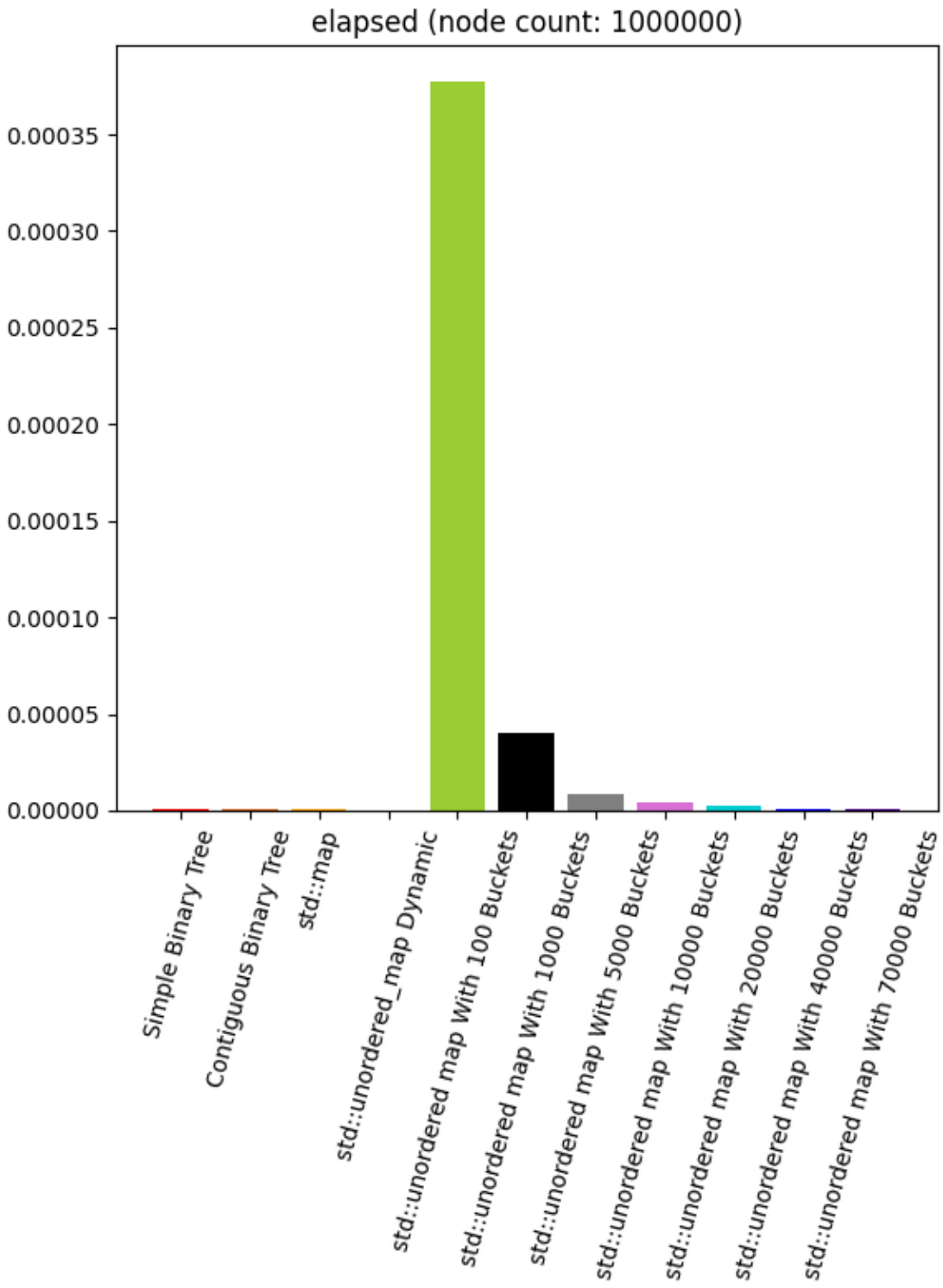

cycles (node count: 1000000)
800000
700000
600000
500000
400000
300000
200000
100000
0
Simple Binary Tree
Contiguous Binary Tree
std::map
std::unordered_map Dynamic
std::unordered map With 100 Buckets
std::unordered map With 1000 Buckets
std::unordered map With 5000 Buckets
std::unordered map With 10000 Buckets
std::unordered map With 20000 Buckets
std::unordered map With 40000 Buckets
std::unordered map With 70000 Buckets

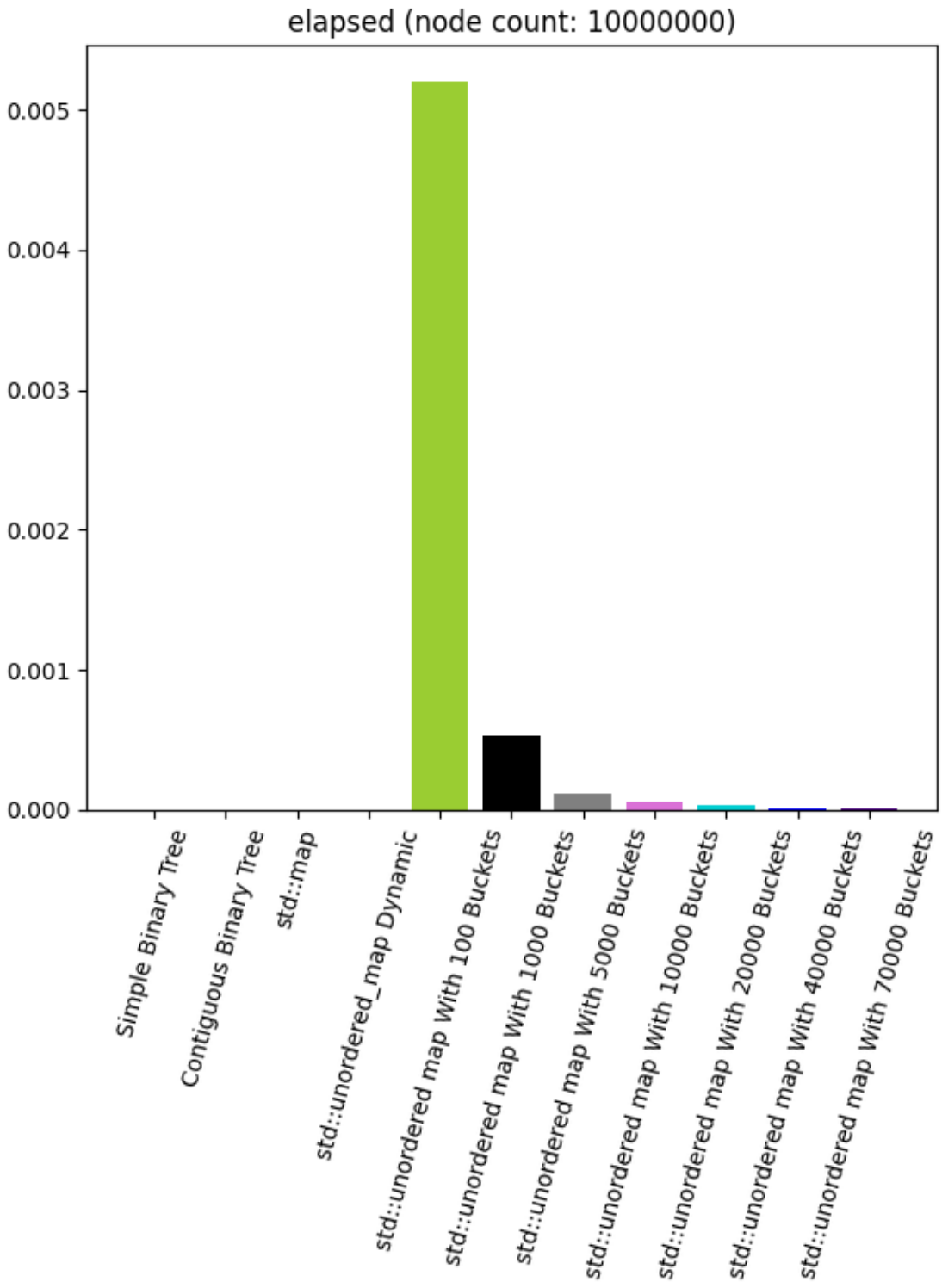

cycles (node count: 10000000)
1e7
1.0
0.8
0.6
0.4
0.2
0.0
Simple Binary Tree
Contiguous Binary Tree
std::map
std::unordered_map Dynamic
std::unordered map With 100 Buckets
std::unordered map With 1000 Buckets
std::unordered map With 5000 Buckets
std::unordered map With 10000 Buckets
std::unordered map With 20000 Buckets
std::unordered map With 40000 Buckets
std::unordered map With 70000 Buckets

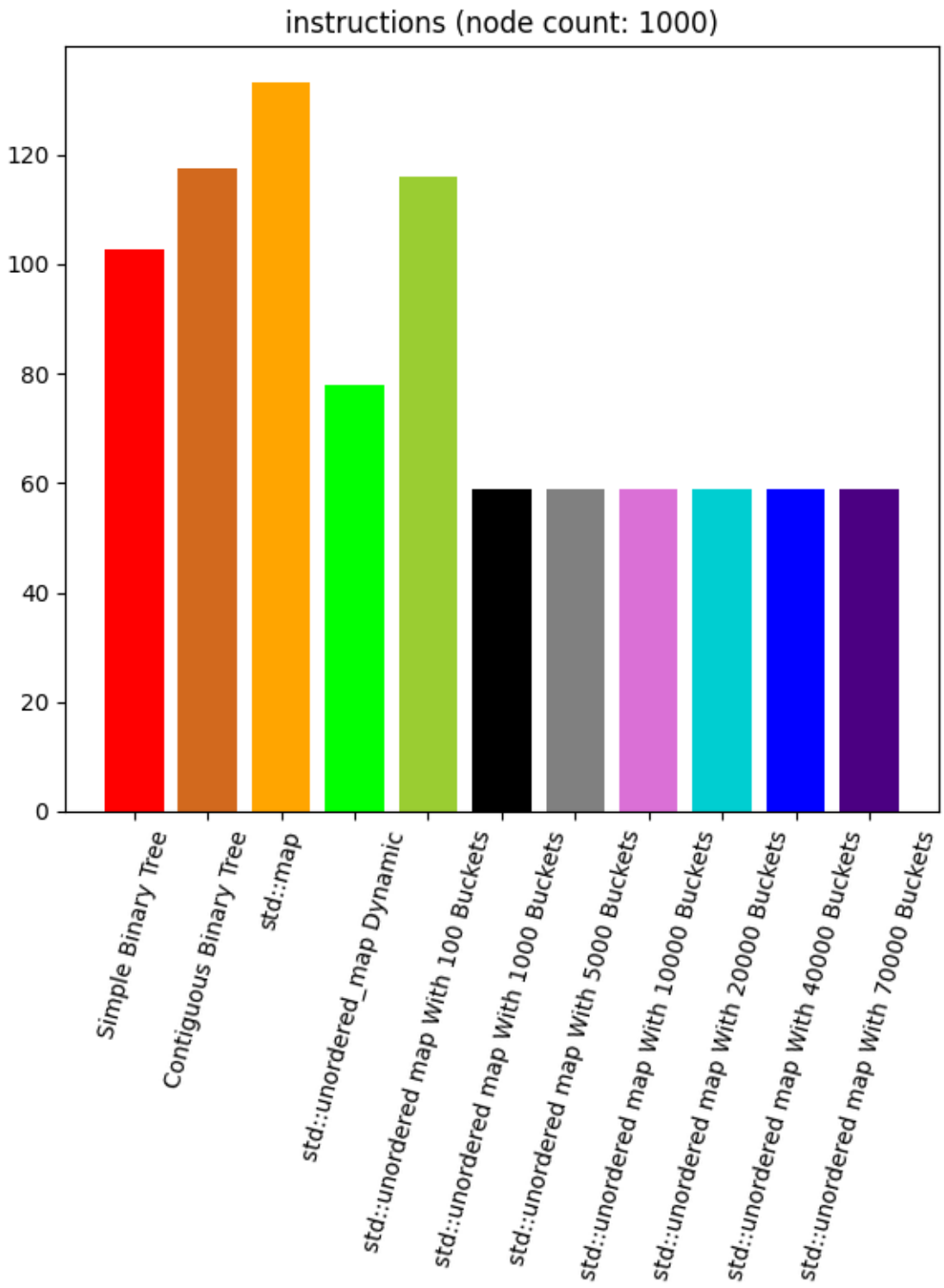

elapsed (node count: 1000)
1e−8
0
1
2
3
4
5
6
Simple Binary Tree
Contiguous Binary Tree
std::map
std::unordered_map Dynamic
std::unordered map With 100 Buckets
std::unordered map With 1000 Buckets
std::unordered map With 5000 Buckets
std::unordered map With 10000 Buckets
std::unordered map With 20000 Buckets
std::unordered map With 40000 Buckets
std::unordered map With 70000 Buckets

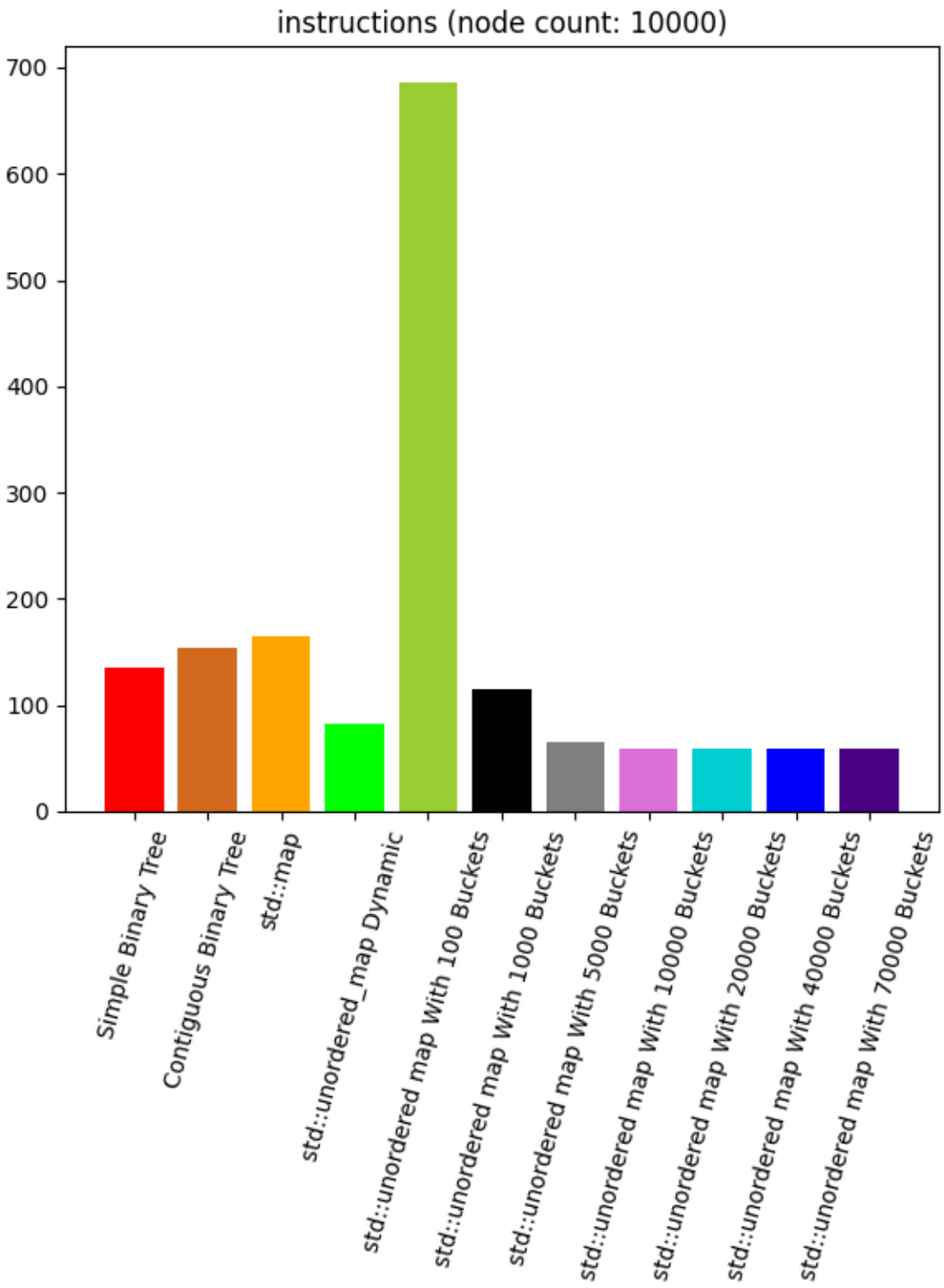

elapsed (node count: 10000)
1e−7
2.5
2.0
1.5
1.0
0.5
0.0
Simple Binary Tree
Contiguous Binary Tree
std::map
std::unordered_map Dynamic
std::unordered map With 100 Buckets
std::unordered map With 1000 Buckets
std::unordered map With 5000 Buckets
std::unordered map With 10000 Buckets
std::unordered map With 20000 Buckets
std::unordered map With 40000 Buckets
std::unordered map With 70000 Buckets

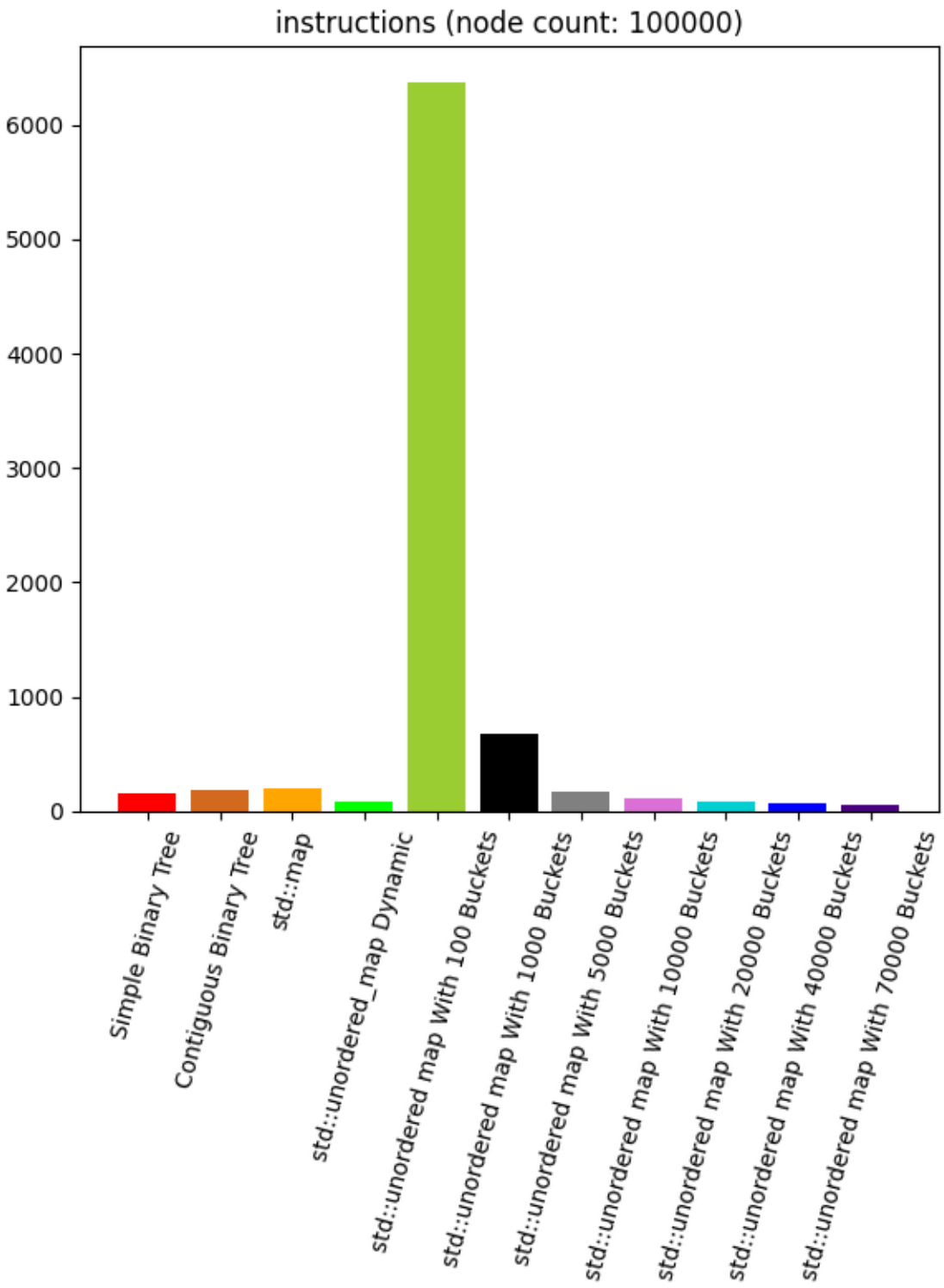

elapsed (node count: 100000)
1e−6
0
1
2
3
4
5
6
7
Simple Binary Tree
Contiguous Binary Tree
std::map
std::unordered_map Dynamic
std::unordered map With 100 Buckets
std::unordered map With 1000 Buckets
std::unordered map With 5000 Buckets
std::unordered map With 10000 Buckets
std::unordered map With 20000 Buckets
std::unordered map With 40000 Buckets
std::unordered map With 70000 Buckets

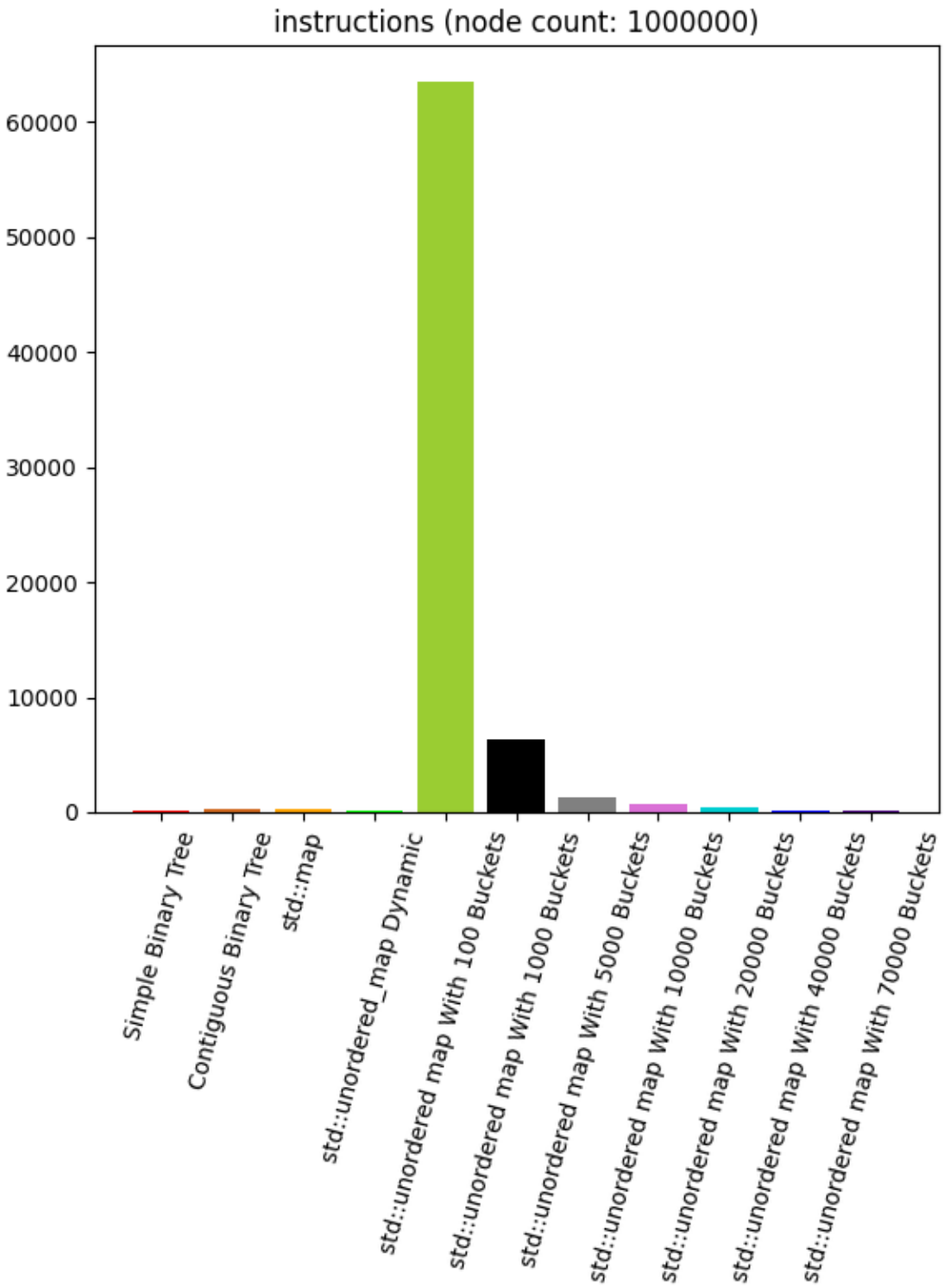

elapsed (node count: 1000000)
0.00035
0.00030
0.00025
0.00020
0.00015
0.00010
0.00005
0.00000
Simple Binary Tree
Contiguous Binary Tree
std::map
std::unordered_map Dynamic
std::unordered map With 100 Buckets
std::unordered map With 1000 Buckets
std::unordered map With 5000 Buckets
std::unordered map With 10000 Buckets
std::unordered map With 20000 Buckets
std::unordered map With 40000 Buckets
std::unordered map With 70000 Buckets

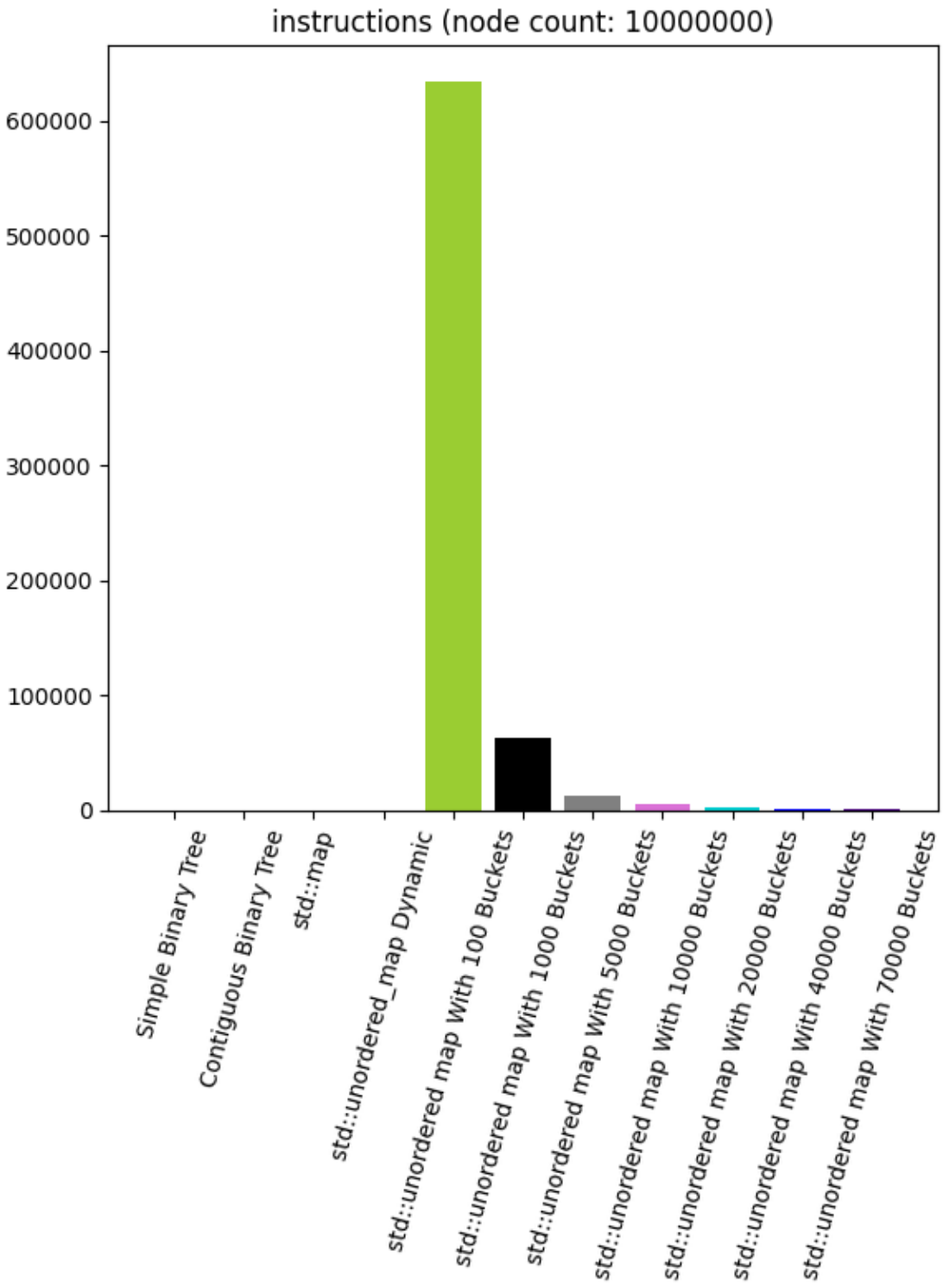

elapsed (node count: 10000000)
0.005
0.004
0.003
0.002
0.001
0.000
Simple Binary Tree
Contiguous Binary Tree
std::map
std::unordered_map Dynamic
std::unordered map With 100 Buckets
std::unordered map With 1000 Buckets
std::unordered map With 5000 Buckets
std::unordered map With 10000 Buckets
std::unordered map With 20000 Buckets
std::unordered map With 40000 Buckets
std::unordered map With 70000 Buckets

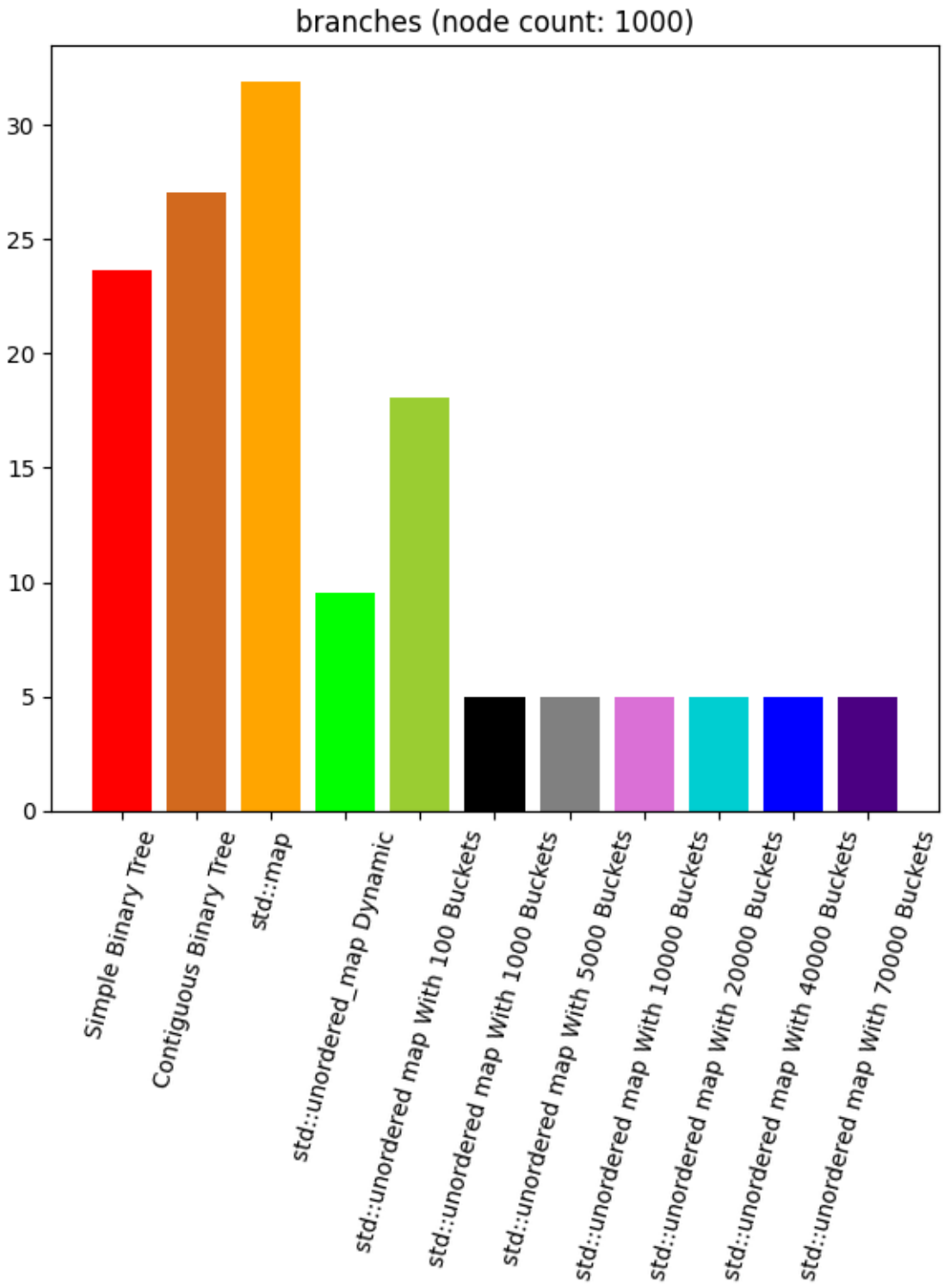

instructions (node count: 1000)
0
20
40
60
80
100
120
Simple Binary Tree
Contiguous Binary Tree
std::map
std::unordered_map Dynamic
std::unordered map With 100 Buckets
std::unordered map With 1000 Buckets
std::unordered map With 5000 Buckets
std::unordered map With 10000 Buckets
std::unordered map With 20000 Buckets
std::unordered map With 40000 Buckets
std::unordered map With 70000 Buckets

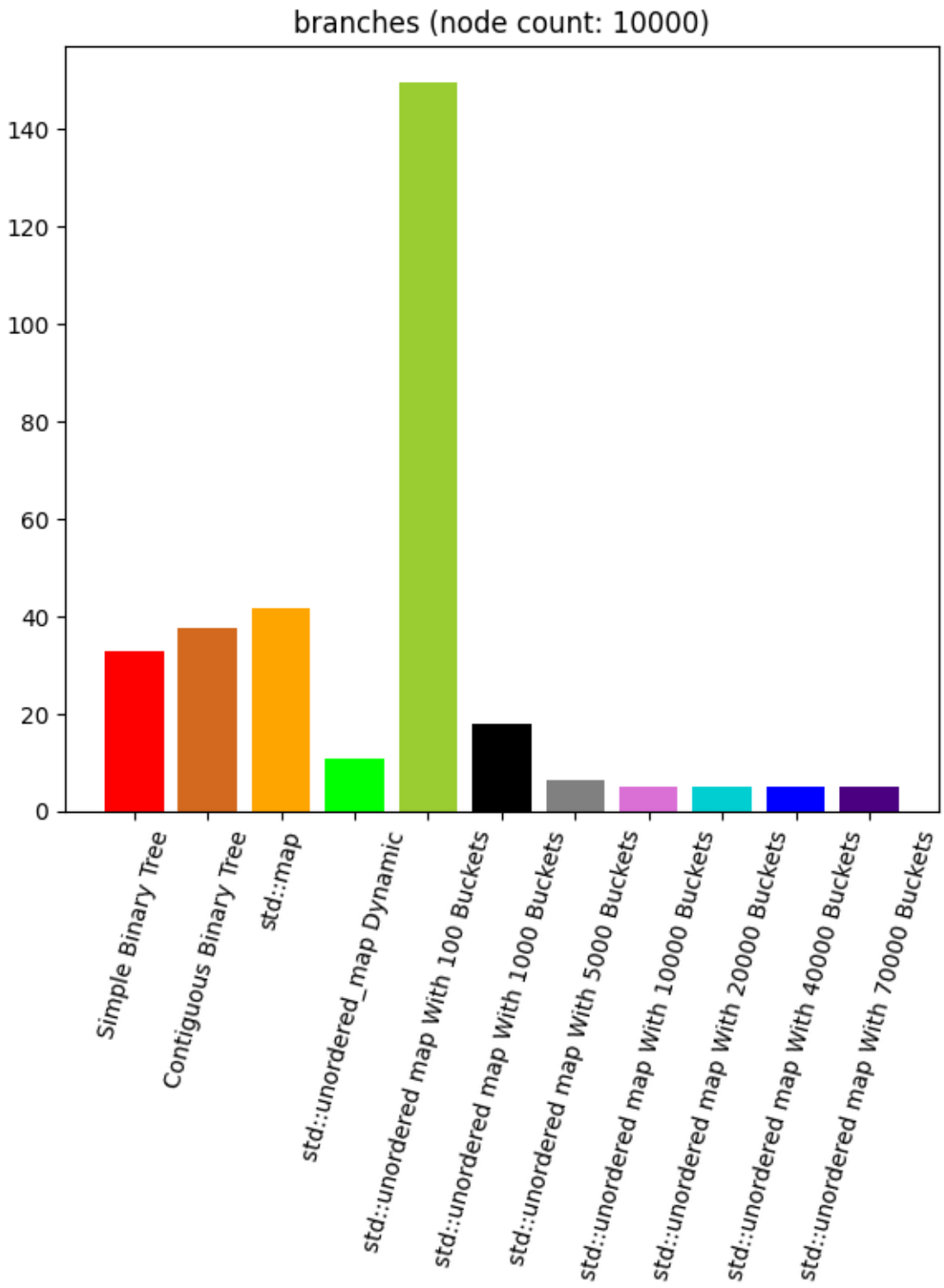

instructions (node count: 10000)
700
600
500
400
300
200
100
0
Simple Binary Tree
Contiguous Binary Tree
std::map
std::unordered_map Dynamic
std::unordered map With 100 Buckets
std::unordered map With 1000 Buckets
std::unordered map With 5000 Buckets
std::unordered map With 10000 Buckets
std::unordered map With 20000 Buckets
std::unordered map With 40000 Buckets
std::unordered map With 70000 Buckets

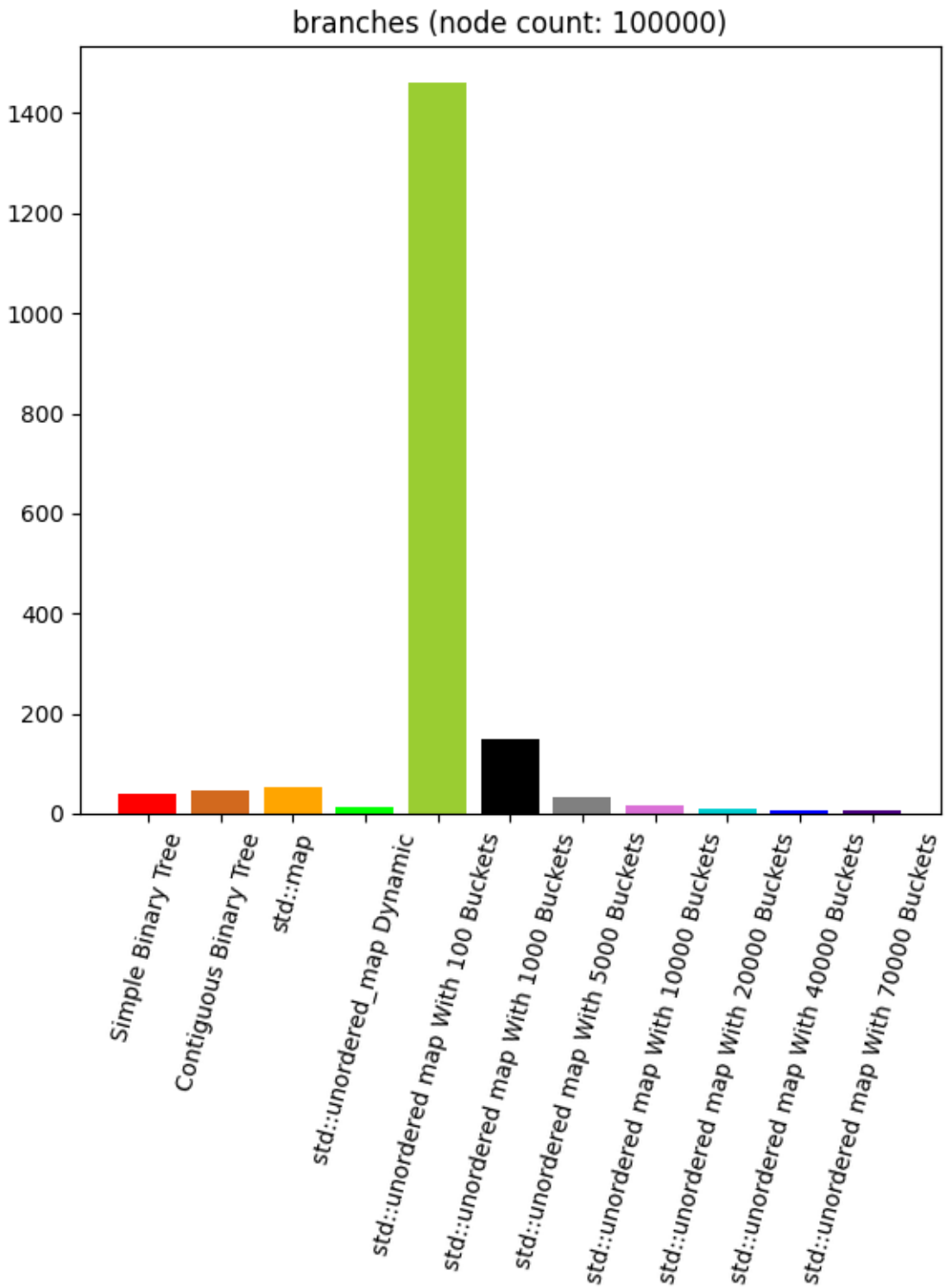

instructions (node count: 100000)
6000
5000
4000
3000
2000
1000
0
Simple Binary Tree
Contiguous Binary Tree
std::map
std::unordered_map Dynamic
std::unordered map With 100 Buckets
std::unordered map With 1000 Buckets
std::unordered map With 5000 Buckets
std::unordered map With 10000 Buckets
std::unordered map With 20000 Buckets
std::unordered map With 40000 Buckets
std::unordered map With 70000 Buckets

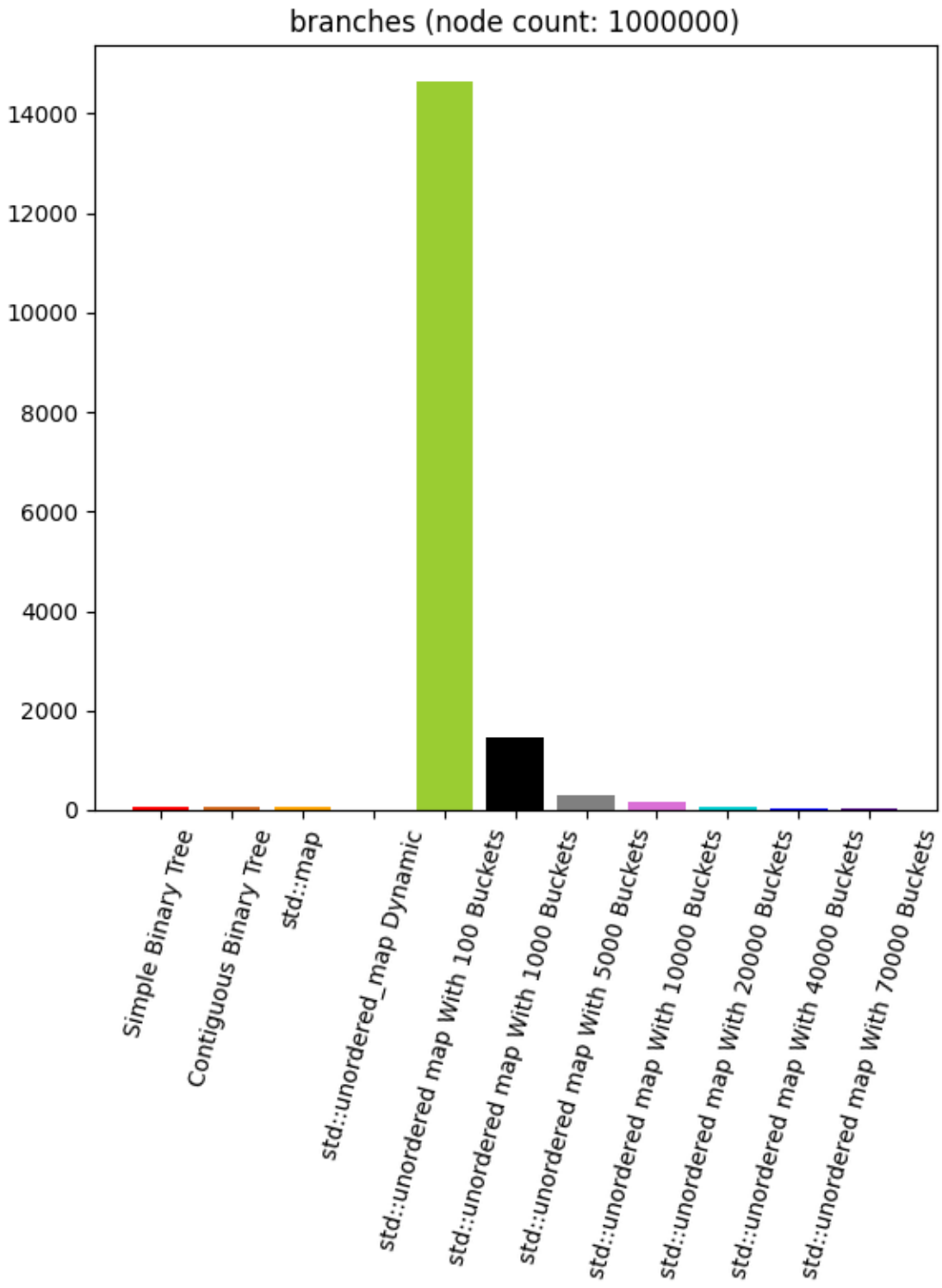

instructions (node count: 1000000)
60000
50000
40000
30000
20000
10000
0
Simple Binary Tree
Contiguous Binary Tree
std::map
std::unordered_map Dynamic
std::unordered map With 100 Buckets
std::unordered map With 1000 Buckets
std::unordered map With 5000 Buckets
std::unordered map With 10000 Buckets
std::unordered map With 20000 Buckets
std::unordered map With 40000 Buckets
std::unordered map With 70000 Buckets

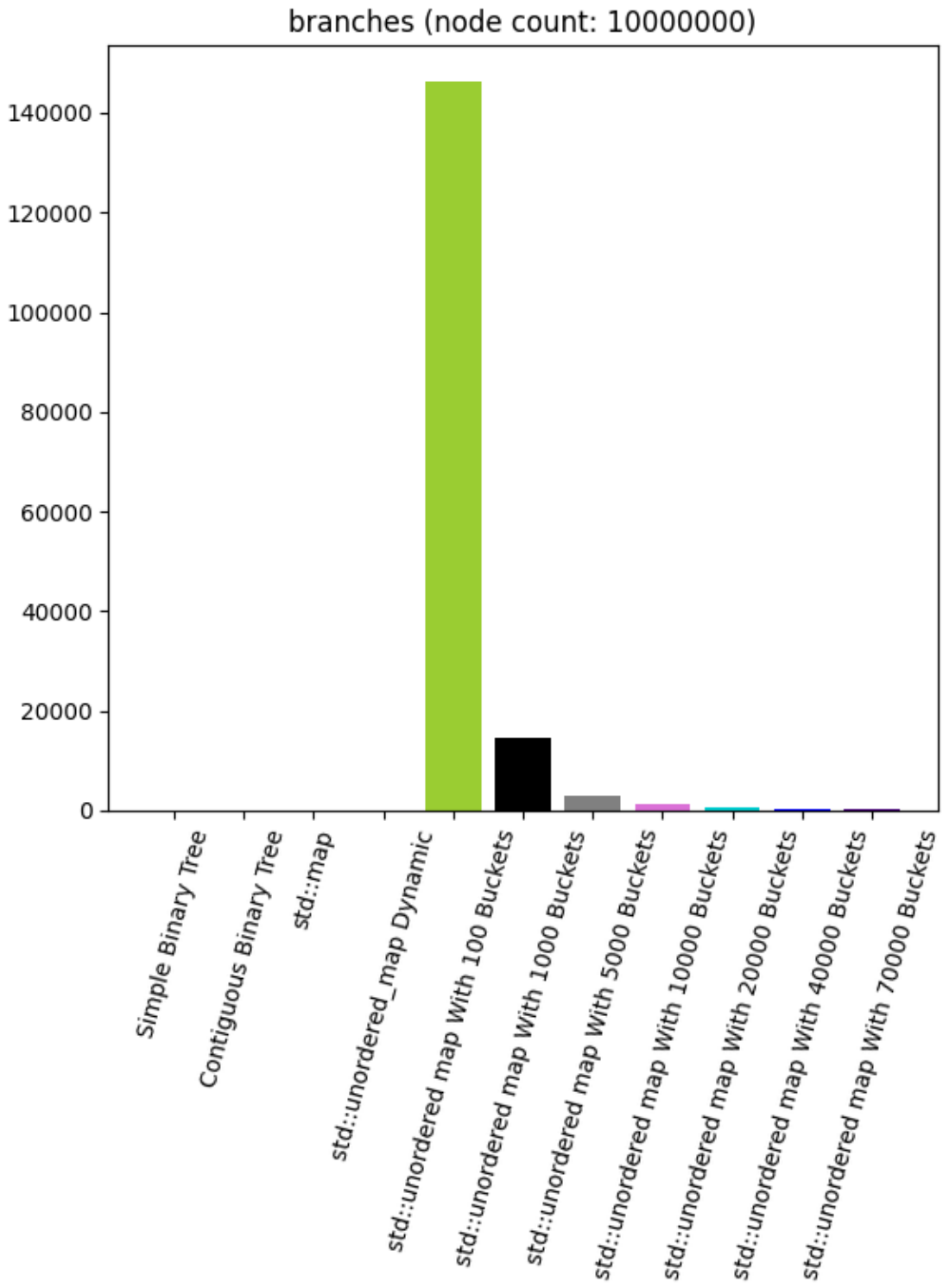

instructions (node count: 10000000)
600000
500000
400000
300000
200000
100000
0
Simple Binary Tree
Contiguous Binary Tree
std::map
std::unordered_map Dynamic
std::unordered map With 100 Buckets
std::unordered map With 1000 Buckets
std::unordered map With 5000 Buckets
std::unordered map With 10000 Buckets
std::unordered map With 20000 Buckets
std::unordered map With 40000 Buckets
std::unordered map With 70000 Buckets

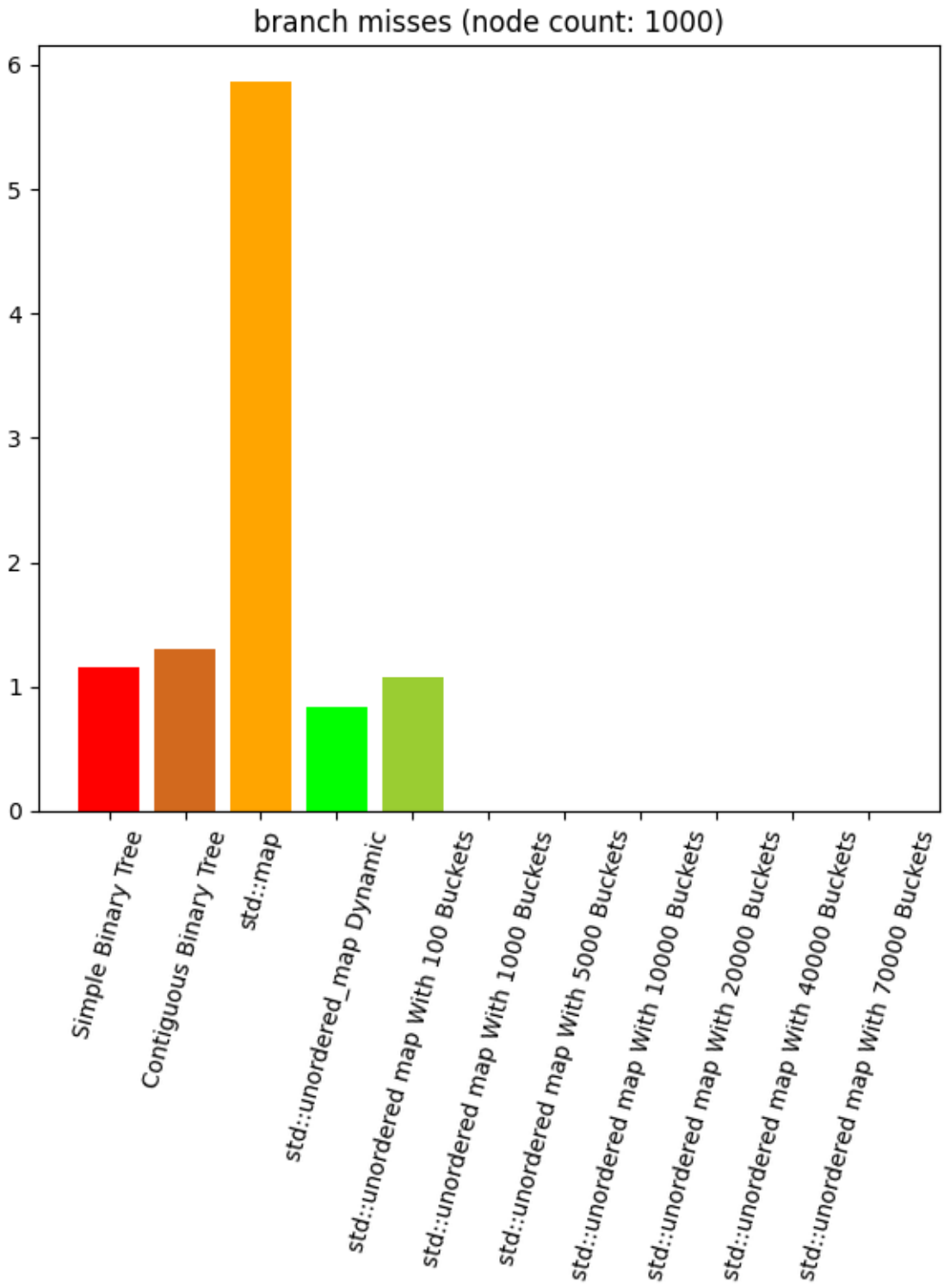

branches (node count: 1000)
0
5
10
15
20
25
30
Simple Binary Tree
Contiguous Binary Tree
std::map
std::unordered_map Dynamic
std::unordered map With 100 Buckets
std::unordered map With 1000 Buckets
std::unordered map With 5000 Buckets
std::unordered map With 10000 Buckets
std::unordered map With 20000 Buckets
std::unordered map With 40000 Buckets
std::unordered map With 70000 Buckets

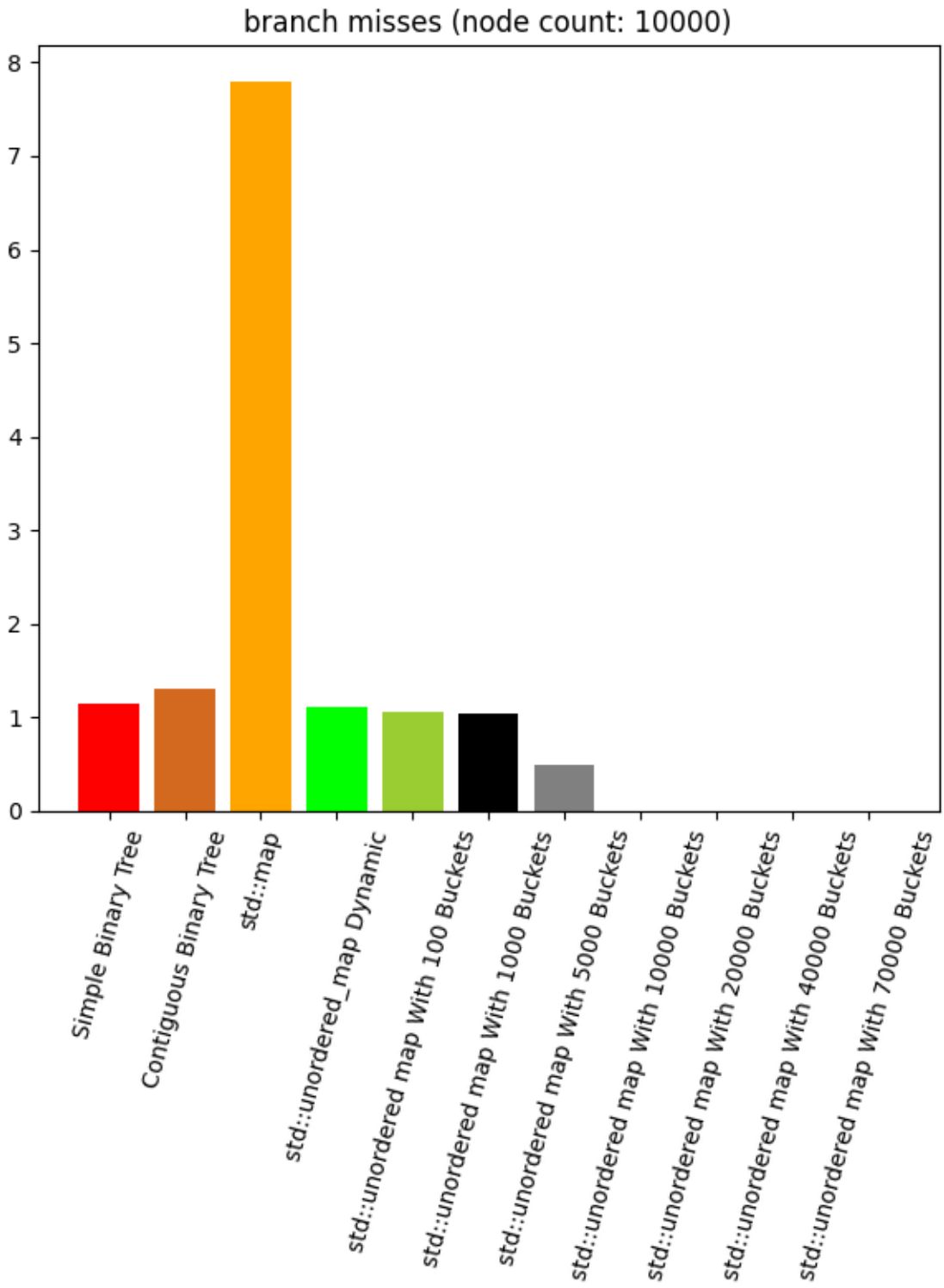

branches (node count: 10000)
0
20
40
60
80
100
120
140
Simple Binary Tree
Contiguous Binary Tree
std::map
std::unordered_map Dynamic
std::unordered map With 100 Buckets
std::unordered map With 1000 Buckets
std::unordered map With 5000 Buckets
std::unordered map With 10000 Buckets
std::unordered map With 20000 Buckets
std::unordered map With 40000 Buckets
std::unordered map With 70000 Buckets

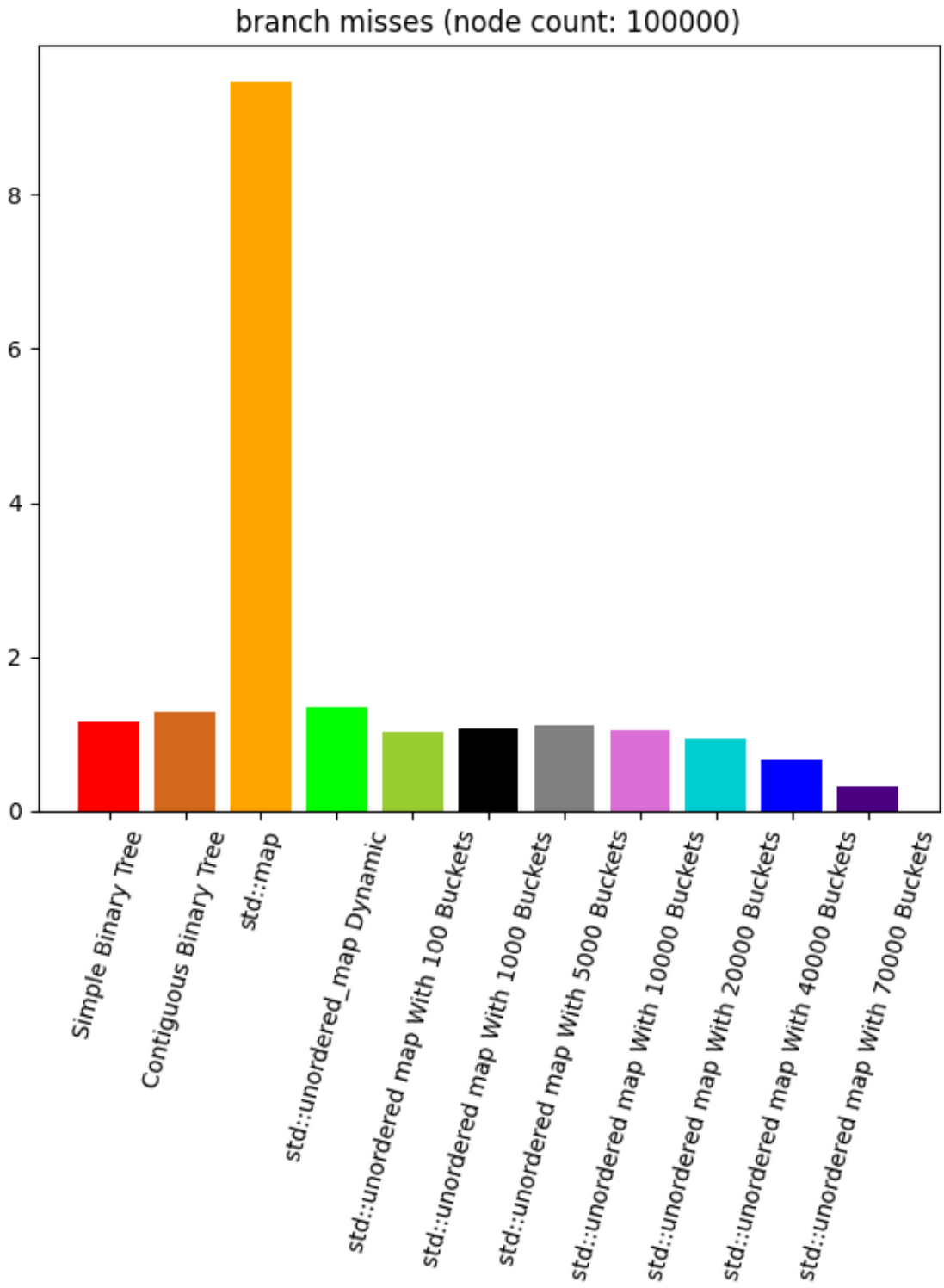

branches (node count: 100000)
1400
1200
1000
800
600
400
200
0
Simple Binary Tree
Contiguous Binary Tree
std::map
std::unordered_map Dynamic
std::unordered map With 100 Buckets
std::unordered map With 1000 Buckets
std::unordered map With 5000 Buckets
std::unordered map With 10000 Buckets
std::unordered map With 20000 Buckets
std::unordered map With 40000 Buckets
std::unordered map With 70000 Buckets

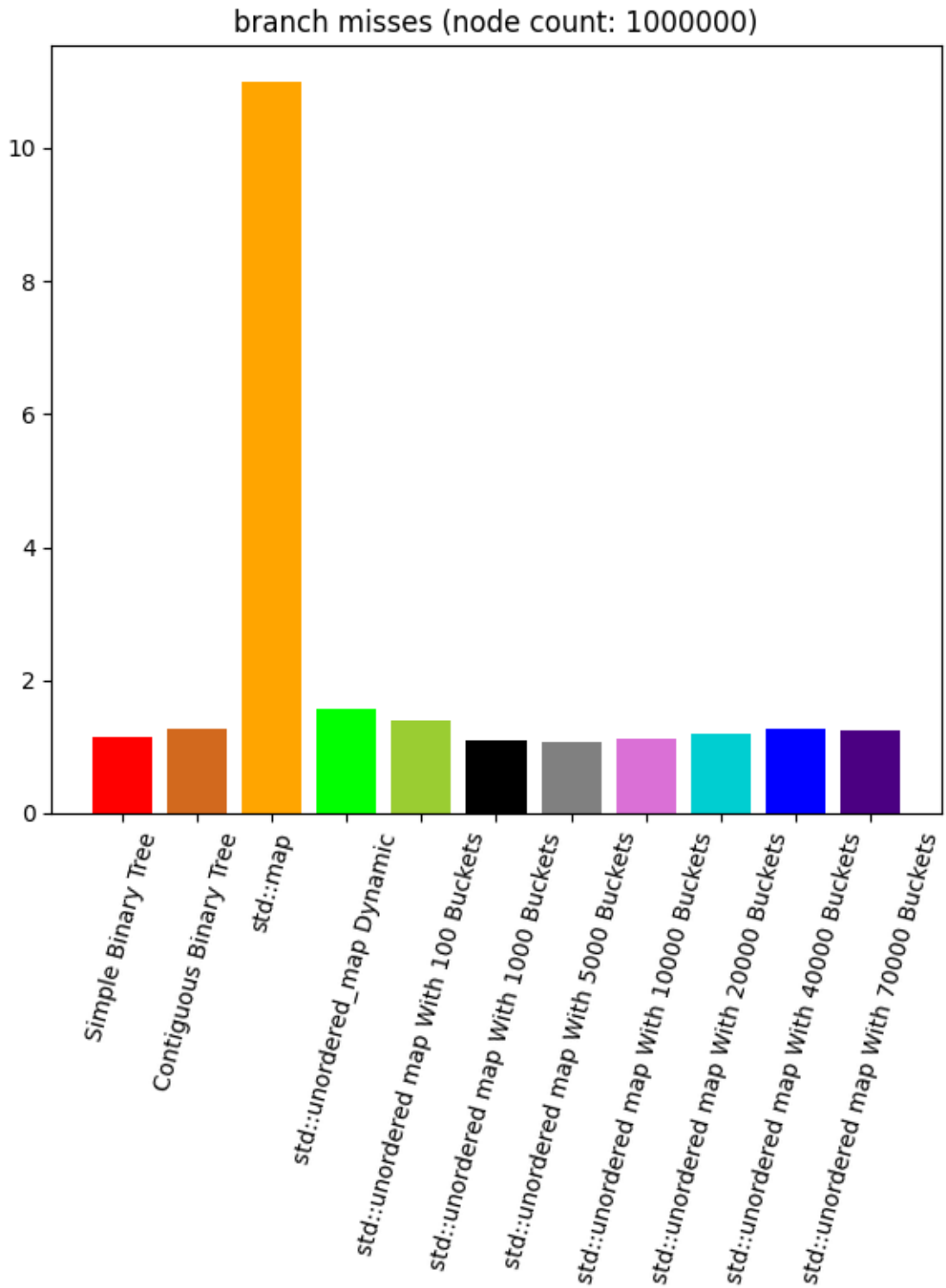

branches (node count: 1000000)
14000
12000
10000
8000
6000
4000
2000
0
Simple Binary Tree
Contiguous Binary Tree
std::map
std::unordered_map Dynamic
std::unordered map With 100 Buckets
std::unordered map With 1000 Buckets
std::unordered map With 5000 Buckets
std::unordered map With 10000 Buckets
std::unordered map With 20000 Buckets
std::unordered map With 40000 Buckets
std::unordered map With 70000 Buckets

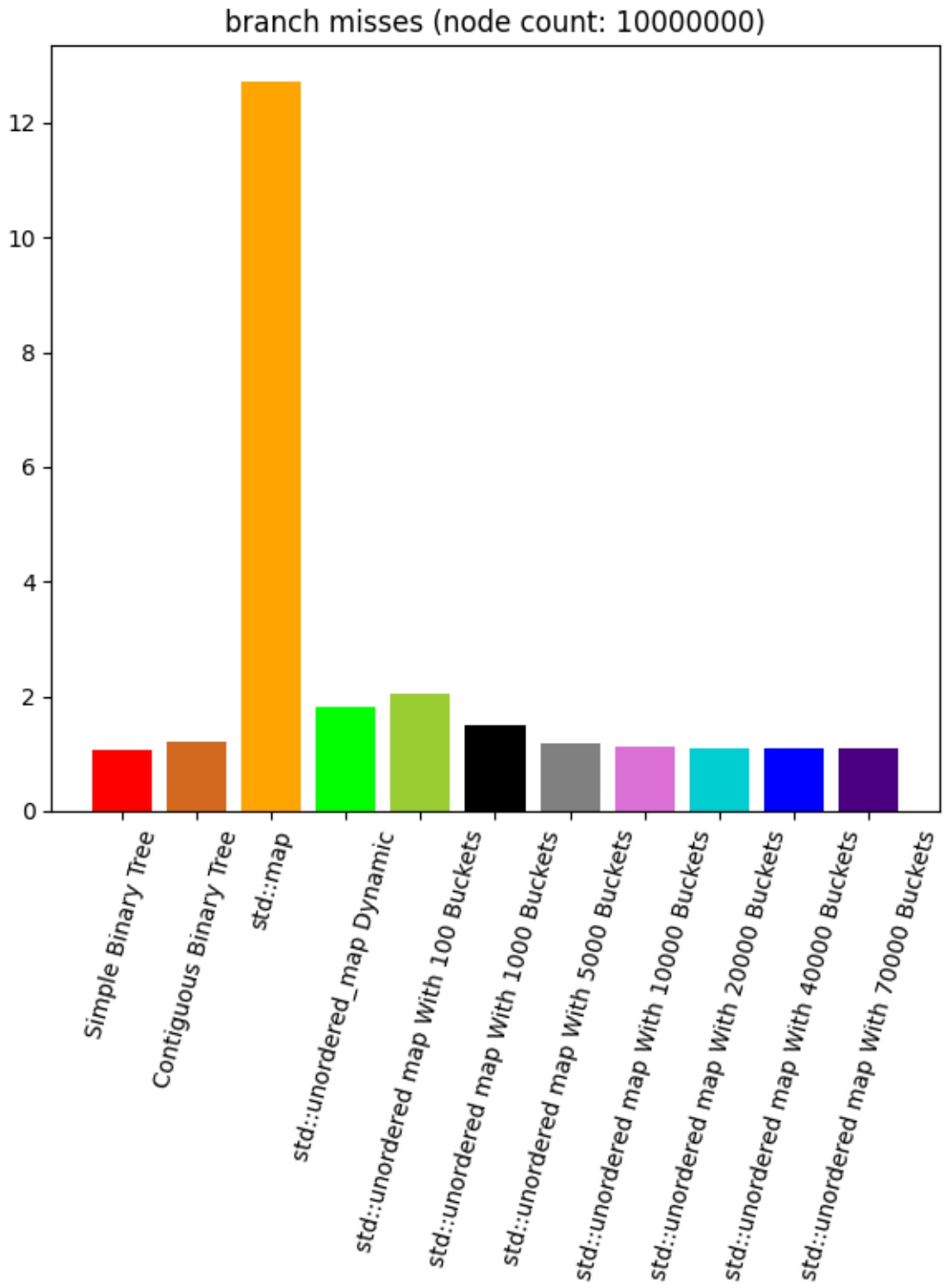

branches (node count: 10000000)
140000
120000
100000
80000
60000
40000
20000
0
Simple Binary Tree
Contiguous Binary Tree
std::map
std::unordered_map Dynamic
std::unordered map With 100 Buckets
std::unordered map With 1000 Buckets
std::unordered map With 5000 Buckets
std::unordered map With 10000 Buckets
std::unordered map With 20000 Buckets
std::unordered map With 40000 Buckets
std::unordered map With 70000 Buckets

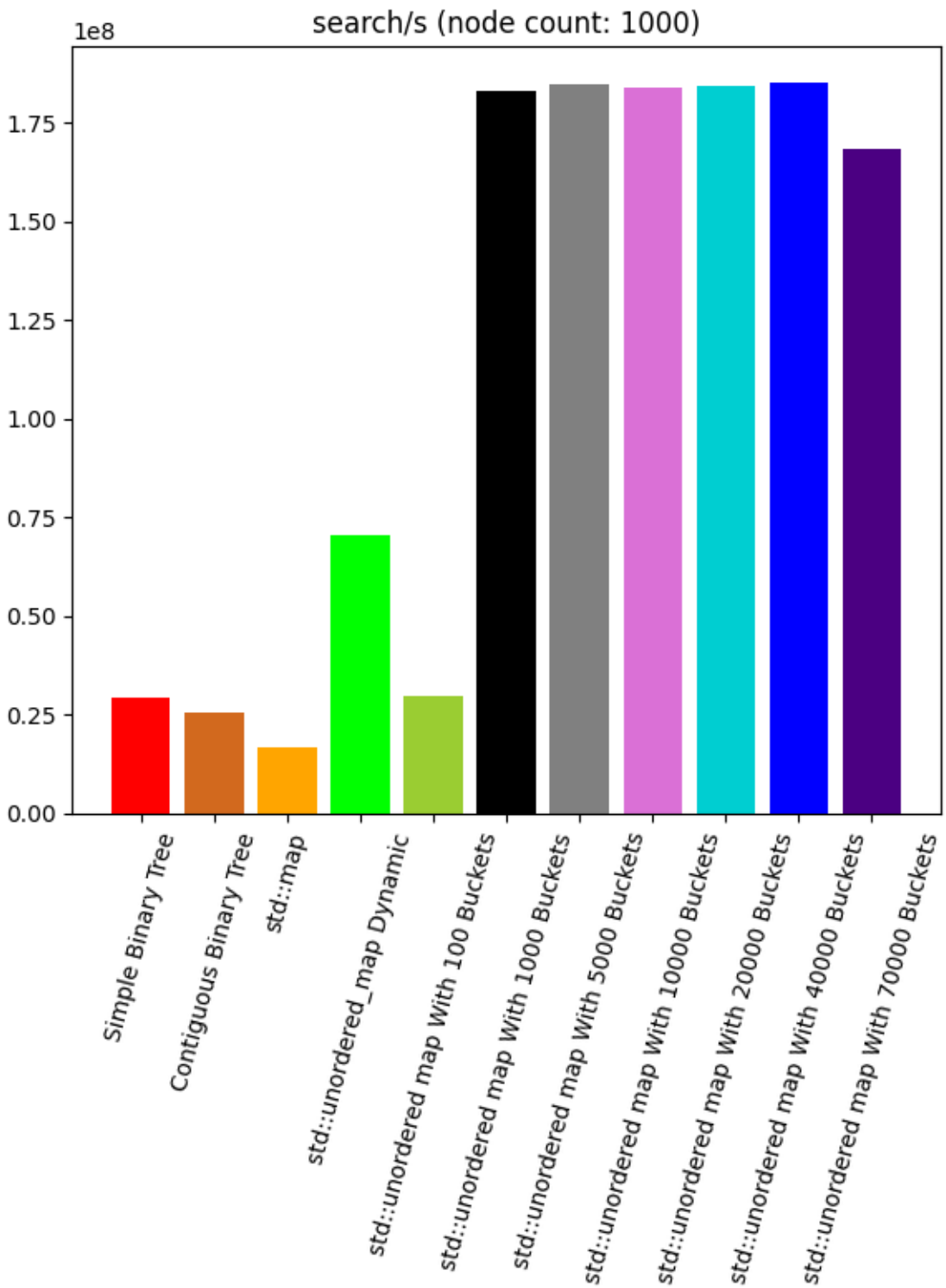

branch misses (node count: 1000)
0
1
2
3
4
5
6
Simple Binary Tree
Contiguous Binary Tree
std::map
std::unordered_map Dynamic
std::unordered map With 100 Buckets
std::unordered map With 1000 Buckets
std::unordered map With 5000 Buckets
std::unordered map With 10000 Buckets
std::unordered map With 20000 Buckets
std::unordered map With 40000 Buckets
std::unordered map With 70000 Buckets

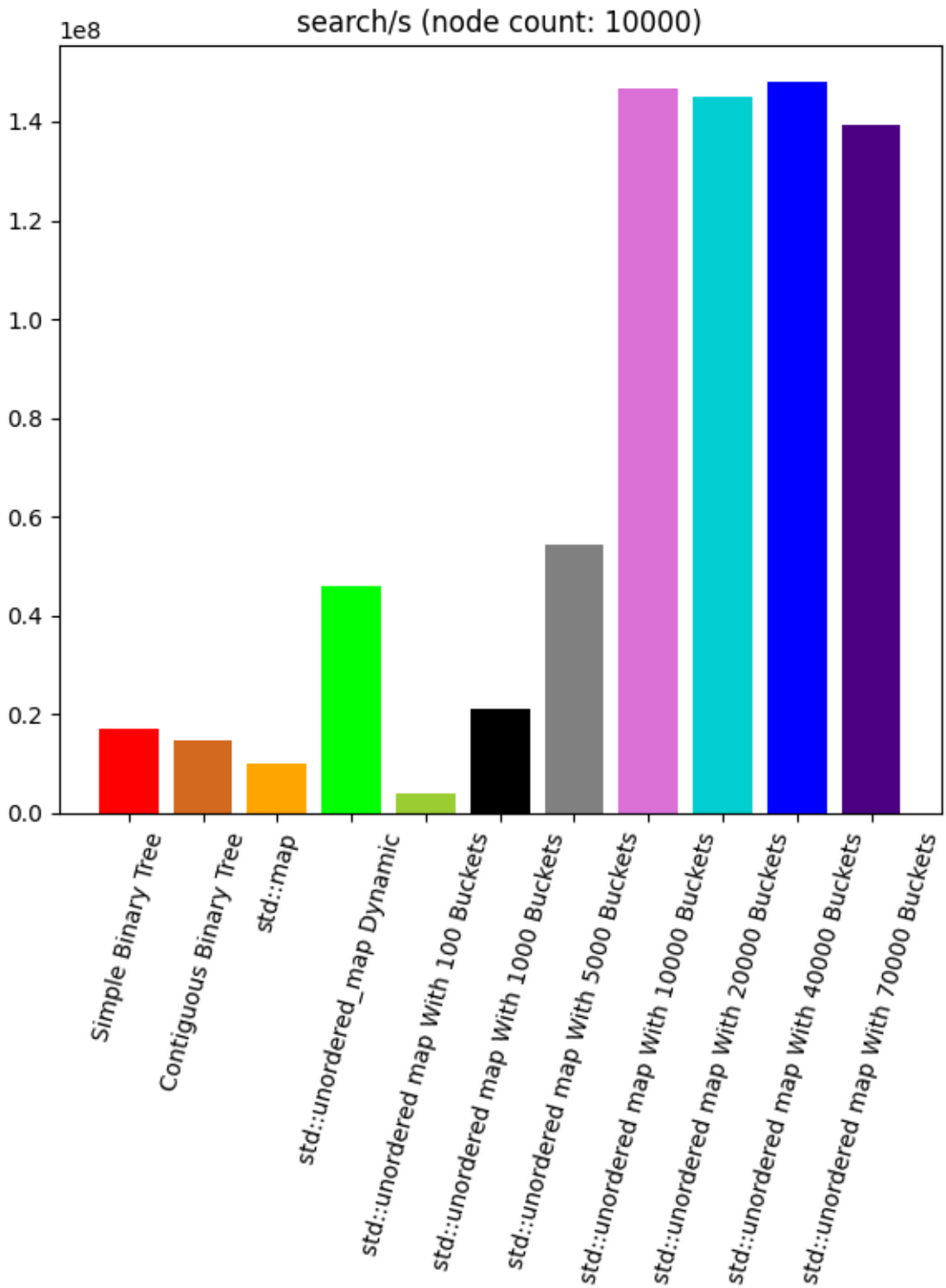

branch misses (node count: 10000)
0
1
2
3
4
5
6
7
8
Simple Binary Tree
Contiguous Binary Tree
std::map
std::unordered_map Dynamic
std::unordered map With 100 Buckets
std::unordered map With 1000 Buckets
std::unordered map With 5000 Buckets
std::unordered map With 10000 Buckets
std::unordered map With 20000 Buckets
std::unordered map With 40000 Buckets
std::unordered map With 70000 Buckets

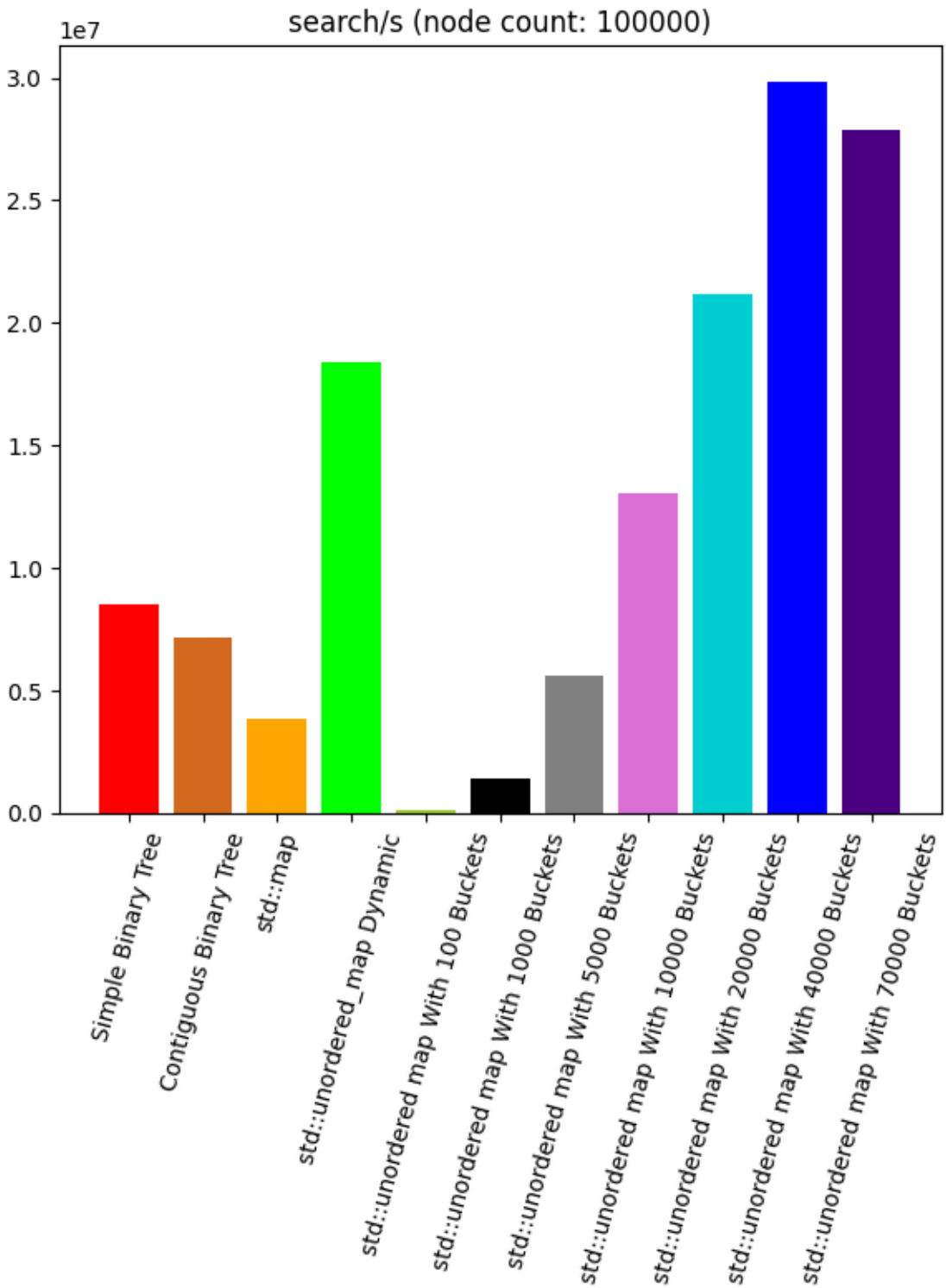

branch misses (node count: 100000)
0
2
4
6
8
Simple Binary Tree
Contiguous Binary Tree
std::map
std::unordered_map Dynamic
std::unordered map With 100 Buckets
std::unordered map With 1000 Buckets
std::unordered map With 5000 Buckets
std::unordered map With 10000 Buckets
std::unordered map With 20000 Buckets
std::unordered map With 40000 Buckets
std::unordered map With 70000 Buckets

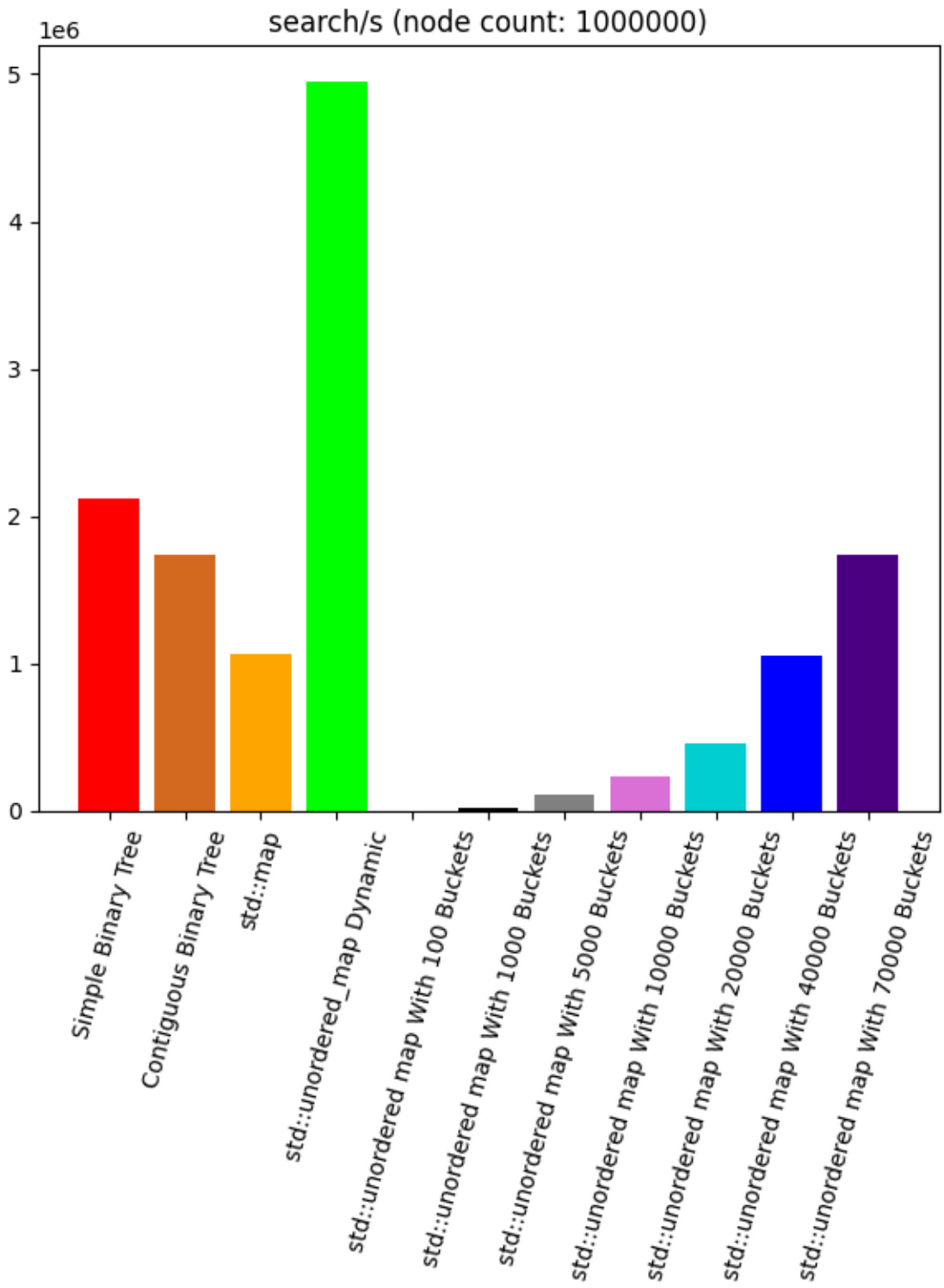

branch misses (node count: 1000000)
0
2
4
6
8
10
Simple Binary Tree
Contiguous Binary Tree
std::map
std::unordered_map Dynamic
std::unordered map With 100 Buckets
std::unordered map With 1000 Buckets
std::unordered map With 5000 Buckets
std::unordered map With 10000 Buckets
std::unordered map With 20000 Buckets
std::unordered map With 40000 Buckets
std::unordered map With 70000 Buckets

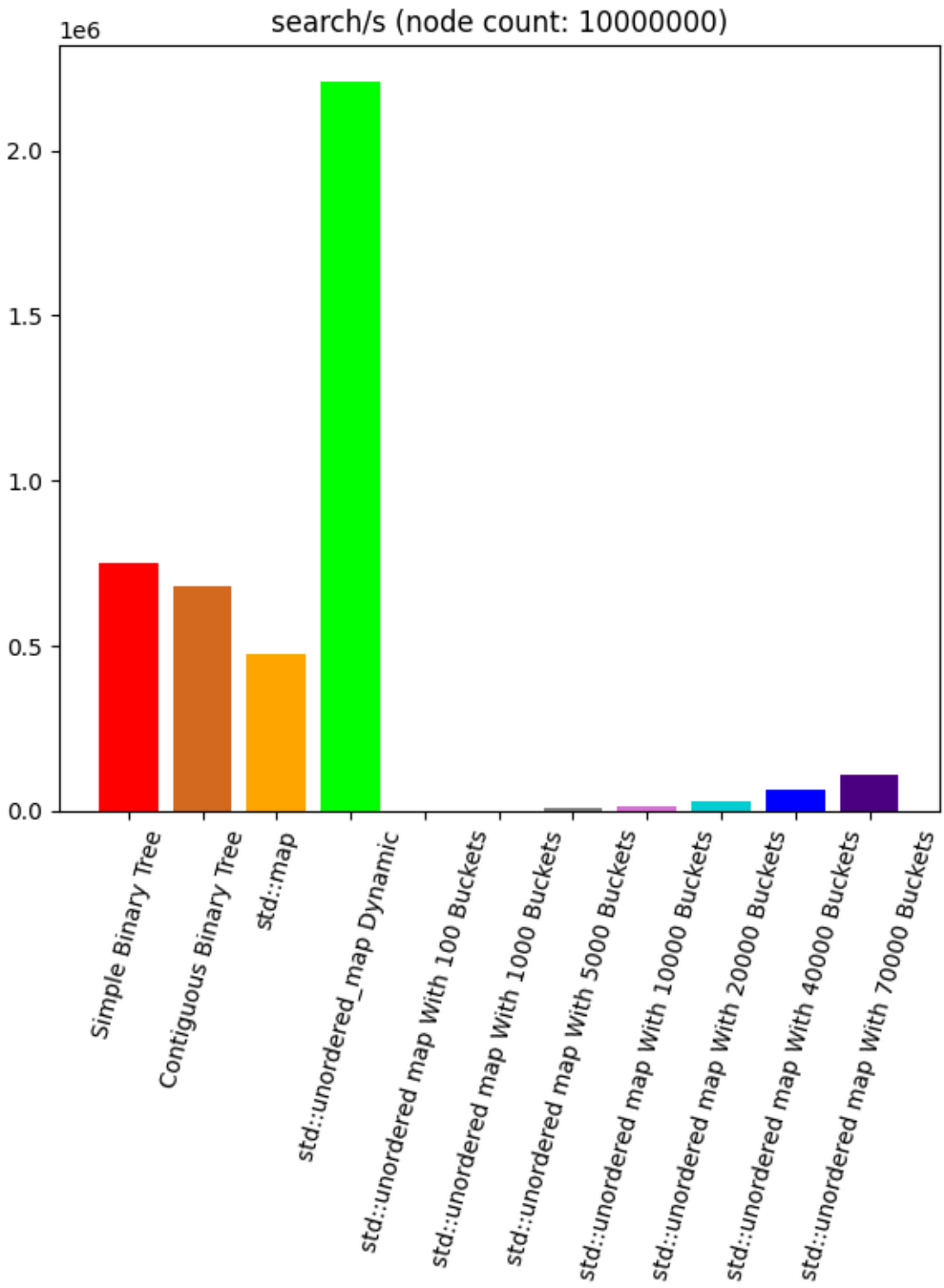

branch misses (node count: 10000000)
0
2
4
6
8
10
12
Simple Binary Tree
Contiguous Binary Tree
std::map
std::unordered_map Dynamic
std::unordered map With 100 Buckets
std::unordered map With 1000 Buckets
std::unordered map With 5000 Buckets
std::unordered map With 10000 Buckets
std::unordered map With 20000 Buckets
std::unordered map With 40000 Buckets
std::unordered map With 70000 Buckets

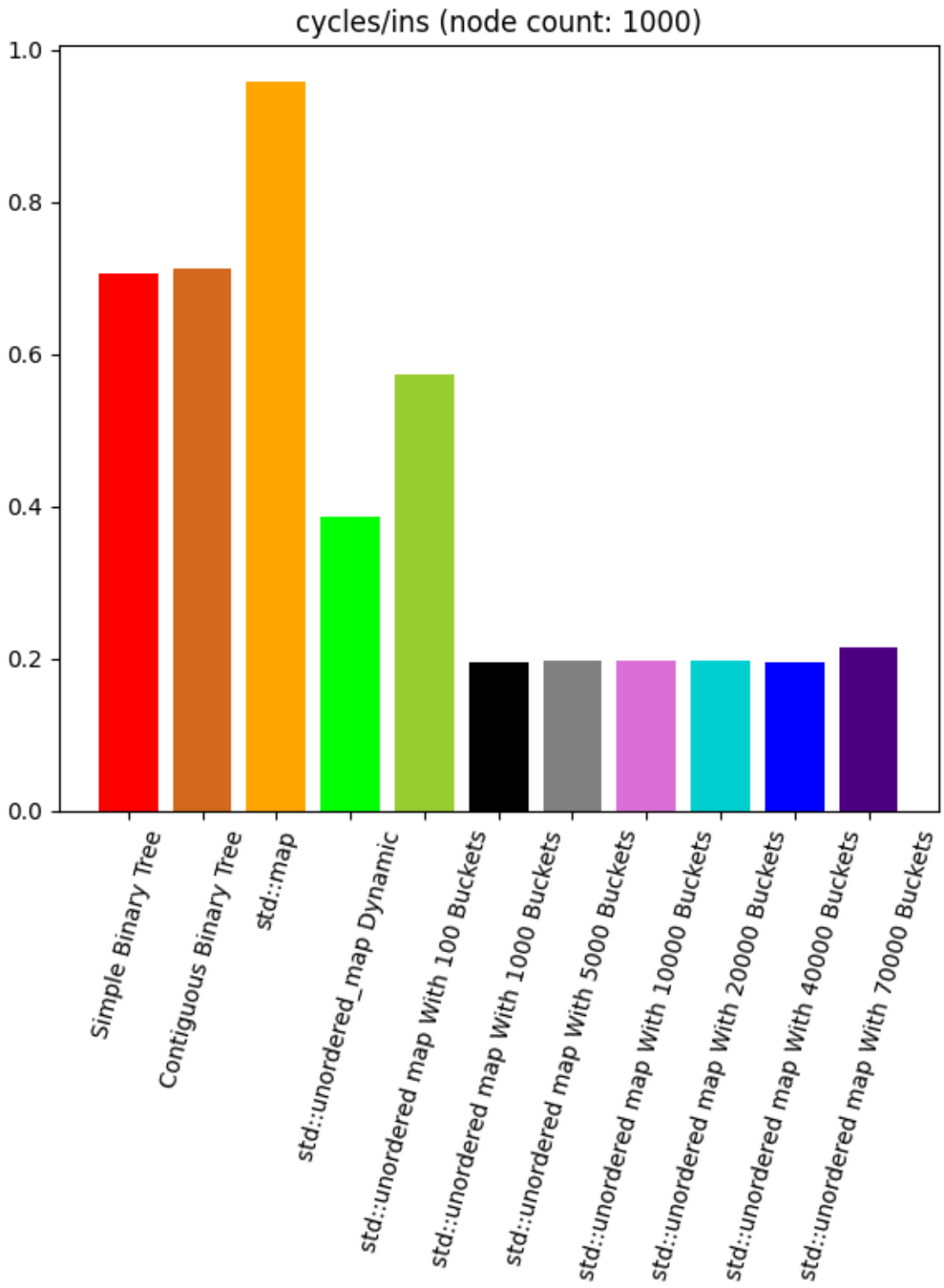

search/s (node count: 1000)
1e8
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
Simple Binary Tree
Contiguous Binary Tree
std::map
std::unordered_map Dynamic
std::unordered map With 100 Buckets
std::unordered map With 1000 Buckets
std::unordered map With 5000 Buckets
std::unordered map With 10000 Buckets
std::unordered map With 20000 Buckets
std::unordered map With 40000 Buckets
std::unordered map With 70000 Buckets

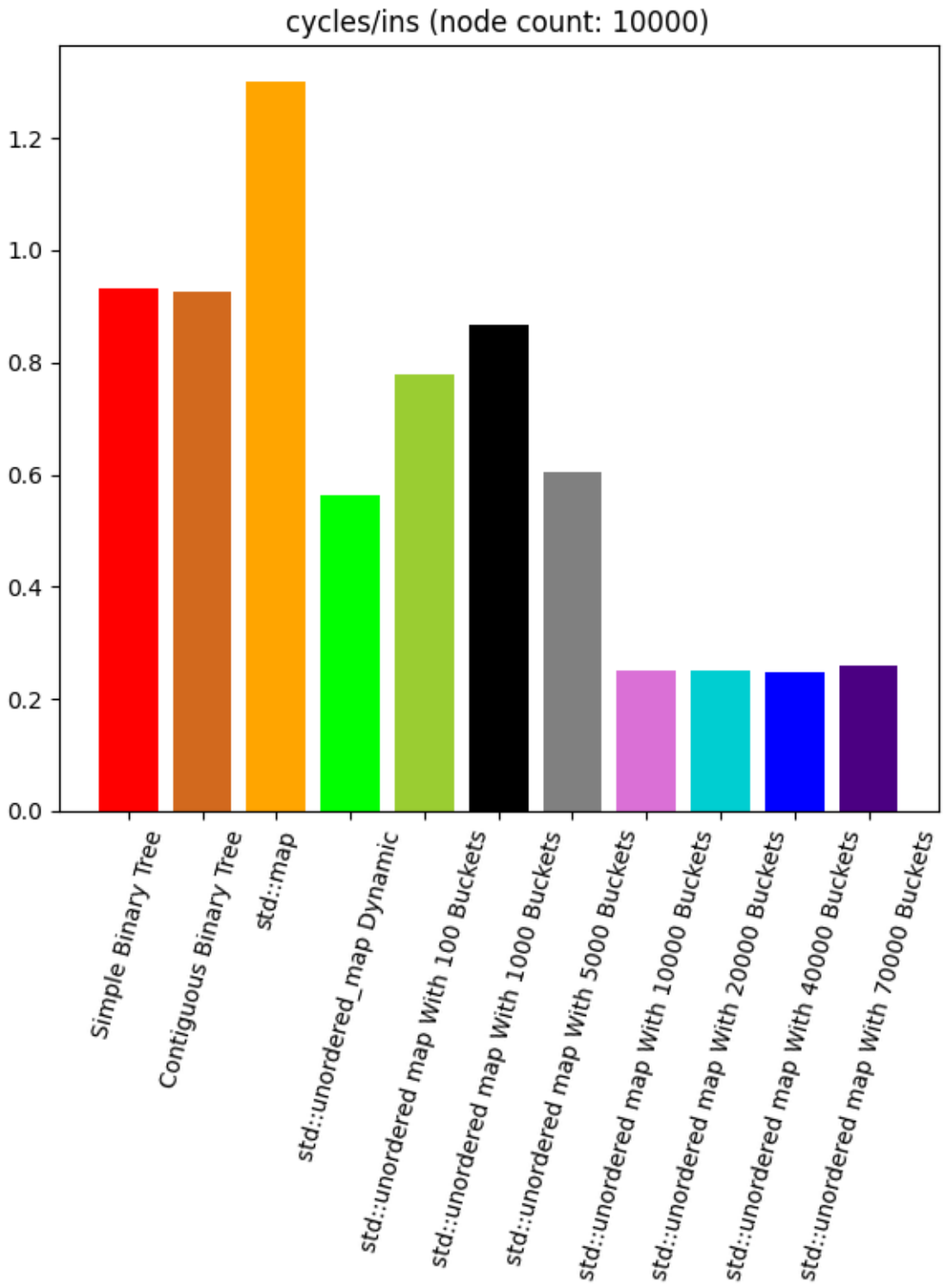

search/s (node count: 10000)
1e8
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
Simple Binary Tree
Contiguous Binary Tree
std::map
std::unordered_map Dynamic
std::unordered map With 100 Buckets
std::unordered map With 1000 Buckets
std::unordered map With 5000 Buckets
std::unordered map With 10000 Buckets
std::unordered map With 20000 Buckets
std::unordered map With 40000 Buckets
std::unordered map With 70000 Buckets

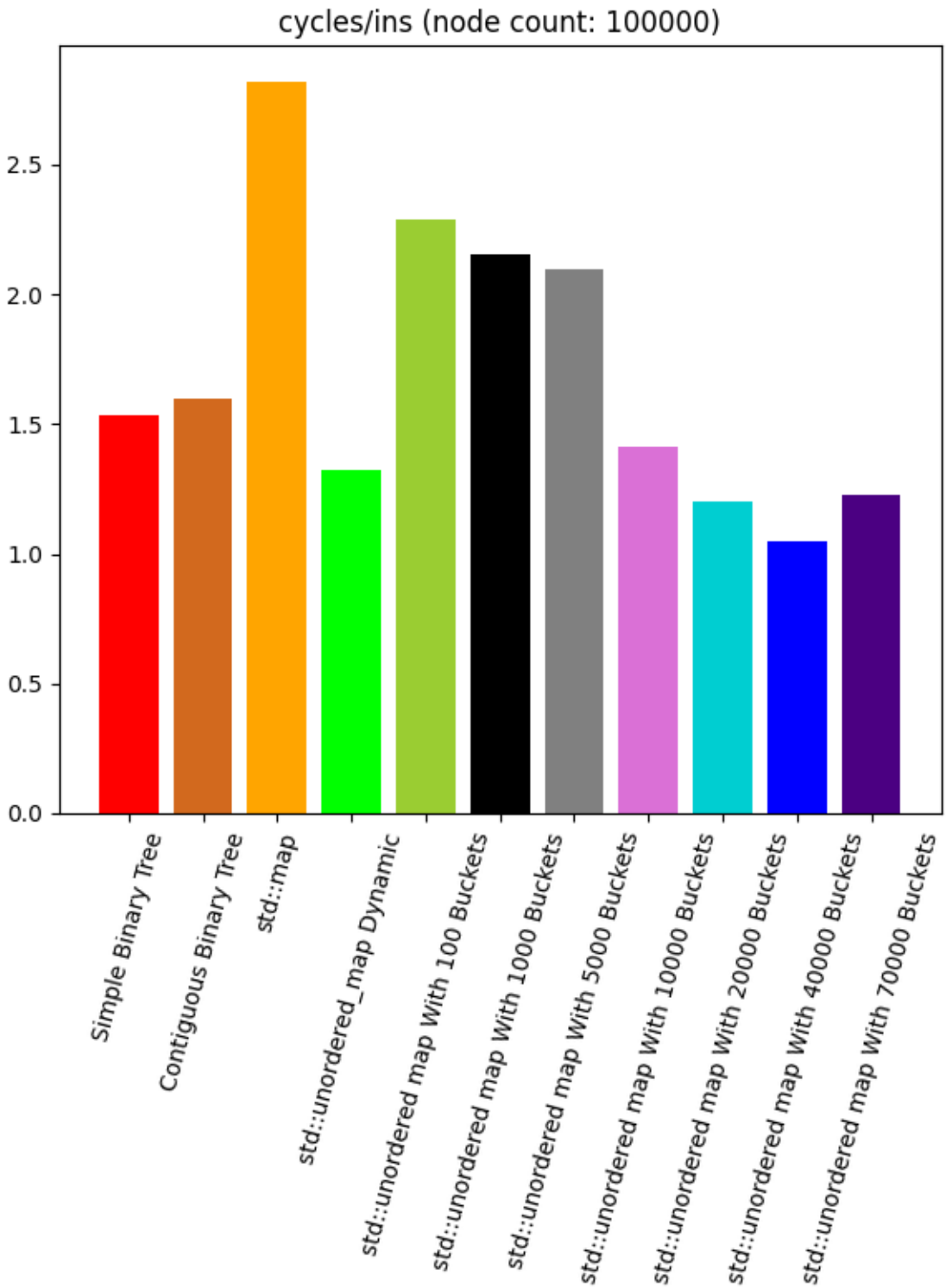

search/s (node count: 100000)
1e7
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Simple Binary Tree
Contiguous Binary Tree
std::map
std::unordered_map Dynamic
std::unordered map With 100 Buckets
std::unordered map With 1000 Buckets
std::unordered map With 5000 Buckets
std::unordered map With 10000 Buckets
std::unordered map With 20000 Buckets
std::unordered map With 40000 Buckets
std::unordered map With 70000 Buckets

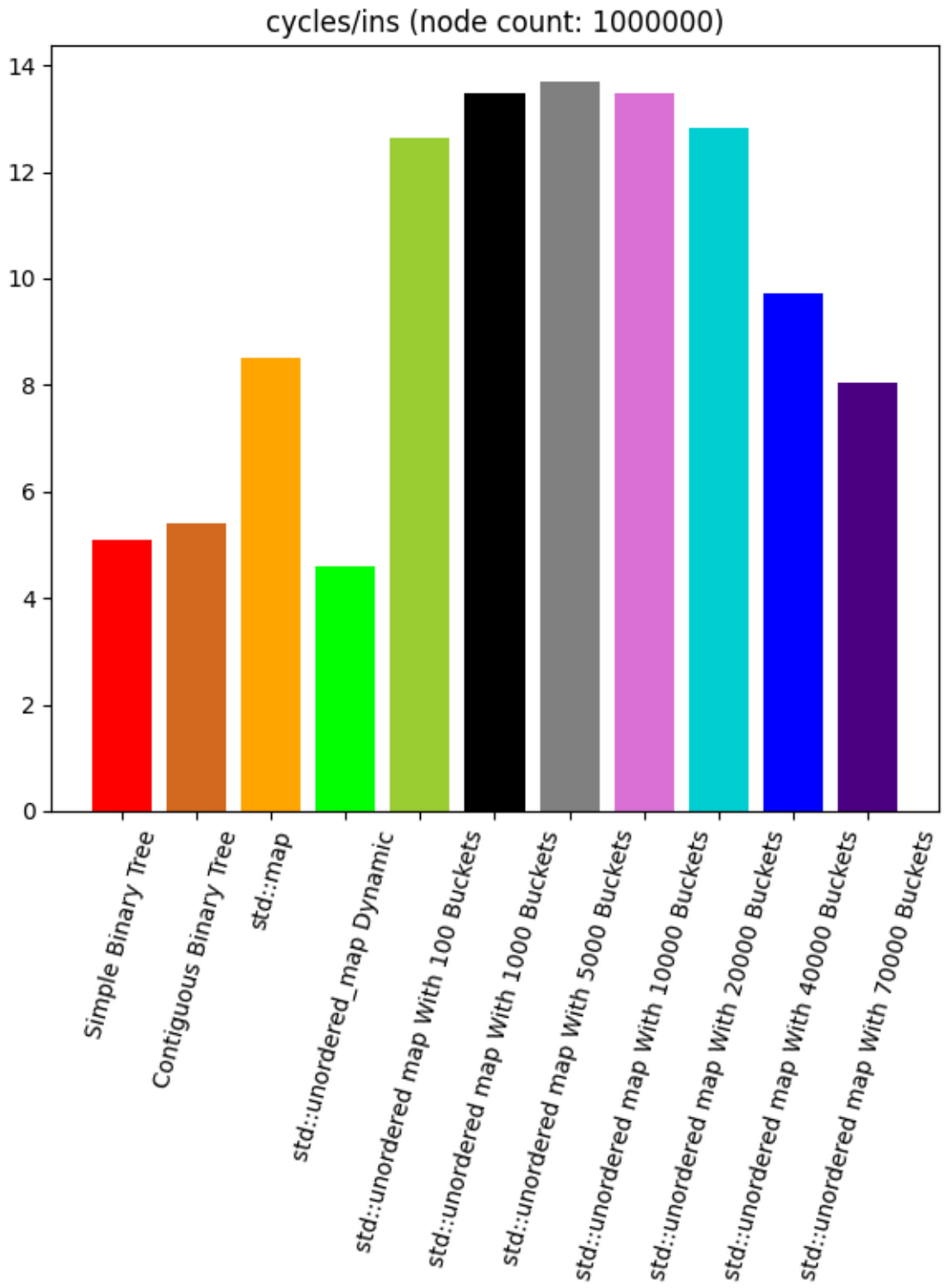

search/s (node count: 1000000)
1e6
5
4
3
2
1
0
Simple Binary Tree
Contiguous Binary Tree
std::map
std::unordered_map Dynamic
std::unordered map With 100 Buckets
std::unordered map With 1000 Buckets
std::unordered map With 5000 Buckets
std::unordered map With 10000 Buckets
std::unordered map With 20000 Buckets
std::unordered map With 40000 Buckets
std::unordered map With 70000 Buckets

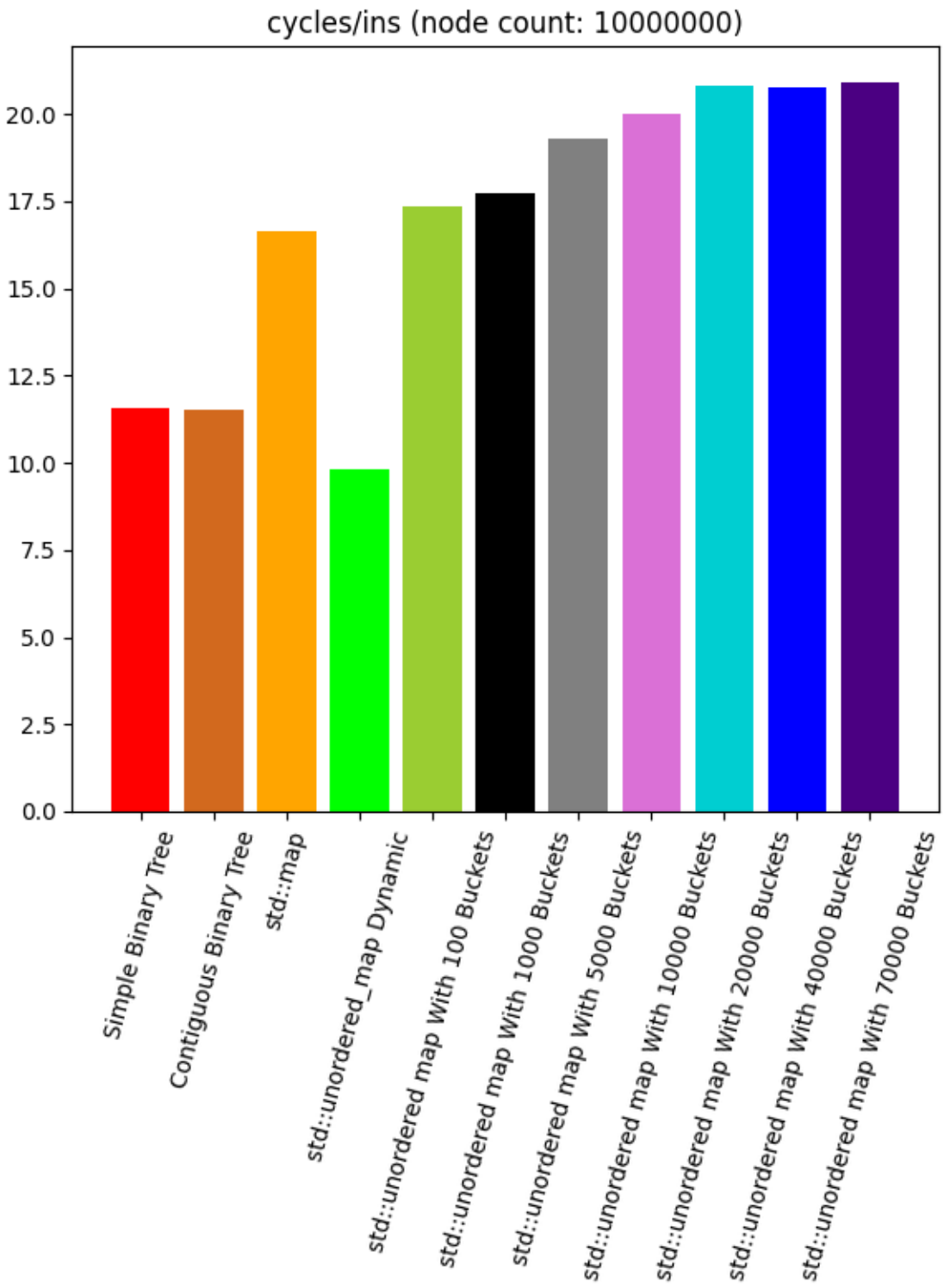

search/s (node count: 10000000)
1e6
2.0
1.5
1.0
0.5
0.0
Simple Binary Tree
Contiguous Binary Tree
std::map
std::unordered_map Dynamic
std::unordered map With 100 Buckets
std::unordered map With 1000 Buckets
std::unordered map With 5000 Buckets
std::unordered map With 10000 Buckets
std::unordered map With 20000 Buckets
std::unordered map With 40000 Buckets
std::unordered map With 70000 Buckets

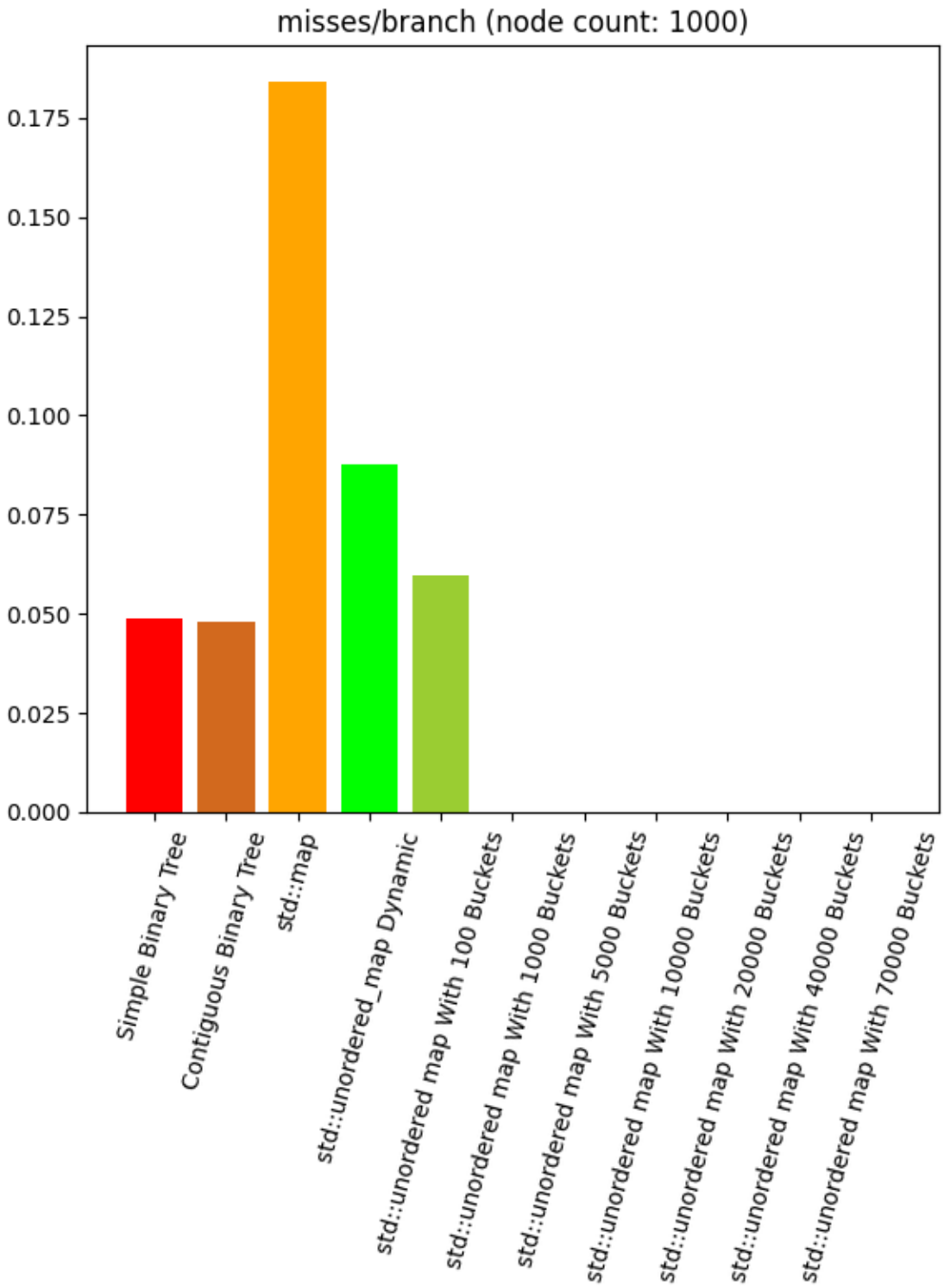

cycles/ins (node count: 1000)
1.0
0.8
0.6
0.4
0.2
0.0
Simple Binary Tree
Contiguous Binary Tree
std::map
std::unordered_map Dynamic
std::unordered map With 100 Buckets
std::unordered map With 1000 Buckets
std::unordered map With 5000 Buckets
std::unordered map With 10000 Buckets
std::unordered map With 20000 Buckets
std::unordered map With 40000 Buckets
std::unordered map With 70000 Buckets

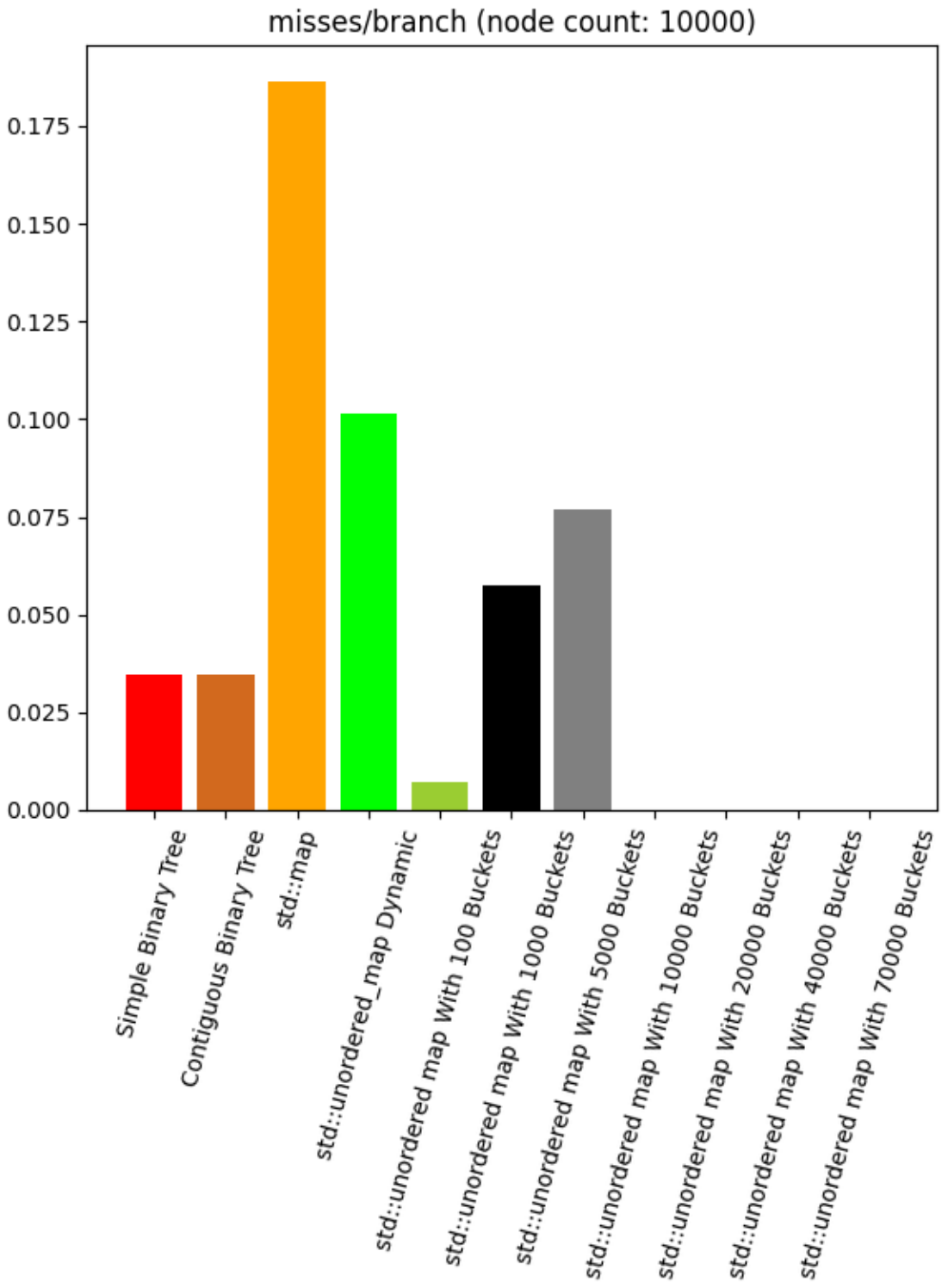

cycles/ins (node count: 10000)
1.2
1.0
0.8
0.6
0.4
0.2
0.0
Simple Binary Tree
Contiguous Binary Tree
std::map
std::unordered_map Dynamic
std::unordered map With 100 Buckets
std::unordered map With 1000 Buckets
std::unordered map With 5000 Buckets
std::unordered map With 10000 Buckets
std::unordered map With 20000 Buckets
std::unordered map With 40000 Buckets
std::unordered map With 70000 Buckets

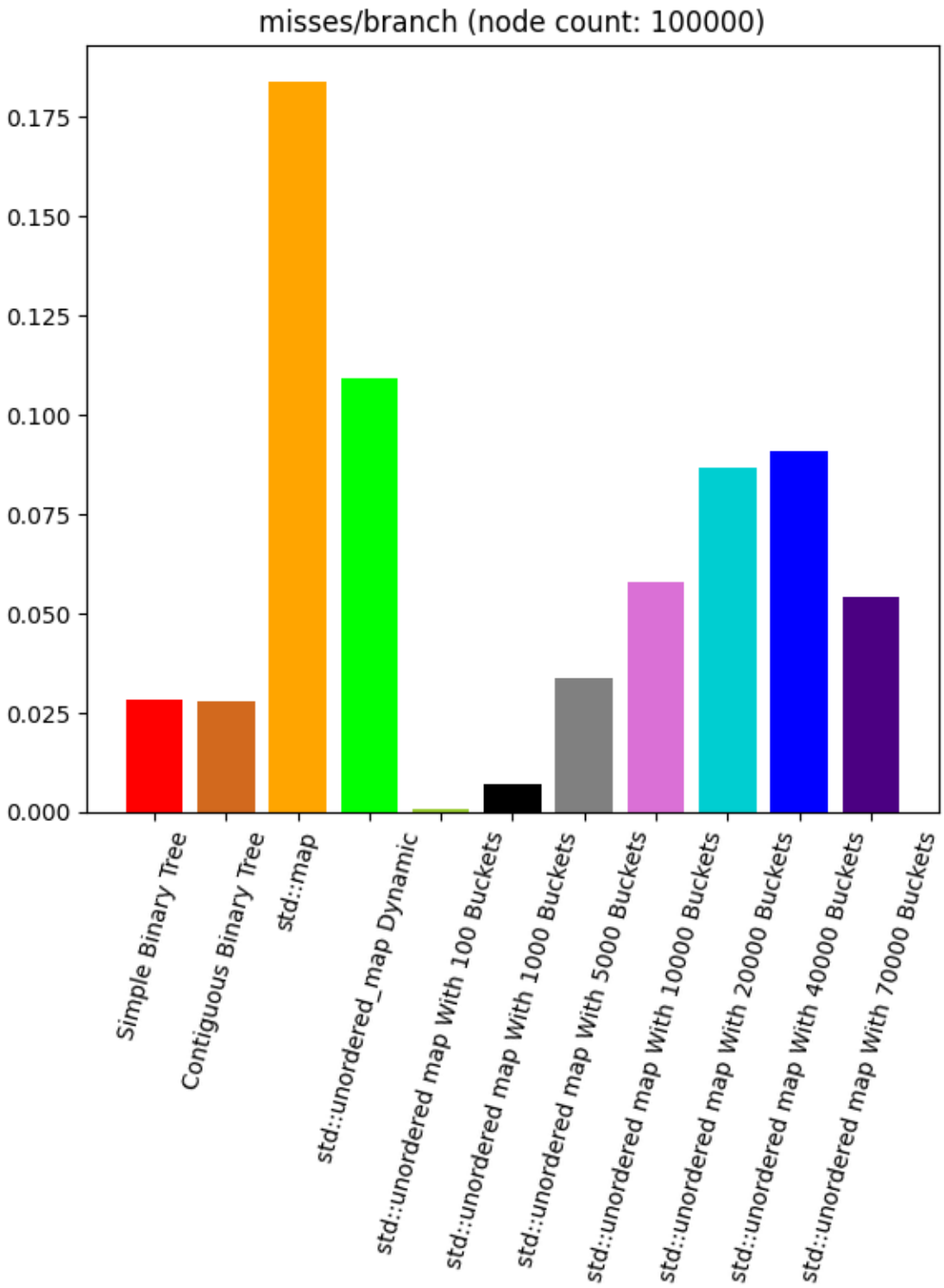

cycles/ins (node count: 100000)
2.5
2.0
1.5
1.0
0.5
0.0
Simple Binary Tree
Contiguous Binary Tree
std::map
std::unordered_map Dynamic
std::unordered map With 100 Buckets
std::unordered map With 1000 Buckets
std::unordered map With 5000 Buckets
std::unordered map With 10000 Buckets
std::unordered map With 20000 Buckets
std::unordered map With 40000 Buckets
std::unordered map With 70000 Buckets

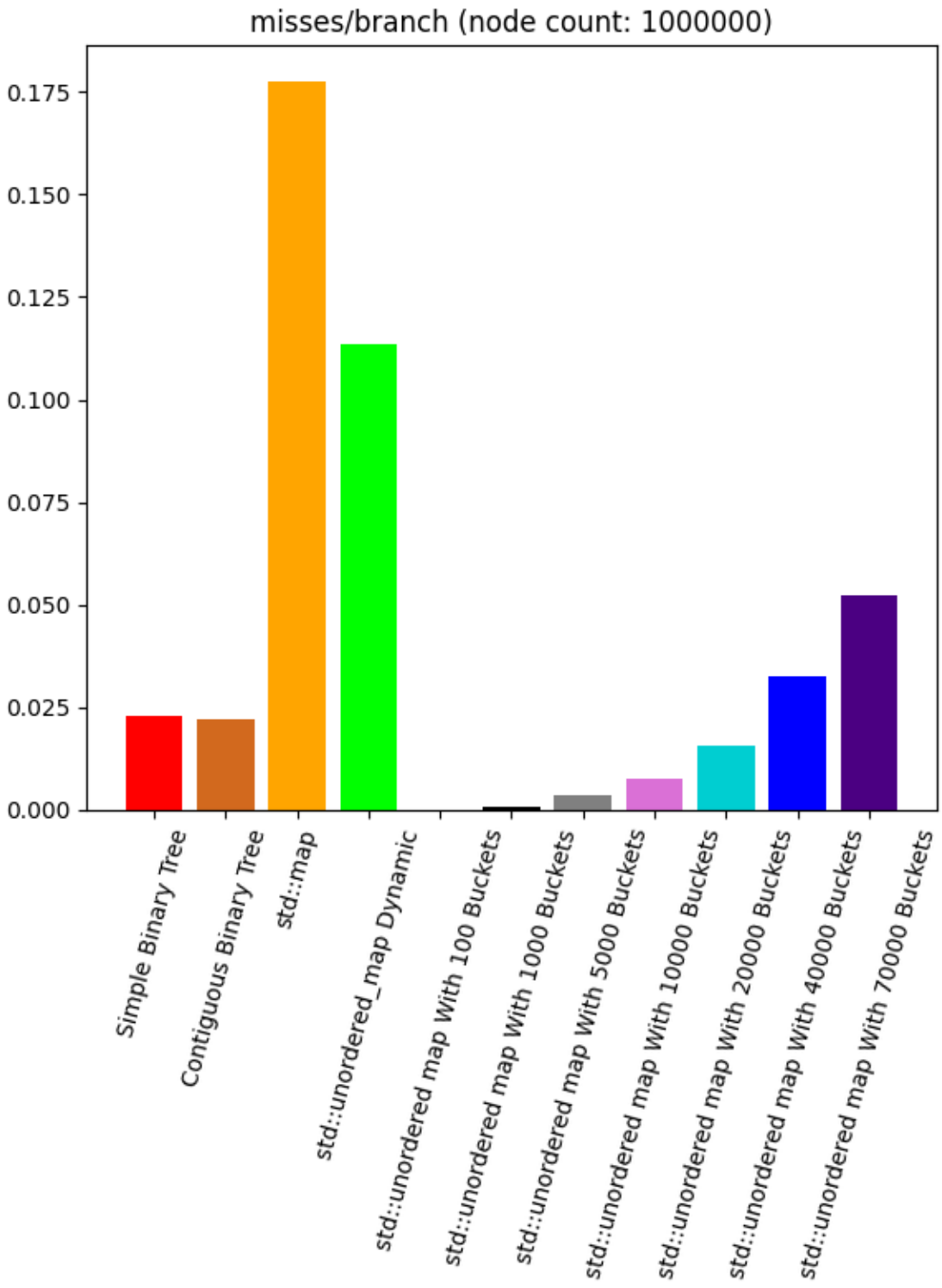

cycles/ins (node count: 1000000)
0
2
4
6
8
10
12
14
Simple Binary Tree
Contiguous Binary Tree
std::map
std::unordered_map Dynamic
std::unordered map With 100 Buckets
std::unordered map With 1000 Buckets
std::unordered map With 5000 Buckets
std::unordered map With 10000 Buckets
std::unordered map With 20000 Buckets
std::unordered map With 40000 Buckets
std::unordered map With 70000 Buckets

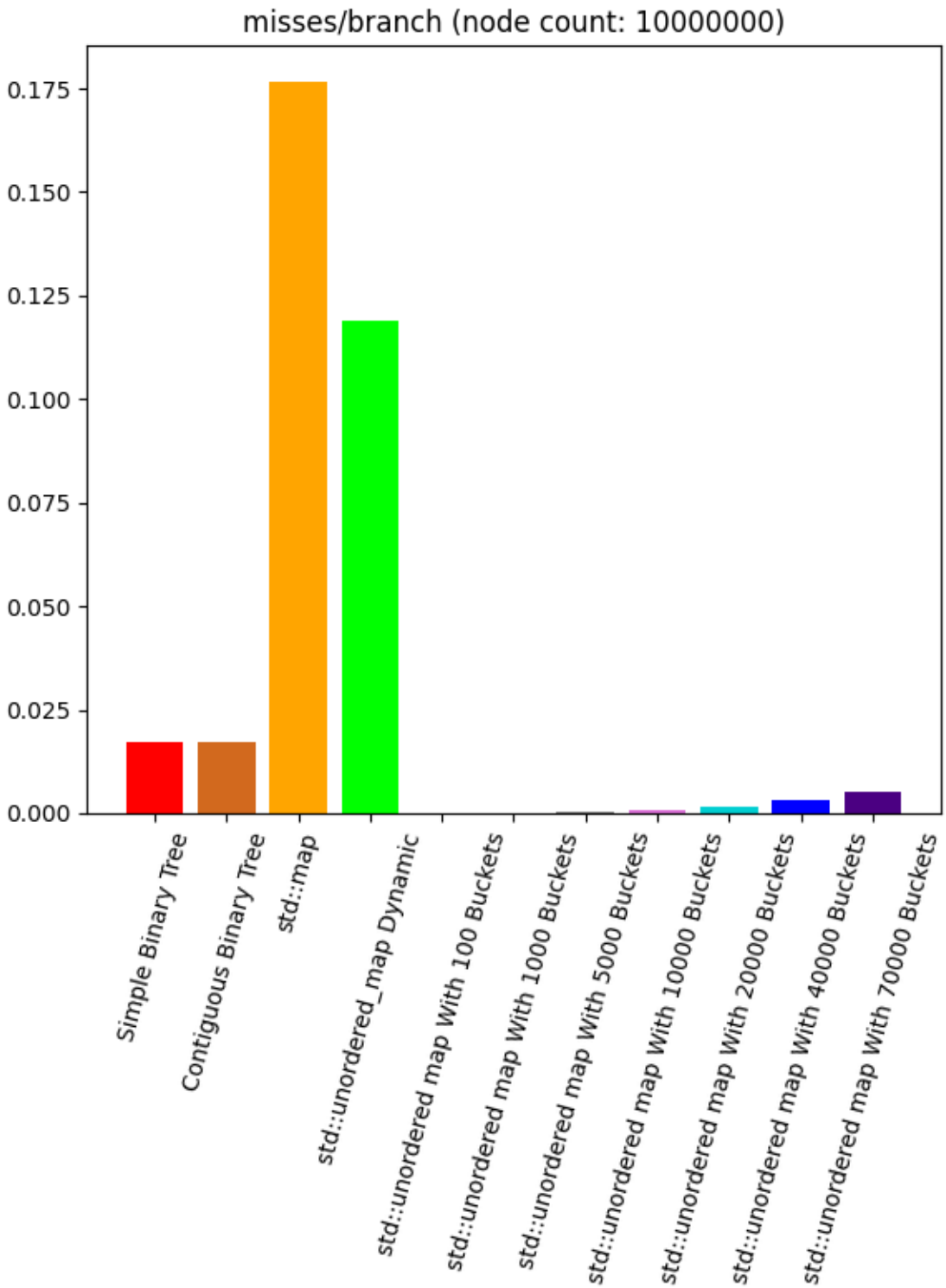

cycles/ins (node count: 10000000)
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
Simple Binary Tree
Contiguous Binary Tree
std::map
std::unordered_map Dynamic
std::unordered map With 100 Buckets
std::unordered map With 1000 Buckets
std::unordered map With 5000 Buckets
std::unordered map With 10000 Buckets
std::unordered map With 20000 Buckets
std::unordered map With 40000 Buckets
std::unordered map With 70000 Buckets


misses/branch (node count: 1000)
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
Simple Binary Tree
Contiguous Binary Tree
std::map
std::unordered_map Dynamic
std::unordered map With 100 Buckets
std::unordered map With 1000 Buckets
std::unordered map With 5000 Buckets
std::unordered map With 10000 Buckets
std::unordered map With 20000 Buckets
std::unordered map With 40000 Buckets
std::unordered map With 70000 Buckets


misses/branch (node count: 10000)
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
Simple Binary Tree
Contiguous Binary Tree
std::map
std::unordered_map Dynamic
std::unordered map With 100 Buckets
std::unordered map With 1000 Buckets
std::unordered map With 5000 Buckets
std::unordered map With 10000 Buckets
std::unordered map With 20000 Buckets
std::unordered map With 40000 Buckets
std::unordered map With 70000 Buckets


misses/branch (node count: 100000)
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
Simple Binary Tree
Contiguous Binary Tree
std::map
std::unordered_map Dynamic
std::unordered map With 100 Buckets
std::unordered map With 1000 Buckets
std::unordered map With 5000 Buckets
std::unordered map With 10000 Buckets
std::unordered map With 20000 Buckets
std::unordered map With 40000 Buckets
std::unordered map With 70000 Buckets


misses/branch (node count: 1000000)
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
Simple Binary Tree
Contiguous Binary Tree
std::map
std::unordered_map Dynamic
std::unordered map With 100 Buckets
std::unordered map With 1000 Buckets
std::unordered map With 5000 Buckets
std::unordered map With 10000 Buckets
std::unordered map With 20000 Buckets
std::unordered map With 40000 Buckets
std::unordered map With 70000 Buckets




```
Simple Binary Tree:
      Nodes        elapsed       search/s  instructions        cycles  \
```

```
0      1000  3.427458e-08  2.917614e+07    102.785523    72.682329
1     10000  5.891518e-08  1.697355e+07    135.377227   126.190191
2    100000  1.173728e-07  8.519864e+06    162.428208   249.452678
3   1000000  4.726596e-07  2.115688e+06    195.497930   992.625215
4  10000000  1.333370e-06  7.499793e+05    236.032083  2729.714110

   cycles/ins   branches  branch misses  misses/branch
0    0.707126  23.653022       1.155964       0.048872
1    0.932138  32.964932       1.149369       0.034866
2    1.535772  40.693813       1.160470       0.028517
3    5.077421  50.142351       1.156322       0.023061
4   11.565013  61.724336       1.071421       0.017358
```

Contiguous Binary Tree:

```
      Nodes       elapsed      search/s  instructions       cycles  \
0      1000  3.916635e-08  2.553212e+07    117.469174    83.646732
1     10000  6.698975e-08  1.492766e+07    154.706531   143.402543
2    100000  1.393964e-07  7.173785e+06    185.632236   296.278390
3   1000000  5.755121e-07  1.737583e+06    223.426288  1204.710302
4  10000000  1.463927e-06  6.830941e+05    269.750490  3106.788457

   cycles/ins   branches  branch misses  misses/branch
0    0.712074  27.032030       1.297485       0.047998
1    0.926933  37.671273       1.307186       0.034700
2    1.596050  46.507213       1.291515       0.027770
3    5.391981  57.305625       1.272271       0.022202
4   11.517267  70.541845       1.203090       0.017055
```

std::map:

```
      Nodes       elapsed      search/s  instructions       cycles  \
0      1000  6.003871e-08  1.665592e+07    133.042920   127.407591
1     10000  1.004480e-07  9.955396e+06    165.253641   214.749389
2    100000  2.604784e-07  3.839089e+06    196.577295   553.584208
3   1000000  9.371887e-07  1.067021e+06    230.310744  1958.170002
4  10000000  2.100107e-06  4.761663e+05    263.138610  4374.780900

   cycles/ins   branches  branch misses  misses/branch
0    0.957643  31.853759       5.860312       0.183976
1    1.299514  41.812118       7.787738       0.186256
2    2.816115  51.459195       9.457451       0.183785
3    8.502295  61.954330      10.987123       0.177342
4   16.625386  72.022487      12.709310       0.176463
```

std::unordered_map Dynamic:

```
      Nodes       elapsed      search/s  instructions      cycles  cycles/ins  \
0      1000  1.422204e-08  7.031342e+07     78.006141   30.139094    0.386368
1     10000  2.177588e-08  4.592237e+07     82.607674   46.450158    0.562298
2    100000  5.428987e-08  1.841964e+07     87.082521  115.244108    1.323390
```

```
3   1000000  2.022141e-07  4.945253e+06     91.901688  422.187819    4.593907
4  10000000  4.532744e-07  2.206169e+06     96.591499  947.366106    9.807966

    branches  branch misses  misses/branch
0   9.550538       0.837206       0.087661
1  10.973162       1.112558       0.101389
2  12.351314       1.351165       0.109394
3  13.850618       1.569914       0.113346
4  15.288927       1.817824       0.118898
```

std::unordered map With 100 Buckets:

```
      Nodes       elapsed      search/s   instructions        cycles  \
0      1000  3.348581e-08  2.986340e+07     115.796341  6.632751e+01
1     10000  2.515556e-07  3.975264e+06     685.293643  5.343246e+02
2    100000  6.930406e-06  1.442917e+05    6363.289132  1.457017e+04
3   1000000  3.770950e-04  2.651852e+03   63440.938372  8.022636e+05
4  10000000  5.199030e-03  1.923436e+02  633408.482744  1.098478e+07

   cycles/ins       branches  branch misses  misses/branch
0    0.572795      18.106830       1.078899       0.059585
1    0.779702     149.529269       1.056974       0.007069
2    2.289722    1459.840348       1.040038       0.000712
3   12.645835   14631.886047       1.404884       0.000096
4   17.342331  146165.991044       2.060842       0.000014
```

std::unordered map With 1000 Buckets:

```
      Nodes       elapsed      search/s  instructions        cycles  \
0      1000  5.470532e-09  1.827976e+08     59.000008  1.156260e+01
1     10000  4.747225e-08  2.106494e+07    115.700782  1.002545e+02
2    100000  7.151728e-07  1.398263e+06    682.444439  1.467782e+03
3   1000000  4.011953e-05  2.492552e+04   6349.355973  8.547832e+04
4  10000000  5.319181e-04  1.879989e+03  63079.744081  1.119849e+06

   cycles/ins      branches  branch misses  misses/branch
0    0.195976      5.000004       0.000016       0.000003
1    0.866498     18.084827       1.041864       0.057610
2    2.150772    148.872130       1.078116       0.007242
3   13.462518   1456.651206       1.090911       0.000749
4   17.752915  14548.623169       1.512393       0.000104
```

std::unordered map With 5000 Buckets:

```
      Nodes       elapsed      search/s  instructions         cycles  \
0      1000  5.410372e-09  1.848302e+08     59.000008      11.625343
1     10000  1.834319e-08  5.451616e+07     65.389119      39.497655
2    100000  1.778785e-07  5.621816e+06    180.243170     378.214801
3   1000000  8.524686e-06  1.173064e+05   1328.338890   18176.946814
4  10000000  1.171406e-04  8.536752e+03  12825.823689  247583.157577
```

```
   cycles/ins     branches  branch misses  misses/branch
0    0.197040     5.000003       0.000016       0.000003
1    0.604040     6.474426       0.498514       0.076997
2    2.098359    32.979570       1.118099       0.033903
3   13.683968   297.930621       1.076593       0.003614
4   19.303490  2951.264943       1.180242       0.000400
```

std::unordered map With 10000 Buckets:

```
      Nodes       elapsed      search/s  instructions         cycles  \
0      1000  5.436593e-09  1.839387e+08     59.000007      11.726358
1     10000  6.819468e-09  1.466390e+08     59.000010      14.708451
2    100000  7.643974e-08  1.308220e+07    115.852081     163.433520
3   1000000  4.325830e-06  2.311695e+05    684.489232    9217.874932
4  10000000  5.970849e-05  1.674804e+04   6377.845597  127604.852718

   cycles/ins     branches  branch misses  misses/branch
0    0.198752     5.000003       0.000012       0.000002
1    0.249296     5.000006       0.000016       0.000003
2    1.410709    18.119744       1.050552       0.057978
3   13.466793   149.346823       1.124490       0.007529
4   20.007517  1463.240337       1.120132       0.000766
```

std::unordered map With 20000 Buckets:

```
      Nodes       elapsed      search/s  instructions        cycles  \
0      1000  5.426845e-09  1.842691e+08     59.000009     11.651648
1     10000  6.891297e-09  1.451106e+08     59.000010     14.845007
2    100000  4.723969e-08  2.116864e+07     84.027574    101.014819
3   1000000  2.192916e-06  4.560139e+05    365.372424   4684.804782
4  10000000  3.101795e-05  3.223940e+04   3182.715080  66259.746748

   cycles/ins    branches  branch misses  misses/branch
0    0.197486    5.000005       0.000014       0.000003
1    0.251610    5.000006       0.000016       0.000003
2    1.202163   10.775597       0.934885       0.086759
3   12.821999   75.702966       1.200855       0.015863
4   20.818623  725.880413       1.085353       0.001495
```

std::unordered map With 40000 Buckets:

```
      Nodes       elapsed      search/s  instructions        cycles  \
0      1000  5.407252e-09  1.849368e+08     59.000007     11.601655
1     10000  6.760227e-09  1.479240e+08     59.000010     14.567762
2    100000  3.354845e-08  2.980764e+07     68.596986     71.932770
3   1000000  9.437248e-07  1.059631e+06    207.136374   2010.413392
4  10000000  1.577713e-05  6.338289e+04   1598.671589  33196.636219

   cycles/ins    branches  branch misses  misses/branch
0    0.196638    5.000002       0.000013       0.000003
1    0.246911    5.000006       0.000016       0.000003
```

```
2    1.048629    7.214686       0.654750       0.090752
3    9.705748   39.185944       1.279602       0.032655
4   20.765138  360.320806       1.083626       0.003007
```

std::unordered map With 70000 Buckets:

```
      Nodes       elapsed      search/s  instructions        cycles  \
0      1000  5.939181e-09  1.683734e+08     59.000009     12.752772
1     10000  7.179683e-09  1.392819e+08     59.000010     15.350045
2    100000  3.590328e-08  2.785261e+07     62.531140     76.610767
3   1000000  5.761556e-07  1.735642e+06    141.690495   1137.582263
4  10000000  9.285400e-06  1.076960e+05    944.922392  19743.928277

   cycles/ins    branches  branch misses  misses/branch
0    0.216149    5.000005       0.000014       0.000003
1    0.260170    5.000005       0.000015       0.000003
2    1.225162    5.814875       0.315988       0.054341
3    8.028642   24.082564       1.257039       0.052197
4   20.894762  209.450579       1.108169       0.005291
```